

XLIFF Version 2.3. Part 2: Extended

Committee Specification Draft 01

16 May 2026

Specification URIs

This stage:

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<https://docs.oasis-open.org/xliff/xliff-core/v2.2/cs01/xliff-extended-v2.2-cs01-part2.pdf>
<https://docs.oasis-open.org/xliff/xliff-core/v2.2/cs01/xliff-extended-v2.2-cs01-part2.xml>

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<https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html> (Authoritative)
<https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.pdf>
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Latest stage:

<https://docs.oasis-open.org/xliff/xliff-core/v2.2/xliff-extended-v2.2-part2.html> (Authoritative)
<https://docs.oasis-open.org/xliff/xliff-core/v2.2/xliff-extended-v2.2-part2.pdf>
<https://docs.oasis-open.org/xliff/xliff-core/v2.2/xliff-extended-v2.2-part2.xml>

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Additional artifacts:

This prose specification is one component of a Work Product that also includes:

- XLIFF Version 2.2. Part 1: Core. <https://docs.oasis-open.org/xliff/xliff-core/v2.2/xliff-core-v2.2-part1.html>
- XLIFF Version 2.2. Part 2: Extended. (this document) <https://docs.oasis-open.org/xliff/xliff-core/v2.2/xliff-extended-v2.2-part2.html>
- XML schemas: <https://docs.oasis-open.org/xliff/xliff-core/v2.2/schemas/>

Related Work:

This specification replaces or supersedes:

- XLIFF Version 2.1. 13 February 2018. OASIS Standard. Latest Stage: <https://docs.oasis-open.org/xliff/xliff-core/v2.1/xliff-core-v2.1.html>

Declared XML Namespaces:

- `urn:oasis:names:tc:xliff:document:2.2`

- urn:oasis:names:tc:xliff:matches:2.0
- urn:oasis:names:tc:xliff:glossary:2.0
- urn:oasis:names:tc:xliff:fs:2.0
- urn:oasis:names:tc:xliff:metadata:2.0
- urn:oasis:names:tc:xliff:resourcedata:2.0
- urn:oasis:names:tc:xliff:sizerestriction:2.0
- urn:oasis:names:tc:xliff:validation:2.0
- urn:oasis:names:tc:xliff:pgs:1.0
- urn:oasis:names:tc:xliff:pvn:2.3

Key words:

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] and [RFC8174] when, and only when, they appear in all capitals, as shown here.

Abstract:

This document is Part 2 of a multi-part specification which defines Version 2.3 of the XML Localisation Interchange File Format (XLIFF). The purpose of this vocabulary is to store localizable data and carry it from one step of the localization process to the other, while allowing interoperability between and among tools.

Status:

This document was last revised or approved by the XLIFF TC on the above date. The level of approval is also listed above. Check the "Latest stage" location noted above for possible later revisions of this document. Any other numbered Versions and other technical work produced by the Technical Committee (TC) are listed at <https://groups.oasis-open.org/communities/tc-community-home2?CommunityKey=3d0f1f56-8477-4b53-9b14-018dc7d3eef>.

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1 Introduction

XLIFF is the XML Localization Interchange File Format designed by a group of multilingual content publishers, software providers, localization service providers, localization tools providers, and researchers. It is intended to give any multilingual content owner a single interchange file format that can be understood by any localization provider, using any conformant localization tool. While the primary focus is on being a lossless interchange format, usage of XLIFF as a processing format is neither encouraged nor discouraged or prohibited.

All text is normative unless otherwise labeled.

1.1 Changes from earlier Versions

XLIFF 2.2 has two main significant differences from the previous version published (XLIFF 2.1):

1. XLIFF 2.2 is presented in two separate documents:
 - XLIFF Version 2.2. Part 1: Core presents the XLIFF core, which is the minimum set of XML elements and attributes required to create a valid XLIFF file.
 - XLIFF Version 2.2. Part 2: Extended presents the XLIFF core as well as the optional modules that were created to store information about specific processes. For example, the [Translation Candidates Module](#) was designed to store translation suggestions and their associated metadata.

In previous versions of XLIFF, the specification was always presented in a single document. This change in the presentation mode was introduced to produce a simplified version (Part 1: Core) of the specification that would be easier to use, especially for those agents who are not interested in implementing the optional modules.

2. XLIFF 2.2 includes the new [Plural, Gender, and Select Module](#), which was designed to store information needed to represent and process messages with variants. The new module description can be found in the document XLIFF Version 2.2. Part 2: Extended.

For a detailed list of changes made between version 2.1 and the current version (2.2) please see the [Specification Change Tracking](#) section in Appendix D.

Note that all changes introduced in version 2.2 were designed to maintain compatibility with versions 2.0 and 2.1.

1.2 Definition of terms

Agent

any application or tool that generates (creates), reads, edits, writes, processes, stores, renders or otherwise handles XLIFF documents.

Agent is the most general application conformance target that subsumes all other specialized user agents disregarding whether they are defined in this specification or not.

Enrich, Enriching

the process of associating module and extension based metadata and resources with the Extracted XLIFF payload

Processing Requirements

- Enriching MAY happen at the time of Extraction.



Note

Extractor knowledge of the native format is not assumed while Enriching.

Enricher, Enricher Agent

any Agent that performs the Enriching process

Extract, Extraction

the process of encoding localizable content from a native content or User Interface format as XLIFF payload, so that localizable parts of the content in the source language are available for translation into the target language along with the necessary context information

Extractor, Extractor Agent

any Agent that performs the Extraction process

Merge, Merging

the process of importing XLIFF payload back to the originating native format, based on the full knowledge of the Extraction mechanism, so that the localized content or User Interface strings replace the source language in the native format

Merger, Merger Agent

an Agent that performs the Merge process



Warning

Unless specified otherwise, any Merger is deemed to have the same knowledge of the native format as the Extractor throughout the specification.

Mergers independent of Extractors can succeed, but it is out of scope of this specification to specify interoperability for merging back without the full Extractor knowledge of the native format.

Modify, Modification

the process of changing core and module XLIFF structural and inline elements that were previously created by other Writers

Processing Requirements

- XLIFF elements MAY be Modified and Enriched at the same time.



Note

Extractor or Enricher knowledge of the native format is not assumed while modifying.

Modifier, Modifier Agent

an Agent that performs the Modification process

Translation, Translate

a rendering of the meaning of the source text, expressed in the target language

Writer, Writer Agent

an Agent that creates, generates, or otherwise writes an XLIFF document for whatever purpose, including but not limited to Extractor, Modifier, and Enricher Agents.



Note

Since XLIFF is intended as an exchange format rather than a processing format, many applications will need to generate XLIFF documents from their internal processing

formats, even in cases when they are processing XLIFF documents created by another Extractor.

1.3 Key concepts

XLIFF Core

The core of XLIFF 2.3 consists of the minimum set of XML elements and attributes required to (a) prepare a document that contains text extracted from one or more files for localization, (b) allow it to be completed with the translation of the extracted text, and (c) allow the generation of translated versions of the original document.

The XML namespace that corresponds to the core subset of XLIFF 2.3 is "urn:oasis:names:tc:xliff:document:2.3".

XLIFF-defined (elements and attributes)

The following is the list of allowed schema URI prefixes for XLIFF-defined elements and attributes:

```
urn:oasis:names:tc:xliff:  
http://www.w3.org/2005/11/its
```

However, the following namespaces are NOT considered XLIFF-defined for the purposes of the XLIFF 2.3 specification:

```
urn:oasis:names:tc:xliff:document:1.0  
urn:oasis:names:tc:xliff:document:1.1  
urn:oasis:names:tc:xliff:document:1.2
```

Elements and attributes from other namespaces are not XLIFF-defined.

XLIFF Document

Any XML document that declares the namespace "urn:oasis:names:tc:xliff:document:2.3" as its main namespace, has `<xliff>` as the root element and complies with the XML Schemas and the declared Constraints that are part of this specification.

XLIFF Module

A module is an OPTIONAL set of XML elements and attributes that stores information about a process applied to an XLIFF Document and the data incorporated into the document as result of that process.

Each official module defined for XLIFF 2.3 has its grammar defined in an independent XML Schema with a separate namespace.

2 Fragment Identification

Because XLIFF documents do not follow the usual behavior of XML documents when it comes to element identifiers, this specification defines how Agents MUST interpret the fragment identifiers in IRIs pointing to XLIFF documents.



Note

Note that some identifiers may change during the localization process. For example `<data>` elements may be re-grouped or not depending on how tools treat identical original data.

Constraints

- A fragment identifier MUST match the following format:

```
<expression>      ::= "#" ["/"] <selector>
                    {<selectorSeparator> <selector>}
<selector>        ::= [<prefix> <prefixSeparator>] <id>
<prefix>          ::= NMTOKEN
<id>              ::= NMTOKEN
<prefixSeparator> ::= "="
<selectorSeparator> ::= "/"
```

- There MUST NOT be two identical prefixes in the expression.
- When used, the following selectors MUST be declared in this order: file selector, group selector and unit selector.
- The selectors for modules or extensions, `<note>`, `<segment>` or `<ignorable>` or source inline elements, target inline elements and `<data>` have the following constraints:
 - Only one of them MAY be used in the expression.
 - The one used MUST be the last selector of the expression.



Warning

Note that due to the above Constraints, referencing fragments using third party namespaces within Modules or extensions (including but not limited to XLIFF Core or the Metadata Module) is not possible. This is to restrict the complexity of the fragment identification mechanism, as it would otherwise have potentially unlimited depth.

2.1 Selectors for Core Elements

- The prefix `f` indicates a `<file>` id and the value of that id is unique among all `<file>` id attribute values within the enclosing `<xliff>` element.
- The prefix `g` indicates a `<group>` id and the value of that id is unique among all `<group>` id attribute values within the enclosing `<file>` element.
- The prefix `u` indicates a `<unit>` id and the value of that id is unique among all `<unit>` id attribute values within the enclosing `<file>` element.
- The prefix `n` indicates a `<note>` id and the value of that id is unique among all `<note>` id attribute values within the immediate enclosing `<file>`, `<group>`, or `<unit>` element.

- The prefix `d` indicates a `<data>` id and the value of that id is unique among all `<data>` id attribute values within the enclosing `<unit>` element.
- The prefix `t` indicates an id for an inline element in the `<target>` element and the value of that id is unique within the enclosing `<unit>` element (with the exception of the matching inline elements in the `<source>`).
- No prefix indicates an id for a `<segment>` or an `<ignorable>` or an inline element in the `<source>` element and the value of that id is unique within the enclosing `<unit>` element (with the exception of the matching inline elements in the `<target>`).

2.2 Selectors for Modules and Extensions

A selector for a module or an extension uses a registered prefix and the value of that id is unique within the immediate enclosing `<file>`, `<group>` or `<unit>` element.

Constraints

- The prefix of a module or an extension MUST be an NMTOKEN longer than 1 character and MUST be defined in the module or extension specification.
- The prefix of a module or an extension MUST be registered with the XLIFF TC.
- A given module or extension namespace URI MUST be associated with a single prefix.
- A prefix MAY be associated with more than one namespace URI (to allow for example different versions of a given module or extension to use the same prefix).

See also the [constraints related to how IDs need to be specified in extensions](#) (which applies for modules as well).

2.3 Relative References

Fragment identifiers that do not start with a character `/` (U+002F) are relative to their location in the document, or to the document being processed.

Any unit, group or file selector missing to resolve the relative reference is obtained from the immediate enclosing unit, group or file elements.

2.4 Examples

Given the following XLIFF document:

```
<xliff xmlns="urn:oasis:names:tc:xliff:document:2.2" version="2.2"
  srcLang="en" trgLang="fr">
  <file id="f1">
    <notes>
      <note id="n1">note for file.</note>
    </notes>
    <unit id="u1">
      <my:elem xmlns:my="myNamespaceURI" id="x1">data</my:elem>
      <notes>
        <note id="n1">note for unit</note>
      </notes>
      <segment id="s1">
        <source><pc id="1">Hello <mrk id="m1" type="term">World</mrk>!</pc>
        </source>
```

```
        <target><pc id="1">Bonjour le <mrk id="m1" type="term">Monde</mrk>
            ! </pc></target>
        </segment>
    </unit>
</file>
</xliff>
```

You can have the following fragment identifiers:

- #f=f1/u=u1/1 refers to the element <pc id="1"> of the source content of the element <unit id="u1">.
- #f=f1/u=u1/t=1 refers to the element <pc id="1"> of the target content of the element <unit id="u1">.
- #f=f1/n=n1 refers to the element <note id="n1"> of the element <file id="f1">.
- #f=f1/u=u1/n=n1 refers to the element <note id="n1"> of the element <unit id="u1">.
- #f=f1/u=u1/s1 refers to the element <segment id="s1"> of the element <unit id="u1">.
- Assuming the extension defined by the namespace URI myNamespaceURI has registered the prefix myprefix, the expression #f=f1/u=u1/myprefix=x1 refers to the element <my:element id="x1"> of the element <unit id="u1">.

3 The Core Specification

XLIFF is a bilingual document format designed for containing text that needs translation, its corresponding translations and auxiliary data that makes the translation process possible.

At creation time, an XLIFF document MAY contain only text in the source language. Translations expressed in the target language MAY be added at a later time.

The root element of an XLIFF document is `<xliff>`. It contains a collection of `<file>` elements. Typically, each `<file>` element contains a set of `<unit>` elements that contain the text to be translated in the `<source>` child of one or more `<segment>` elements. Translations are stored in the `<target>` child of each `<segment>` element.

3.1 General Processing Requirements

- An Agent processing a valid XLIFF Document that contains XLIFF-defined elements and attributes that it cannot handle MUST preserve those elements and attributes.
- An Agent processing a valid XLIFF Document that contains custom elements and attributes that it cannot handle SHOULD preserve those elements and attributes.

3.2 Elements

This section contains a description of all elements used in XLIFF Core.

3.2.1 Tree Structure

Legend:

1 = one
+ = one or more
? = zero or one
* = zero or more

```
<xliff>
|
+---<notes> ?
| |
| +---<note> +
|
+---<mda:metadata> ?
| |
| +---<mda:metagroup> +
|   |
|   +---At least one of (<mda:metagroup> OR <mda:meta>)
|
+---<file> +
|
| +---<skeleton> ?
| |
| | +---<other> *
| |
| +---<other> *
|
+---<notes> ?
```

```

| |
| | +---<note> +
| |
+---At least one of (<unit> OR <group>)
| |
| | +---<unit>
| | |
| | | +---<other> *
| | | |
| | | +---<notes> ?
| | | |
| | | | +---<note> +
| | | |
| | | +---<originalData> ?
| | | |
| | | +---<data> +
| | |
| | | +---At least one of (<segment> OR <ignorable>)
| | | |
| | | | +---<segment>
| | | | |
| | | | | +---<source> 1
| | | | | |
| | | | | +---<target> ?
| | | |
| | | | +---<ignorable>
| | | | |
| | | | | +---<source> 1
| | | | | |
| | | | | +---<target> ?
| | |
| | +---<group>
| | |
| | | +---<other> *
| | | |
| | | +---<notes> ?
| | | |
| | | | +---<note> +
| | | |
| | | +---At least one of (<unit> OR <group>)

```

3.2.2 Structural Elements

The structural elements used in XLIFF Core are: `<xliff>`, `<file>`, `<skeleton>`, `<group>`, `<unit>`, `<segment>`, `<ignorable>`, `<notes>`, `<note>`, `<originalData>`, `<data>`, `<source>` and `<target>`.

3.2.2.1 xliff

Root element for XLIFF documents.

Contains:

- Zero or one `<notes>` element followed by
- Zero or one `<mda:metadata>` element followed by
- One or more `<file>` elements

Attributes:

- [version](#), REQUIRED
- [srcLang](#), REQUIRED
- [trgLang](#), OPTIONAL
- [xml:space](#), OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- The [trgLang](#) attribute is REQUIRED if and only if the XLIFF document contains [<target>](#) elements that are children of [<segment>](#) or [<ignorable>](#).



Note

The use of attributes from XLIFF modules MUST be in accordance with the constraints specified in the corresponding modules.

3.2.2.2 file

Container for localization material extracted from an entire single document, or another high level self contained logical node in a content structure that cannot be described in the terms of documents.



Note

Sub-document artifacts such as particular sheets, pages, chapters and similar are better mapped onto the [<group>](#) element. The [<file>](#) element is intended for the highest logical level. For instance a collection of papers would map to a single XLIFF Document, each paper will be represented with one [<file>](#) element, whereas chapters and subsections will map onto nested [<group>](#) elements.

Contains:

- Zero or one [<skeleton>](#) element followed by
- elements from other namespaces, OPTIONAL
- Zero or one [<notes>](#) element followed by
- One or more [<unit>](#) or [<group>](#) elements in any order.

Attributes:

- [id](#), REQUIRED
- [canResegment](#), OPTIONAL
- [original](#), OPTIONAL
- [translate](#), OPTIONAL
- [srcDir](#), OPTIONAL
- [trgDir](#), OPTIONAL
- [xml:space](#), OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- The following XLIFF Module elements are explicitly allowed by the wildcard [other](#):
 - Zero or one [<mda:metadata>](#) elements
 - Zero or one [<res:resourceData>](#) element
 - Zero or one [<slr:profiles>](#) elements
 - Zero or one [<slr:data>](#) elements
 - Zero or one [<val:validation>](#) elements
 - Zero, one, or more [<its:provenanceRecords>](#) elements

- Module and Extension elements MAY be used in any order.



Note

The use of attributes from XLIFF modules MUST be in accordance with the constraints specified in the corresponding modules.

3.2.2.3 skeleton

Container for non-translatable material pertaining to the parent `<file>` element.

Contains:

Either

- Non-translatable text
- elements from other namespaces

or

- is empty.

Attributes:

- `href`, OPTIONAL

Constraints

- The attribute `href` is REQUIRED if and only if the `<skeleton>` element is empty.

Processing Requirements

- Modifiers and Enrichers processing an XLIFF document that contains a `<skeleton>` element MUST NOT change that element, its attributes, or its content.
- Extractors creating an XLIFF document with a `<skeleton>` element MUST leave the `<skeleton>` element empty if and only if they specify the attribute `href`.

3.2.2.4 group

Provides a way to organize units into a structured hierarchy.

Note that this is especially useful for mirroring a source format's hierarchical structure.

Contains:

- elements from other namespaces, OPTIONAL
- Zero or one `<notes>` element followed by
- Zero, one or more `<unit>` or `<group>` elements in any order.

Attributes:

- `id`, REQUIRED
- `name`, OPTIONAL
- `canResegment`, OPTIONAL
- `translate`, OPTIONAL
- `srcDir`, OPTIONAL
- `trgDir`, OPTIONAL
- `type`, OPTIONAL
- `xml:space`, OPTIONAL

- attributes from other namespaces, OPTIONAL

Constraints

- The following XLIFF Module elements are explicitly allowed by the wildcard other:
 - Zero or one `<mda:metadata>` elements
 - Zero or one `<slr:data>` elements
 - Zero or one `<val:validation>` elements
 - Zero, one, or more `<its:provenanceRecords>` elements
- Module and Extension elements MAY be used in any order.



Note

The use of attributes from XLIFF modules MUST be in accordance with the constraints specified in the corresponding modules.

3.2.2.5 unit

Static container for a dynamic structure of elements holding the extracted translatable source text, aligned with the translated text.

Contains:

- elements from other namespaces, OPTIONAL
- Zero or one `<notes>` elements followed by
- Zero or one `<originalData>` element followed by
- One or more `<segment>` or `<ignorable>` elements in any order.

Attributes:

- `id`, REQUIRED
- `name`, OPTIONAL
- `canResegment`, OPTIONAL
- `translate`, OPTIONAL
- `srcDir`, OPTIONAL
- `trgDir`, OPTIONAL
- `xml:space`, OPTIONAL
- `type`, OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- A `<unit>` MUST contain at least one `<segment>` element.
- The following XLIFF Module elements are explicitly allowed by the wildcard other:
 - Zero or one `<mtc:matches>` elements
 - Zero or one `<gls:glossary>` elements
 - Zero or one `<mda:metadata>` elements
 - Zero or one `<res:resourceData>` elements
 - Zero or one `<slr:data>` elements
 - Zero or one `<val:validation>` elements
 - Zero, one, or more `<its:locQualityIssues>` elements
 - Zero, one, or more `<its:provenanceRecords>` elements
- Module and Extension elements MAY be used in any order.



Note

The use of attributes from XLIFF modules MUST be in accordance with the constraints specified in the corresponding modules.

3.2.2.6 segment

This element is a container to hold in its aligned pair of children elements the minimum portion of translatable source text and its translation in the given [Segmentation](#).

Contains:

- One [<source>](#) element followed by
- Zero or one [<target>](#) element

Attributes:

- [id](#), OPTIONAL
- [canResegment](#), OPTIONAL
- [state](#), OPTIONAL
- [subState](#), OPTIONAL
- attributes from the namespace `urn:oasis:names:tc:xliff:pgs:1.0`, OPTIONAL, provided that the Constraints specified in the [Plural, Gender, and Select Module](#) are met.

3.2.2.7 ignorable

Part of the extracted content that is not included in a segment (and therefore not translatable). For example tools can use `<ignorable>` to store the white space and/or codes that are between two segments.

Contains:

- One [<source>](#) element followed by
- Zero or one [<target>](#) element

Attributes:

- [id](#), OPTIONAL

3.2.2.8 notes

Collection of comments.

Contains:

- One or more [<note>](#) elements

3.2.2.9 note

This is an XLIFF specific way how to present end user readable comments and annotations. A note can contain information about [<source>](#), [<target>](#), [<unit>](#), [<group>](#), [<file>](#) or [<xliff>](#) elements.

Contains:

- Text

Attributes:

- [id](#), OPTIONAL
- [appliesTo](#), OPTIONAL
- [category](#), OPTIONAL

- **priority**, OPTIONAL
- **ref**, OPTIONAL
- attributes from other namespaces, OPTIONAL

Example:

```
<unit id="18">
  <notes>
    <note id="1" ref="#18-0">Change text to lower case</note>
  </notes>
  <segment state="initial" id="18-0">
    <source>Create Memories from Existing Translations</source>
  </segment>
</unit>
```



Note

When the **ref** attribute points to a **<segment>** element, by default the **<note>** content applies to its **<source>** child, unless the optional **appliesTo** attribute is set to target.



Note

The use of attributes from XLIFF modules MUST be in accordance with the constraints specified in the corresponding modules.

3.2.2.10 originalData

Unit-level collection of original data for the inline codes.

Contains:

- One or more **<data>** elements

3.2.2.11 data

Storage for the original data of an inline code.

Contains:

- Non-translatable text
- Zero, one or more **<cp>** elements.

Non-translatable text and **<cp>** elements MAY appear in any order.

Attributes:

- **id**, REQUIRED
- **dir**, OPTIONAL
- **xml:space**, OPTIONAL, the value is restricted to preserve on this element

3.2.2.12 source

Portion of text to be translated.

Contains:

- Text

- Zero, one or more `<cp>` elements
- Zero, one or more `<ph>` elements
- Zero, one or more `<pc>` elements
- Zero, one or more `<sc>` elements
- Zero, one or more `<ec>` elements
- Zero, one or more `<mrk>` elements
- Zero, one or more `<sm>` elements
- Zero, one or more `<em>` elements

Text and inline elements may appear in any order.

Attributes:

- `xml:lang`, OPTIONAL
- `xml:space`, OPTIONAL

Constraints

- When a `<source>` element is a child of `<segment>` or `<ignorable>`, the explicit or inherited value of the OPTIONAL `xml:lang` attribute MUST be equal to the value of the `srcLang` attribute of the enclosing `<xliff>` element.

3.2.2.13 target

The translation of the sibling `<source>` element.

Contains:

- Text
- Zero, one or more `<cp>` elements
- Zero, one or more `<ph>` elements
- Zero, one or more `<pc>` elements
- Zero, one or more `<sc>` elements
- Zero, one or more `<ec>` elements
- Zero, one or more `<mrk>` elements
- Zero, one or more `<sm>` elements
- Zero, one or more `<em>` elements

Text and inline elements may appear in any order.

Attributes:

- `xml:lang`, OPTIONAL
- `xml:space`, OPTIONAL
- `order`, OPTIONAL

Constraints

- When a `<target>` element is a child of `<segment>` or `<ignorable>`, the explicit or inherited value of the OPTIONAL `xml:lang` MUST be equal to the value of the `trgLang` attribute of the enclosing `<xliff>` element.

3.2.3 Inline Elements

The XLIFF Core inline elements at the `<source>` or `<target>` level are: `<cp>`, `<ph>`, `<pc>`, `<sc>`, `<ec>`, `<mrk>`, `<sm>` and `<em>`.

The elements at the `<unit>` level directly related to inline elements are: `<originalData>` and `<data>`.

3.2.3.1 cp

Represents a Unicode character that is invalid in XML.

Contains:

This element is always empty.

Parents:

<data>, <mrk>, <source>, <target> and <pc>

Attributes:

- **hex**, REQUIRED

Example:

```
<unit id="1">
  <segment>
    <source>Ctrl+C=<cp hex="0003"/></source>
  </segment>
</unit>
```

The example above shows a character U+0003 (Control C) as it has to be represented in XLIFF.

Processing Requirements

- Writers MUST encode all invalid XML characters of the content using <cp>.
- Writers MUST NOT encode valid XML characters of the content using <cp>.

3.2.3.2 ph

Represents a standalone code of the original format.

Contains:

This element is always empty.

Parents:

<source>, <target>, <pc> and <mrk>

Attributes:

- **canCopy**, OPTIONAL
- **canDelete**, OPTIONAL
- **canReorder**, OPTIONAL
- **copyOf**, OPTIONAL
- **disp**, OPTIONAL
- **equiv**, OPTIONAL
- **id**, REQUIRED.
- **dataRef**, OPTIONAL
- **subFlows**, OPTIONAL
- **subType**, OPTIONAL
- **type**, OPTIONAL
- attributes from other namespaces, OPTIONAL

Example:

```

<unit id="1">
  <originalData>
    <data id="d1">%d</data>
    <data id="d2">&lt;br/></data>
  </originalData>
  <segment>
    <source>Number of entries: <ph id="1" dataRef="d1" /><ph id="2"
      dataRef="d2"/>(These entries are only the ones matching the
      current filter settings)</source>
  </segment>
</unit>

```

Constraints

- The following XLIFF Module attributes are explicitly allowed by the wildcard other:
 - attributes from the namespace `urn:oasis:names:tc:xliff:fs:2.0`, OPTIONAL, provided that the Constraints specified in the [Format Style Module](#) are met.
 - attributes from the namespace `urn:oasis:names:tc:xliff:sizerestriction:2.0`, OPTIONAL, provided that the Constraints specified in the [Size and Length Restriction Module](#) are met.
- No other attributes MUST be used.

Processing Requirements

- Extractors MUST NOT use the `<ph>` element to represent spanning codes.

Rationale: Using a standalone placeholder code for a spanning code does not allow for controlling the span (for instance tag order and data integrity) when modifying inline content and is in direct contradiction to the business logic described in [Representation of the codes](#) and normative statements included in [Usage of <pc> and <sc>/<ec>](#)



Note

It is possible although not advised to use `<ph>` to mask non translatable inline content. The preferred way of protecting portions of inline content from translation is the Core [Translate Annotation](#). See also discussion in the *ITS Module section on representing translatability inline*.

3.2.3.3 pc

Represents a well-formed spanning original code.

Contains:

- Text
- Zero, one or more `<cp>` elements
- Zero, one or more `<ph>` elements
- Zero, one or more `<pc>` elements
- Zero, one or more `<sc>` elements
- Zero, one or more `<ec>` elements
- Zero, one or more `<mrk>` elements
- Zero, one or more `<sm>` elements
- Zero, one or more `<em>` elements

Text and inline elements may appear in any order.

Parents:

- <source>
- <target>
- <pc>
- <mrk>

Attributes:

- canCopy, OPTIONAL
- canDelete, OPTIONAL
- canOverlap, OPTIONAL
- canReorder, OPTIONAL
- copyOf, OPTIONAL
- dispEnd, OPTIONAL
- dispStart, OPTIONAL
- equivEnd, OPTIONAL
- equivStart, OPTIONAL
- id, REQUIRED
- dataRefEnd, OPTIONAL
- dataRefStart, OPTIONAL
- subFlowsEnd, OPTIONAL
- subFlowsStart, OPTIONAL
- subType, OPTIONAL
- type, OPTIONAL
- dir, OPTIONAL
- attributes from other namespaces, OPTIONAL

Example:

```
<unit id="1">
  <originalData>
    <data id="1">&lt;B&gt;</data>
    <data id="2">&lt;/B&gt;</data>
  </originalData>
  <segment>
    <source><pc id="1" dataRefStart="1" dataRefEnd="2">Important</pc>
      text</source>
    </segment>
  </unit>
```

Constraints

- The following XLIFF Module attributes are explicitly allowed by the wildcard other:
 - attributes from the namespace urn:oasis:names:tc:xliff:fs:2.0, OPTIONAL, provided that the Constraints specified in the [Format Style Module](#) are met.
 - attributes from the namespace urn:oasis:names:tc:xliff:sizerestriction:2.0, OPTIONAL, provided that the Constraints specified in the [Size and Length Restriction Module](#) are met.
- No other attributes MUST be used.

Processing Requirements

- Extractors MUST NOT use the <pc> element to represent standalone codes.

Rationale: Using a spanning code for a standalone code can easily result in having text inside a span where the original format does not allow it.

3.2.3.4 sc

Start of a spanning original code.

Contains:

This element is always empty.

Parents:

[<source>](#), [<target>](#), [<pc>](#) and [<mrk>](#)

Attributes:

- [canCopy](#), OPTIONAL
- [canDelete](#), OPTIONAL
- [canOverlap](#), OPTIONAL
- [canReorder](#), OPTIONAL
- [copyOf](#), OPTIONAL
- [dataRef](#), OPTIONAL
- [dir](#), OPTIONAL
- [disp](#), OPTIONAL
- [equiv](#), OPTIONAL
- [id](#), REQUIRED
- [isolated](#), OPTIONAL
- [subFlows](#), OPTIONAL
- [subType](#), OPTIONAL
- [type](#), OPTIONAL
- attributes from other namespaces, OPTIONAL

Example:

```
<unit id="1">
  <segment>
    <source><sc id="1" type="fmt" subType="xlf:b"/>
      First sentence. </source>
    </segment>
    <segment>
      <source>Second sentence.<ec startRef="1" type="fmt"
        subType="xlf:b"/></source>
    </segment>
  </unit>
```

Constraints

- The following XLIFF Module attributes are explicitly allowed by the wildcard other:
 - attributes from the namespace [urn:oasis:names:tc:xliff:fs:2.0](#), OPTIONAL, provided that the Constraints specified in the [Format Style Module](#) are met.
 - attributes from the namespace [urn:oasis:names:tc:xliff:sizerestriction:2.0](#), OPTIONAL, provided that the Constraints specified in the [Size and Length Restriction Module](#) are met.
- No other attributes MUST be used.
- The values of the attributes [canCopy](#), [canDelete](#), [canReorder](#) and [canOverlap](#) MUST be the same as the values the ones in the [<ec>](#) element corresponding to this start code.

- If the OPTIONAL attribute `isolated` is present, its value MUST be set to yes when the `<ec>` element corresponding to this start marker is not in the same `<unit>`. When the corresponding `<ec>` element is present in the same `<unit>`, the attribute value MUST be set to no.

Processing Requirements

- Extractors MUST NOT use the `<sc>` / `<ec>` pair to represent standalone codes.

Rationale: Using a spanning code for a standalone code can easily result in having text inside a span where the original format does not allow it.

3.2.3.5 `ec`

End of a spanning original code.

Contains:

This element is always empty.

Parents:

`<source>`, `<target>`, `<pc>` and `<mrk>`

Attributes:

- `canCopy`, OPTIONAL
- `canDelete`, OPTIONAL
- `canOverlap`, OPTIONAL
- `canReorder`, OPTIONAL
- `copyOf`, OPTIONAL
- `dataRef`, OPTIONAL
- `dir`, OPTIONAL
- `disp`, OPTIONAL
- `equiv`, OPTIONAL
- `id`, OPTIONAL
- `isolated`, OPTIONAL
- `startRef`, OPTIONAL
- `subFlows`, OPTIONAL
- `subType`, OPTIONAL
- `type`, OPTIONAL
- attributes from other namespaces, OPTIONAL

Example:

```
<unit id="1">
  <originalData>
    <data id="d1">\b </data>
    <data id="d2">\i </data>
    <data id="d3">\b0 </data>
    <data id="d4">\i0 </data>
  </originalData>
  <segment>
    <source>Text in <sc id="1" dataRef="d1"/>bold <sc id="2"
      dataRef="d2"/> and<ec startRef="1" dataRef="d3"/>
      italics<ec startRef="2" dataRef="d4"/>. </source>
    </segment>
  </unit>
```

Constraints

- The following XLIFF Module attributes are explicitly allowed by the wildcard other:
 - attributes from the namespace `urn:oasis:names:tc:xliff:fs:2.0`, OPTIONAL, provided that the Constraints specified in the [Format Style Module](#) are met.
 - attributes from the namespace `urn:oasis:names:tc:xliff:sizerestriction:2.0`, OPTIONAL, provided that the Constraints specified in the [Size and Length Restriction Module](#) are met.
- No other attributes MUST be used.
- The values of the attributes `canCopy`, `canDelete` and `canOverlap` MUST be the same as the values the ones in the `<sc>` element corresponding to this end code.
- The value of the attribute `canReorder` MUST be no if the value of `canReorder` is `firstNo` in the `<sc>` element corresponding to this end code.
- The attribute `isolated` MUST be set to yes if and only if the `<sc>` element corresponding to this end code is not in the same `<unit>` and set to no otherwise.
- If and only if the attribute `isolated` is set to yes, the attribute `id` MUST be used instead of the attribute `startRef` that MUST be used otherwise.
- If and only if the attribute `isolated` is set to yes, the attribute `dir` MAY be used, otherwise the attribute `dir` MUST NOT be used on the `<ec>` element.

Processing Requirements

- Extractors MUST NOT use the `<sc>` / `<ec>` pair to represent standalone codes.

Rationale: Using a spanning code for a standalone code can easily result in having text inside a span where the original format does not allow it.

3.2.3.6 mrk

Represents an annotation pertaining to the marked span.

Contains:

- Text
- Zero, one or more `<cp>` elements
- Zero, one or more `<ph>` elements
- Zero, one or more `<pc>` elements
- Zero, one or more `<sc>` elements
- Zero, one or more `<ec>` elements
- Zero, one or more `<mrk>` elements
- Zero, one or more `<sm>` elements
- Zero, one or more `<em>` elements

Text and inline elements may appear in any order.

Parents:

`<source>`, `<target>`, `<pc>` and `<mrk>`

Attributes:

- `id`, REQUIRED
- `translate`, OPTIONAL
- `type`, OPTIONAL

- [ref](#), OPTIONAL
- [value](#), OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- The [\[XML namespace\]](#) MUST NOT be used at this extension point.
- The following XLIFF Module attributes are explicitly allowed by the wildcard other:
 - attributes from the namespace `urn:oasis:names:tc:xliff:fs:2.0`, OPTIONAL, provided that the Constraints specified in the [Format Style Module](#) are met.
 - attributes from the namespace `urn:oasis:names:tc:xliff:sizerestriction:2.0`, OPTIONAL, provided that the Constraints specified in the [Size and Length Restriction Module](#) are met.
 - attributes from the namespace `http://www.w3.org/2005/11/its`, OPTIONAL, provided that the Constraints specified in the *ITS Module* are met.
 - attributes from the namespace `urn:oasis:names:tc:xliff:itsm:2.1`, OPTIONAL, provided that the Constraints specified in the *ITS Module* are met.

See the [Annotations section](#) for more details and examples on how to use the `<mrk>` element.

3.2.3.7 sm

Start marker of an annotation where the spanning marker cannot be used for well-formedness reasons.

Contains:

This element is always empty.

Parents:

[<source>](#), [<target>](#), [<pc>](#) and [<mrk>](#)

Attributes:

- [id](#), REQUIRED
- [translate](#), OPTIONAL
- [type](#), OPTIONAL
- [ref](#), OPTIONAL
- [value](#), OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- The [\[XML namespace\]](#) MUST NOT be used at this extension point.
- The following XLIFF Module attributes are explicitly allowed by the wildcard other:
 - attributes from the namespace `urn:oasis:names:tc:xliff:fs:2.0`, OPTIONAL, provided that the Constraints specified in the [Format Style Module](#) are met.
 - attributes from the namespace `urn:oasis:names:tc:xliff:sizerestriction:2.0`, OPTIONAL, provided that the Constraints specified in the [Size and Length Restriction Module](#) are met.
 - attributes from the namespace `http://www.w3.org/2005/11/its`, OPTIONAL, provided that the Constraints specified in the *ITS Module* are met.
 - attributes from the namespace `urn:oasis:names:tc:xliff:itsm:2.1`, OPTIONAL, provided that the Constraints specified in the *ITS Module* are met.

See the [Annotations section](#) for more details and examples on how to use the `<sm>` element.

3.2.3.8 em

End marker of an annotation where the spanning marker cannot be used for well-formedness reasons.

Contains:

This element is always empty.

Parents:

[<source>](#), [<target>](#), [<pc>](#) and [<mrk>](#)

Attributes:

- [startRef](#), REQUIRED

See the [Annotations section](#) for more details and examples on how to use the [](#) element.

3.3 Attributes

This section lists all the various attributes used in XLIFF core elements.

3.3.1 XLIFF Attributes

The attributes defined in XLIFF 2.3 are: [appliesTo](#), [canCopy](#), [canDelete](#), [canOverlap](#), [canReorder](#), [canResegment](#), [category](#), [copyOf](#), [dataRef](#), [dataRefEnd](#), [dataRefStart](#), [dir](#), [disp](#), [dispEnd](#), [dispStart](#), [equiv](#), [equivEnd](#), [equivStart](#), [hex](#), [href](#), [id](#), [isolated](#), [name](#), [order](#), [original](#), [priority](#), [ref](#), [srcDir](#), [srcLang](#), [startRef](#), [state](#), [subFlows](#), [subFlowsEnd](#), [subFlowsStart](#), [subState](#), [subType](#), [trgLang](#), [translate](#), [trgDir](#), [type](#), [value](#) and [version](#).

3.3.1.1 appliesTo

Comment target - indicates the element to what the content of the note applies.

Value description: source or target.

Default value: undefined.

Used in: [<note>](#).

3.3.1.2 canCopy

Replication editing hint - indicates whether or not the inline code can be copied.

Value description: yes if the code can be copied, no if the code is not intended to be copied.

Default value: yes.

Used in: [<pc>](#), [<sc>](#), [<ec>](#), [<ph>](#).

3.3.1.3 canDelete

Deletion editing hint - indicates whether or not the inline code can be deleted.

Value description: yes if the code can be deleted, no if the code is not allowed to be deleted.

Default value: yes.

Used in: [<pc>](#), [<sc>](#), [<ec>](#), [<ph>](#).

3.3.1.4 canOverlap

Code can overlap - indicates whether or not the spanning code where this attribute is used can enclose partial spanning codes (i.e. a start code without its corresponding end code, or an end code without its corresponding start code).

Value description: yes or no.

Default value: default values for this attribute depend on the element in which it is used:

- When used in `<pc>`: no.
- When used in `<sc>` or `<ec>`: yes.

Used in: `<pc>`, `<sc>` and `<ec>`

Example:

```
<unit id="1">
  <originalData>
    <data id="1">\i1 </data>
    <data id="2">\i0 </data>
    <data id="3">{\b </data>
    <data id="4">}</data>
  </originalData>
  <segment>
    <source><pc id="1" dataRefStart="3" dataRefEnd="4" canOverlap="no">
      Bold, <sc id="2" dataRef="1" canOverlap="yes"/>both</pc>,
      italics<ec startRef="2" dataRef="2"/></source>
    </segment>
  </unit>
```

3.3.1.5 canReorder

Re-ordering editing hint - indicates whether or not the inline code can be re-ordered. See [Editing Hints section](#) for more details.

Value description: yes in case the code can be re-ordered, firstNo when the code is the first element of a sequence that cannot be re-ordered, no when it is another element of such a sequence.

Default value: yes.

Used in: `<pc>`, `<sc>`, `<ec>`, `<ph>`.

For the normative Usage Description see Constraints and Processing Requirements in the [Editing Hints section](#).

3.3.1.6 canResegment

Can resegment - indicates whether or not the source text in the scope of the given canResegment flag can be reorganized into a different structure of `<segment>` elements within the same parent `<unit>`.

Value description: yes or no.

Default value: default values for this attribute depend on the element in which it is used:

- When used in `<file>`:

The value yes.

- When used in any other element:

The value of the [canResegment](#) attribute of its parent element.

Used in: [<file>](#) [<group>](#) [<unit>](#), and [<segment>](#).

3.3.1.7 category

Category - provides a way to categorize notes.

Value description: Text.

Default value: undefined

Used in: [<note>](#).

3.3.1.8 copyOf

Reference to base code - holds the [id](#) of the base code of a copied code.

Value description: NMTOKEN. The [id](#) value of the base code of which this code is a copy.

Default value: undefined

Used in: [<ph>](#), [<pc>](#), [<sc>](#), [<ec>](#).

Example:

```
<unit id="1">
  <segment>
    <source>Äter <pc id="1">katter möss</pc>?</source>
    <target>Do <pc id="1">cats</pc> eat <pc id="2" copyOf="1">
      mice</pc>? </target>
    </segment>
  </unit>
```

3.3.1.9 dataRef

Original data reference - holds the identifier of the [<data>](#) element that contains the original data for a given inline code.

Value description: An [\[XML Schema Datatypes\]](#) NMTOKEN that MUST be the value of the [id](#) attribute of one of the [<data>](#) element listed in the same [<unit>](#) element.

Default value: undefined.

Used in: [<ph>](#), [<sc>](#), [<ec>](#).

Example:

```
<unit id="1">
  <originalData>
    <data id="d1">{0}</data>
  </originalData>
  <segment>
    <source>Error in '<ph id="1" dataRef="d1"/>'.</source>
    <target>Erreur dans '<ph id="1" dataRef="d1"/>'.</target>
```

```
</segment>
</unit>
```

The example above shows a `<ph>` element that has its original data stored outside the content, in a `<data>` element.

3.3.1.10 dataRefEnd

Original data reference - holds the identifier of the `<data>` element that contains the original data for the end marker of a given inline code.

Value description: An [XML Schema Datatypes] NMTOKEN that MUST be the value of the `id` attribute of one of the `<data>` element listed in the same `<unit>` element.

Default value: undefined.

Used in: `<pc>`.

Example:

```
<unit id="1">
  <originalData>
    <data id="d1">&lt;EM></data>
    <data id="d2">&lt;/EM></data>
  </originalData>
  <segment>
    <source><pc id="1" dataRefStart="d1" dataRefEnd="d2">
      Efficiency</pc> is the operative word here.</source>
    <target><pc id="1" dataRefStart="d1" dataRefEnd="d2">
      Efficacité</pc> est le mot clé ici.</target>
  </segment>
</unit>
```

The example above shows two `<pc>` elements with their original data stored outside the content, in two `<data>` elements.

3.3.1.11 dataRefStart

Original data reference - holds the identifier of the `<data>` element that contains the original data for the start marker of a given inline code.

Value description: An [XML Schema Datatypes] NMTOKEN that MUST be the value of the `id` attribute of one of the `<data>` element listed in the same `<unit>` element.

Default value: undefined.

Used in: `<pc>`.

Example:

```
<unit id="1">
  <originalData>
    <data id="d1">&lt;EM></data>
    <data id="d2">&lt;/EM></data>
  </originalData>
  <segment>
    <source><pc id="1" dataRefStart="d1" dataRefEnd="d2">
      Efficiency</pc> is the operative word here.</source>
```

```

    <target><pc id="1" dataRefStart="d1" dataRefEnd="d2">
      Efficacité</pc> est le mot clé ici.</target>
    </segment>
  </unit>

```

The example above shows two `<pc>` elements with their original data stored outside the content, in two `<data>` elements.

3.3.1.12 dir

Directionality - indicates the directionality of content.

Value description: `ltr` (Left-To-Right), `rtl` (Right-To-Left), or `auto` (determined heuristically, based on the first strong directional character in scope, see [UAX #9]).

Default value: default values for this attribute depend on the element in which it is used:

- When used in a `<pc>`, `<sc>`, or `<ec>` element that has a `<source>` element as its parent:
The value of the `srcDir` attribute of the `<unit>` element, in which the elements are located.
- When used in a `<pc>`, `<sc>`, or `<ec>` element that has a `<target>` element as its parent:
The value of the `trgDir` attribute of the `<unit>` element, in which the elements are located.
- When used in a `<pc>`, `<sc>`, or `<ec>` element that has a `<pc>` element as its parent:
The value of the `dir` attribute of the parent `<pc>` element.
- When used in `<data>`:
The value `auto`.

Used in: `<data>`, `<pc>`, `<sc>`, and `<ec>`.

3.3.1.13 disp

Display text - holds an alternative user-friendly display representation of the original data of the inline code.

Value description: Text.

Default value: undefined

Used in: `<ph>`, `<sc>`, `<ec>`.

Example:

```

<unit id="1">
  <originalData>
    <data id="d1">{1}</data>
  </originalData>
  <segment>
    <source>Welcome back <ph id="1" disp="[UserName]" dataRef="d1"/>!
    </source>
  </segment>
</unit>

```




Note

To provide a plain text equivalent of the code, use the [equiv](#) attribute.

3.3.1.14 dispEnd

Display text - holds an alternative user-friendly display representation of the original data of the end marker of an inline code.

Value description: Text.

Default value: undefined

Used in: [<pc>](#).

Example:

```

<unit id="1">
  <originalData>
    <data id="d1">\cf1\ul\b\f1\fs24 </data>
    <data id="d2">\cf0\ulnone\b0\f0\fs22 </data>
  </originalData>
  <segment>
    <source>Example of <pc id="1" dataRefStart="d1" dataRefEnd="d2"
      dispStart="&lt;span>" dispEnd="&lt;/span>">
      formatted text</pc>.</source>
    </segment>
  </unit>

```

In the example above, the [dispStart](#) and [dispEnd](#) attributes provide a more user-friendly representation of the original formatting codes.



Note

To provide a plain text equivalent of the code, use the [equivEnd](#) attribute.

3.3.1.15 dispStart

Display text - holds an alternative user-friendly display representation of the original data of the start marker of an inline code.

Value description: Text.

Default value: undefined

Used in: [<pc>](#).

Example:

```

<unit id="1">
  <originalData>
    <data id="d1">\cf1\ul\b\f1\fs24 </data>
    <data id="d2">\cf0\ulnone\b0\f0\fs22 </data>
  </originalData>
  <segment>
    <source>Example of <pc id="1" dataRefStart="d1" dataRefEnd="d2"
      dispStart="&lt;span>" dispEnd="&lt;/span>">

```

```
formatted text</pc>.</source>
</segment>
</unit>
```

In the example above, the `dispStart` and `dispEnd` attributes provide a more user-friendly representation of the original formatting codes.



Note

To provide a plain text equivalent of the code, use the `equivStart` attribute.

3.3.1.16 equiv

Equivalent text - holds a plain text representation of the original data of the inline code that can be used when generating a plain text representation of the content.

Value description: Text.

Default value: an empty string.

Used in: `<ph>`, `<sc>`, `<ec>`.

Example:

```
<unit id="1">
  <originalData>
    <data id="d1">&#x26;</data>
  </originalData>
  <segment>
    <source>Open <ph id="1" equiv="" dataRef="d1"/>File</source>
  </segment>
</unit>
```

In this example the `equiv` attribute of the `<ph>` element is used to indicate that the original data of the code can be ignored in the text representation of the string. This could, for instance, help a spell-checker tool to process the content as "Open File".



Note

To provide a user-friendly representation, use the `disp` attribute.

3.3.1.17 equivEnd

Equivalent text - holds a plain text representation of the original data of the end marker of an inline code that can be used when generating a plain text representation of the content.

Value description: Text.

Default value: an empty string

Used in: `<pc>`.

Example:

```
<unit id="1">
  <originalData>
    <data id="d1">&lt;span class="link" onclick="linkTo('dbId5345')">
```

```

    </data>
    <data id="d2">&lt;/span></data>
  </originalData>
  <segment>
    <source>The jam made of <pc id="1" dataRefStart="d1" equivStart=""
      dataRefEnd="d2" equivEnd="">lingonberries</pc> is quite
      tasty.</source>
    </segment>
  </unit>

```



Note

To provide a user-friendly representation, use the [dispEnd](#) attribute.

3.3.1.18 equivStart

Equivalent text - holds a plain text representation of the original data of the start marker of an inline code that can be used when generating a plain text representation of the content.

Value description: Text.

Default value: an empty string

Used in: [<pc>](#).

Example:

```

<unit id="1">
  <originalData>
    <data id="d1">&lt;span class="link" onclick="linkTo('dbId5345')">
      </data>
    <data id="d2">&lt;/span></data>
  </originalData>
  <segment>
    <source>The jam made of <pc id="1" dataRefStart="d1" equivStart=""
      dataRefEnd="d2" equivEnd="">lingonberries</pc> is quite
      tasty.</source>
    </segment>
  </unit>

```



Note

To provide a user-friendly representation, use the [dispStart](#) attribute.

3.3.1.19 hex

Hexadecimal code point - holds the value of a Unicode code point that is invalid in XML.

Value description: A canonical representation of the hexBinary [\[XML Schema Datatypes\]](#) data type: Two hexadecimal digits to represent each octet of the Unicode code point. The allowed values are any of the values representing code points invalid in XML, between hexadecimal 0000 and 10FFFF (both included).

Default value: undefined

Used in: [<cp>](#).

Example:

```
<cp hex="001A"/><cp hex="0003"/>
```

The example above shows a character U+001A and a character U+0003 as they have to be represented in XLIFF.

3.3.1.20 href

Hyperlink reference - a pointer to the location of an external skeleton file pertaining to the enclosing `<file>` element.

Value description: IRI.

Default value: undefined

Used in: `<skeleton>`.

3.3.1.21 id

Identifier - a character string used to identify an element.

Value description: NMTOKEN. The scope of the values for this attribute depends on the element, in which it is used.

- When used in a `<file>` element:
The value MUST be unique among all `<file>` id attribute values within the enclosing `<xliff>` element.
- When used in `<group>` elements:
The value MUST be unique among all `<group>` id attribute values within the enclosing `<file>` element.
- When used in `<unit>` elements:
The value MUST be unique among all `<unit>` id attribute values within the enclosing `<file>` element.
- When used in `<note>` elements:
The value MUST be unique among all `<note>` id attribute values within the immediate enclosing `<file>`, `<group>`, or `<unit>` element.
- When used in `<data>` elements:
The value MUST be unique among all `<data>` id attribute values within the enclosing `<unit>` element.
- When used in `<segment>`, `<ignorable>`, `<mrk>`, `<sm>`, `<pc>`, `<sc>`, `<ec>`, or `<ph>` elements:
 - The inline elements enclosed by a `<target>` element MUST use the duplicate id values of their corresponding inline elements enclosed within the sibling `<source>` element if and only if those corresponding elements exist.
 - Except for the above exception, the value MUST be unique among all of the above within the enclosing `<unit>` element.



Note

All of the above defined uniqueness scopes ignore Module and Extension data. It would be impossible to impose those uniqueness requirements onto Module or Extension data. As Core only Modifiers could inadvertently cause conflicts with Modules or Extensions based data they cannot access. Modules and Extensions reusing Core need to specify their own uniqueness scopes for the [xlf:id](#). In general, Modules and Extensions are advised to mimic the Core uniqueness requirement within their specific wrapper elements enclosing the reused Core elements or attributes, yet Module or Extensions are free to set wider uniqueness scopes if it makes business sense.

Default value: undefined

Used in: [<file>](#), [<group>](#), [<unit>](#), [<note>](#), [<segment>](#), [<ignorable>](#), [<data>](#), [<sc>](#), [<ec>](#), [<ph>](#), [<pc>](#), [<mrk>](#) and [<sm>](#).

3.3.1.22 isolated

Orphan code flag - indicates if the start or end marker of a spanning inline code is not in the same [<unit>](#) as its corresponding end or start code.

Value description: yes if this start or end code is not in the same [<unit>](#) as its corresponding end or start code, no if both codes are in the same [<unit>](#).

Default value: no.

Used in: [<sc>](#), [<ec>](#).

Example:

```
<file id="f2" xmlns:abc="urn:abc">
  <unit id="1">
    <mtc:matches>
      <mtc:match id="tc01" ref="seg2">
        <source><sc id="1" isolated="yes"/>Warning:</source>
        <target><sc id="1" isolated="yes"/>Attention :</target>
      </mtc:match>
    </mtc:matches>
    <segment id="seg2">
      <source><pc id="1">Warning: File not found.</pc></source>
    </segment>
  </unit>
</file>
```

In the example above the [<sc>](#) elements have their [isolated](#) attribute set to yes because they do not have their corresponding [<ec>](#) elements.

3.3.1.23 name

Resource name - the original identifier of the resource corresponding to the extracted [<unit>](#) or [<group>](#).

For example: the key in the key/value pair in a Java properties file, the ID of a string in a Windows string table, the index value of an entry in a database table, etc.

Value description: Text.

Default value: undefined.

Used in: [<unit>](#) and [<group>](#).

3.3.1.24 order

target order - indicates the order, in which to compose the target content parts.

Value description: A positive integer.

Default value: implicit, see below

When order is not explicitly set, the [<target> order](#) corresponds to its sibling [<source>](#), i.e. it is not being moved anywhere when composing target content of the enclosing [<unit>](#) and the implicit [order](#) value is of that position within the [<unit>](#).

Used in: [<target>](#).

Constraints

- The value of the [order](#) attribute MUST be unique within the enclosing [<unit>](#) element.
- The value of each of the [order](#) attributes used within a [<unit>](#) element MUST NOT be higher than N, where N is the number of all current [<segment>](#) and [<ignorable>](#) children of the said [<unit>](#) element.

See the [Segments Order](#) section for the normative usage description.

3.3.1.25 original

Original file - a pointer to the location of the original document from which the content of the enclosing [<file>](#) element is extracted.

Value description: IRI.

Default value: undefined

Used in: [<file>](#).

3.3.1.26 priority

Priority - provides a way to prioritize notes.

Value description: Integer 1-10.

Default value: 1

Used in: [<note>](#).



Note

Note that 1 is the highest priority that can be interpreted as an alert, e.g. an [\[ITS\] Localization Note](#) of the type alert. The best practice is to use only one alert per an annotated element, and the full scale of 2-10 can be used for prioritizing notes of lesser importance than the alert.

3.3.1.27 ref

Reference - holds a reference for the associated element.

Value description: A value of the [\[XML Schema Datatypes\]](#) type anyURI. The semantics of the value depends on where the attribute is used:

- When used in a [<note>](#) element, the URI value is referring to a [<segment>](#), [<source>](#) or [<target>](#) element within the same enclosing [<unit>](#).

When used in a [term annotation](#), the URI value is referring to a resource providing information about the term.

- When used in a [translation candidates annotation](#), the URI value is referring to an external resource providing information about the translation candidate.
- When used in a [comment annotation](#), the value is referring to a [<note>](#) element within the same enclosing [<unit>](#).
- When used in a [custom annotation](#), the value is defined by each custom annotation.

Default value: undefined

Used in: [<note>](#), [<mrk>](#) and [<sm>](#).

Example:

```
<unit id="1">
  <segment>
    <source>The <pc id="1">ref</pc> attribute of a term
      annotation holds a <mrk id="m1" type="term"
      ref="http://dbpedia.org/page/Uniform_Resource_Identifier">
      URI</mrk> pointing to more information about the given
      term.</source>
    </segment>
  </unit>
```

3.3.1.28 srcDir

Source directionality - indicates the directionality of the source content.

Value description: `ltr` (Left-To-Right), `rtl` (Right-To-Left), `,` or `auto` (determined heuristically, based on the first strong directional character in scope, see [\[UAX #9\]](#)).

Default value: default values for this attribute depend on the element in which it is used:

- When used in [<file>](#):
The value `auto`.
- When used in any other element:
The value of the [srcDir](#) attribute of its parent element.

Used in: [<file>](#), [<group>](#), and [<unit>](#).

3.3.1.29 srcLang

Source language - the code of the language, in which the text to be translated is expressed.

Value description: A language code as described in [\[BCP 47\]](#).

Default value: undefined

Used in: [<xliff>](#).

3.3.1.30 startRef

Start code or marker reference - The [id](#) of the [<sc>](#) element or the [<sm>](#) element a given [<ec>](#) element or [](#) element corresponds.

Value description: NMTOKEN.

Default value: undefined

Used in: [<ec>](#), [](#).

Example:

```
<unit id="1">
  <segment>
    <source><sc id="1"/>Bold, <sc id="2"/>both
      <ec startRef="1"/>, italics<ec startRef="2"/></source>
    </segment>
  </unit>
```

3.3.1.31 state

State - indicates the state of the translation of a segment.

Value description: The value MUST be set to one of the following values:

`initial` - indicates the segment is in its initial state.

`translated` - indicates the segment has been translated.

`reviewed` - indicates the segment has been reviewed.

`final` - indicates the segment is finalized and ready to be used.

The 4 defined states constitute a simple linear state machine that advances in the above given order. No particular workflow or process is prescribed, except that the three states more advanced than the default `initial` assume the existence of a translation within the segment. One can further specify the state of the translation using the [subState](#) attribute.

Default value: `initial`

Used in: [<segment>](#)

Processing Requirements

- When the optional [state](#) attribute is added to a [<segment>](#) element, its value MUST be set to `initial` if the element doesn't have a [<target>](#) child. All valid values MAY be used when a [<target>](#) child is present.
- Writers updating the attribute [state](#) MUST also update or delete [subState](#).



Note

[state](#) is an OPTIONAL attribute of segments with a default value and segmentation can change as the XLIFF roundtrip progresses, hence implementers don't have to make explicit use of the attribute. However setting of the attribute is advantageous if a workflow needs to make use of Advanced Validation methods.

3.3.1.32 subFlows

Sub-flows list - holds a list of [id](#) attributes corresponding to the `<unit>` elements that contain the sub-flows for a given inline code.

Value description: A list of NMTOKEN values separated by spaces. Each value corresponds to the [id](#) attribute of a `<unit>` element.

Default value: undefined

Used in: `<ph>`, `<sc>`, `<ec>`.

Example:

See the example in the [Sub-Flows section](#).

3.3.1.33 subFlowsEnd

Sub-flows list - holds a list of [id](#) attributes corresponding to the `<unit>` elements that contain the sub-flows for the end marker of a given inline code.

Value description: A list of NMTOKEN values separated by spaces. Each value corresponds to the [id](#) attribute of a `<unit>` element.

Default value: undefined

Used in: `<pc>`.

Example:

See the example in the [Sub-Flows section](#).

3.3.1.34 subFlowsStart

Sub-flows list - holds a list of [id](#) attributes corresponding to the `<unit>` elements that contain the sub-flows for the start marker of a given inline code.

Value description: A list of NMTOKEN values separated by spaces. Each value corresponds to the [id](#) attribute of a `<unit>` element.

Default value: undefined

Used in: `<pc>`.

Example:

See the example in the [Sub-Flows section](#).

3.3.1.35 subState

subState - indicates a user-defined status for the `<segment>` element.

Value description:

The value is composed of a prefix and a sub-value separated by a character : (U+003A).

The prefix is a string uniquely identifying a collection of values for a specific authority. The sub-value is any string value defined by an authority.

The prefix xlf is reserved for this specification.

Other prefixes and sub-values MAY be defined by the users.

Default value: undefined

Used in: [<segment>](#)

Constraints

- If the attribute [subState](#) is used, the attribute [state](#) MUST be explicitly set.

Processing Requirements

- Writers updating the attribute [state](#) MUST also update or delete [subState](#).

3.3.1.36 subType

subType - indicates the secondary level type of an inline code.

Value description:

The value is composed of a prefix and a sub-value separated by a character : (U+003A).

The prefix is a string uniquely identifying a collection of sub-values for a specific authority. The sub-value is any string value defined by the authority.

The prefix xlf is reserved for this specification, and the following sub-values are defined:

xlf:lb - Line break
xlf:pb - Page break
xlf:b - Bold
xlf:i - Italics
xlf:u - Underlined
xlf:var - Variable

Other prefixes and sub-values MAY be defined by the users.

Default value: undefined

Used in: [<pc>](#), [<sc>](#), [<ec>](#) and [<ph>](#)

Constraints

- If the attribute [subType](#) is used, the attribute [type](#) MUST be specified as well.
- The reserved xlf: prefixed values map onto the [type](#) attribute values as follows:

For xlf:b, xlf:i, xlf:u, xlf:lb, and xlf:pb, the REQUIRED value of the [type](#) attribute is `fmt`.
For xlf:var, the REQUIRED value of the [type](#) attribute is `ui`.

Processing Requirements

- Modifiers updating the attribute [type](#) MUST also update or delete [subType](#).

3.3.1.37 trgLang

Target language - the code of the language, in which the translated text is expressed.

Value description: A language code as described in [\[BCP 47\]](#).

Default value: undefined

Used in: `<xliff>`.

3.3.1.38 translate

Translate - indicates whether or not the source text in the scope of the given `translate` flag is intended for translation.

Value description: yes or no.

Default value: default values for this attribute depend on the element in which it is used:

- When used in `<file>`:

The value yes.

- When used in any other admissible structural element (`<group>` or `<unit>`):

The value of the `translate` attribute of its parent element.

- When used in annotations markers `<mrk>` or `<sm>`:

The value of the `translate` attribute of the innermost `<mrk>` or `<unit>` element, in which the marker in question is located.

Used in: `<file>` `<group>` `<unit>`, `<mrk>` and `<sm>`.

3.3.1.39 trgDir

Target directionality - indicates the directionality of the target content.

Value description: `ltr` (Left-To-Right), `rtl` (Right-To-Left), or `auto` (determined heuristically, based on the first strong directional character in scope, see [UAX #9]).

Default value: default values for this attribute depend on the element in which it is used:

- When used in `<file>`:

The value `auto`.

- When used in any other element:

The value of the `trgDir` attribute of its parent element.

Used in: `<file>`, `<group>`, and `<unit>`.

3.3.1.40 type

Type - indicates the type of an element.

Value description: Allowed values for this attribute depend on the element in which it is used.

- When used in `<pc>`, `<sc>`, `<ec>` or `<ph>`:

The value MUST be set to one of the following values:

`fmt` - Formatting (e.g. a `<b>` element in HTML)

`ui` - User interface element

`quote` - Inline quotation (as opposed to a block citation)

`link` - Link (e.g. an `<a>` element in HTML)

`image` - Image or graphic

`other` - Type of element not covered by any of the other top-level types.

Example:

```
<segment>
  <source><pc id="q1" type="quote">Blázen,
    chce dobýt to nu v takovém po así</pc>, dodal slovy svého
    oblíbeného imaginárního autora.</source>
  <target><pc id="q1" type="quote">Madman, he wants to conquer the
    pole in this weather</pc>, offered he the words of his
    favourite imaginary playwright.</target>
</segment>
```

One can further specify the type of a code using the `subType` attribute.

Default value: undefined

- When used in `<mrk>` or `<sm>`:

One of the following values: generic, comment, term, or a user-defined value that is composed of a prefix and a sub-value separated by a character : (U+003A).

The prefix is a string uniquely identifying a collection of sub-values for a specific authority. The sub-value is any string value defined by the authority.

Default value: generic

- When used in `<group>` or `<unit>`:

A value that is composed of a prefix and a sub-value separated by a character : (U+003A).

The prefix is a string uniquely identifying a collection of sub-values for a specific authority. The sub-value is any string value defined by the authority. The prefix `xlf` is reserved.

Default value: undefined

Used in: `<group>`, `<unit>`, `<pc>`, `<sc>`, `<ec>`, `<mrk>`, `<ph>` and `<sm>`.

Processing Requirements

- Modifiers updating the attribute `type` on `<pc>`, `<sc>`, `<ec>`, or `<ph>` MUST also update or delete `subType`.

3.3.1.41 value

Value - holds a value for the associated annotation.

Value description: Text.

- When used in a [term annotation](#), the value is a definition of the term.
- When used in a [comment annotation](#), the value is the text of the comment.
- When used in a [custom annotation](#), the value is defined by each custom annotation.

Default value: undefined

Used in: `<mrk>` and `<sm>`.

3.3.1.42 version

XLIFF Version - is used to specify the version of the XLIFF Document. This corresponds to the version number of the XLIFF specification that the XLIFF document adheres to. For this specification, the version is 2.2.

Value description: 2.0, 2.1 or 2.2

Used in: `<xliff>`.

3.3.2 XML namespace

The attributes from XML namespace used in XLIFF 2.3 are: `xml:lang` and `xml:space`.

3.3.2.1 xml:lang

Language - the `xml:lang` attribute specifies the language variant of the text of a given element. For example: `xml:lang="fr-FR"` indicates the French language as spoken in France.

Value description: A language code as described in [BCP 47].

Default value: default values for this attribute depend on the element in which it is used:

- When used in a `<source>` element:
The value set in the `srcLang` attribute of the enclosing `<xliff>` element.
- When used in a `<target>` element:
The value set in the `trgLang` attribute of the enclosing `<xliff>` element.
- When used in any other element:
The value of the `xml:lang` attribute of its parent element.

Used in: `<source>`, `<target>` and where extension attributes are allowed.

3.3.2.2 xml:space

White spaces - the `xml:space` attribute specifies how white spaces (ASCII spaces, tabs and line-breaks) are to be treated.

Value description: `default` or `preserve`. The value `default` signals that an application's default white-space processing modes are acceptable for this element; the value `preserve` indicates the intent that applications preserve all the white space. This declared intent is considered to apply to all elements within the content of the element where it is specified, unless overridden with another instance of the `xml:space` attribute. For more information see the section on `xml:space` in the [XML] specification.

Default value: default values for this attribute depend on the element in which it is used:

- When used in `<data>`:
The value `preserve`.
- When used in `<xliff>`:
The value `default`.
- When used in any other element:
The value of the `xml:space` attribute of its parent element.

Used in: `<xliff>`, `<file>`, `<group>`, `<unit>`, `<source>`, `<target>`, and `<data>`.

3.4 CDATA sections

CDATA sections (`<![CDATA[ . . . ]>`) are allowed in XLIFF content, but on output they MAY be changed into normal escaped content.

Note that avoiding CDATA sections is considered a best practice from the internationalization viewpoint [\[XML I18N BP\]](#).

Processing Requirements

- Agents MUST process CDATA sections.
- Writers MAY preserve the original CDATA sections.

3.5 XML Comments

XML comments (`<!-- . . . -->`) are allowed in XLIFF content, but they are ignored in the parsed content.

For example:

```
<source>Text content <!--IMPORTANT-->that is important</source>
```

and

```
<source>Text content that is important</source>
```

are identical after parsing and correspond to the same following parsed content:

```
Text content that is important
```

To annotate a section of the content with a comment that is recognized and preserved by XLIFF user agents, use the `<note>` element, or the `<mrk>` element.

Processing Requirements

- Agents MUST ignore XML comments. That is the XLIFF parsed content is the same whether or not there is an XML comment in the document.
- Writers MAY preserve XML comments on output.

3.6 XML Processing Instructions

XML Processing Instructions [\[XML\]](#) (see specifically <http://www.w3.org/TR/REC-xml/#sec-pi>) are an XML mechanism to "allow documents to contain instructions for applications." XML Processing Instructions are allowed in XLIFF content but they are ignored in the parsed content in the same sense as XML Comments.

Processing Requirements

- Agents MUST NOT use Processing Instructions as a means to implement a feature already specified in XLIFF Core or Modules.
- Writers SHOULD preserve XML Processing Instructions in an XLIFF document.



Warning

Note that Agents using Processing Instructions to implement XLIFF Core or Module features are not compliant XLIFF applications disregarding whether they are otherwise conformant.



Warning

Although this specification encourages XLIFF Agents to preserve XML Processing Instructions, it is not and cannot be, for valid processing reasons, an absolute protection and it is for instance highly unlikely that Processing Instructions could survive an XLIFF roundtrip at the `<segment>` level or lower. Hence implementers are discouraged from using XML Processing Instructions at the `<segment>` and lower levels.

3.7 Inline Content

The XLIFF inline content defines how to encode the content extracted from the original source. The content includes the following types of data:

- **Text** -- Textual content.
- **Inline codes** -- Sequences of content that are not linguistic text, such as formatting codes, variable placeholders, etc.

For example: the element `<b>` in HTML, or the placeholder `{0}` in a Java string.

- **Annotations** -- Markers that delimit a span of the content and carry or point to information about the specified content.

For example: a flag indicating that a given section of text is not intended for translation, or an element indicating that a given expression in the text is a term associated with a definition.

There are two elements that contain inline markup in XLIFF: `<source>` and `<target>`.

In some cases, data directly associated with inline elements MAY also be stored at the `<unit>` level in an `<originalData>` element.

3.7.1 Text

The XLIFF inline markup does not prescribe how to represent normal text, besides that it MUST be valid XML.

3.7.1.1 Characters invalid in XML

Because the content represented in XLIFF can be extracted from anywhere, including software resources and other material that can contain control characters, XLIFF needs to be able to represent all Unicode code points [Unicode].

However, XML does not have the capability to represent all Unicode code points [Unicode], and does not provide any official mechanism to escape the forbidden code points.

To remedy this, the inline markup provides the `<cp>` element.

The syntax and semantic of `<cp>` in XLIFF are similar to the ones of `<cp>` in the Unicode Locale Data Markup Language [LDML].

3.7.2 Inline Codes

The specification takes into account two types of codes:

Original code

An original code is a code that exists in the original document being extracted into XLIFF.

Added code

An added code is a code that does not exist in the original document, but has been added to the content at some point after extraction.

Any code (original or added) belongs to one of the two following categories:

Standalone

A standalone code is a code that corresponds to a single position in the content. An example of such code is the `<br/>` element in HTML.

Spanning

A spanning code is a code that encloses a section of the content using a start and an end marker. There are two kinds of spanning codes:

- Codes that can overlap, that is: they can enclose a non-closing or a non-opening spanning code. Such codes do not have an XML-like behavior. For example the RTF code `\b1... \b0` is a spanning code that is allowed to overlap.
- Codes that cannot overlap, that is: they cannot enclose a partial spanning code and have an XML-like behavior at the same time. An example of such code is the `<emphasis>...</emphasis>` element in DocBook.

When the opening or closing marker of a spanning code does not have its corresponding closing or opening marker in the same unit, it is an orphan code.

3.7.2.1 Representation of the codes

Spanning codes present a set of challenges in XLIFF:

First, because the code format of the original data extracted to XLIFF does not need to be XML, spanning codes can overlap.

For example, in the following RTF content, the format markers are in a sequence: start bold, start italics, end bold, end italics. This does not translate into a well-formed mapping.

```
Text in \b bold \i and\b0 italics\i0
```

Another challenge is the possible effect of segmentation: A spanning code can start in one segment and end in another.

For example, in the following HTML content, the segmentation splits the text independently of the codes so the starting and ending tags of the `<B>...</B>` element end up in different parts of the `<unit>` element:

```
[Sentence <B>one. ][Sentence two.][ ][Sentence</B> three.]
```

Finally, a third potential cause of complication is that the start or the end markers of a spanning code can become orphans if their segment is used outside of its original `<unit>`.

For example, an entry with bold text can be broken down into two segments:

```
Segment 1 = "<b>Warning found: "  
Segment 2 = "The file is read-only</b>"
```


And later, one of the segments can be re-used outside its original `<unit>`, for instance as a translation candidate:

```
New segment = "<b>Warning found - see log</b>"
Fuzzy match = "<b>Warning found: "
```

Because of these use cases, the representation of a spanning code cannot always be mapped to a similar spanning element in XLIFF.

When taking into account these issues, the possible use cases and their corresponding XLIFF representations are as follow:

Table 1. Inline code use cases

Use Case	Example of Representation
Standalone code	<code><ph id='1' /></code>
Well-formed spanning code	<code><pc id='1'>text</pc></code>
Start marker of spanning code	<code><sc id='1' /></code>
End marker of spanning code	<code><ec startRef='1' /></code>
Orphan start marker of spanning code	<code><sc id='1' isolated='yes' /></code>
Orphan end marker of spanning code	<code><ec id='1' isolated='yes' /></code>

3.7.2.2 Usage of `<pc>` and `<sc>/<ec>`

A spanning code MUST be represented using a `<sc>` element and a `<ec>` element if the code is not well-formed or orphan.

For example, the following RTF content has two spans of formatting:

```
Text in \b bold \i and\b0 italics\i0
```

They can only be represented using two pairs of `<sc>` and `<ec>` elements:

```
<unit id="1">
  <originalData>
    <data id="d1">\b </data>
    <data id="d2">\i </data>
    <data id="d3">\b0 </data>
    <data id="d4">\i0 </data>
  </originalData>
  <segment>
    <source>Text in <sc id="1" dataRef="d1"/>bold <sc id="2"
      dataRef="d2"/> and<ec startRef="1" dataRef="d3"/>
      italics<ec startRef="2" dataRef="d4"/>. </source>
    </segment>
  </unit>
```

If the spanning code is well-formed it MAY be represented using either a single `<pc>` element or using a pair of `<sc>` and a `<ec>` elements.

For example, the following RTF content has a single span of formatting:

```
Text in \b bold\b0 .
```

It can be represented using either notations:

Text in `<pc id="1" canOverlap="yes" dataRefStart="c1" dataRefEnd="c2">bold</pc>`.

Text in `<sc id="1" dataRef="c1"/>bold<ec startRef="1" dataRef="c2"/>`.

Processing Requirements

- When both the `<pc>` and the `<sc>/<ec>` representations are possible, Extractors and Modifiers MAY use either one as long as all the information of the inline code (e.g. original data, sub-flow indicators, etc.) are preserved.
- When converting representation between a pair of `<sc>` and `<ec>` elements and a `<pc>` element or vice-versa, Modifiers MUST map their attributes as shown in the following table:

Table 2. Mapping between attributes

<code><pc></code> attributes	<code><sc></code> attributes	<code><ec></code> attributes
id	id	startRef / id (see <code><ec></code>)
type	type	type
subType	subType	subType
dispStart	disp	
dispEnd		disp
equivStart	equiv	
equivEnd		equiv
subFlowsStart	subFlows	
subFlowsEnd		subFlows
dataRefStart	dataRef	
dataRefEnd		dataRef
	isolated	isolated
canCopy	canCopy	canCopy
canDelete	canDelete	canDelete
canReorder	canReorder	canReorder
copyOf	copyOf	copyOf
canOverlap	canOverlap	canOverlap
dir	dir	dir

- Agents MUST be able to handle any of the above two types of inline code representation.

3.7.2.3 Storage of the original data

Most of the time, inline codes correspond to an original construct in the format from which the content was extracted. This is the original data.

XLIFF tries to abstract and normalize as much as possible the extracted content because this allows a better re-use of the material across projects. Some tools require access to the original data in order to create the translated document back into its original format. Others do not.

3.7.2.3.1 No storage of the original data

In this option, the original data of the inline code is not preserved inside the XLIFF document.

The tool that created the initial XLIFF document is responsible for providing a way to re-create the original format properly when merging back the content.

For example, for the following HTML content:

This **naked mole rat** is **pretty ugly**.

one possible XLIFF representation is the following:

```
<unit id="1">
  <segment>
    <source>This <pc id="1">naked mole rat</pc> is
      <pc id="2">pretty ugly</pc>.</source>
    <target>Cet <pc id="1">hétérocéphale</pc> est
      <pc id="2">plutôt laid</pc>.</target>
  </segment>
</unit>
```

3.7.2.3.2 Storage of the original data

In this option, the original data of the inline code is stored in a structure that resides outside the content (i.e. outside `<source>` or `<target>`) but still inside the `<unit>` element.

The structure is an element `<originalData>` that contains a list of `<data>` entries uniquely identified within the `<unit>` by an `id` attribute. In the content, each inline code using this mechanism includes a `dataRef` attribute that points to a `<data>` element where its corresponding original data is stored.

For example, for the following HTML content:

This **naked mole rat** is **pretty ugly**.

The following XLIFF representation stores the original data:

```
<unit id="1">
  <originalData>
    <data id="d1">&lt;B></data>
    <data id="d2">&lt;/B></data>
  </originalData>
  <segment>
    <source>This <pc id="1" dataRefStart="d1" dataRefEnd="d2"> naked
      mole rat</pc> is <pc id="2" dataRefStart="d1"
      dataRefEnd="d2"> pretty ugly</pc>.</source>
    <target>Cet <pc id="1" dataRefStart="d1" dataRefEnd="d2">
      hétérocéphale</pc> est <pc id="2" dataRefStart="d1"
      dataRefEnd="d2"> plutôt laid</pc>.</target>
  </segment>
</unit>
```



Note

This mechanism allows to re-use identical original data by pointing to the same `<data>` element.

3.7.2.4 Adding Codes

When processing content, there are possible cases when new inline codes need to be added.

For example, in the following HTML help content, the text has the name of a button in bold:

Press the **Emergency Stop** button to interrupt the count-down sequence.

In the translated version, the original label needs to remain in English because the user interface, unlike the help, is not translated. However, for convenience, a translation is also provided and emphasized using another style. That new formatting needs to be added:

Appuyez sur le bouton **Emergency Stop** (*Arrêt d'urgence*) pour interrompre le compte à rebours.

Having to split a single formatted span of text into several separate parts during translation, can serve as another example. For instance, the following sentence in Swedish uses bold on the names of two animals:

Äter **katter möss**?

But the English translation separates the two names and therefore needs to duplicate the bold codes.

Do **cats** eat **mice**?

Processing Requirements

- Modifiers MAY add inline codes.
- The **id** value of the added code MUST be different from all **id** values in both source and target content of the unit where the new code is added.
- Mergers MAY ignore added inline codes when merging the translated content back into the original format.

There are several ways to add codes:

3.7.2.4.1 Duplicating an existing code

One way to create a new code is to duplicate an existing one (called the base code).

If the base code is associated with some original data: the new code simply uses the same data.

For example, the translation in the following unit, the second inline code is a duplicate of the first one:

```
<unit id="1">
  <originalData>
    <data id="d1">&lt;b></data>
    <data id="d2">&lt;/b></data>
  </originalData>
  <segment>
    <source>Äter <pc id="1" dataRefStart="d1" dataRefEnd="d2">katter
      möss</pc>?</source>
    <target>Do <pc id="1" dataRefStart="d1" dataRefEnd="d2">
      cats</pc> eat <pc id="2" dataRefStart="d1"
```

```

      dataRefEnd="d2">mice</pc>?</target>
    </segment>
  </unit>

```

If the base code has no associated data, the new code MUST use the [copyOf](#) attribute to indicate the id of the base code. This allows the merging tool to know what original data to re-use.

For example, the translation in the following unit, the second inline code is a duplicate of the first one:

```

<unit id="1">
  <segment>
    <source>Esznek <pc id="1">a magyarok svéd húsgombócot
    </pc>?</source>
    <target>Do <pc id="1">Hungarians</pc> eat <pc id="2"
    copyOf="1">Swedish meatballs</pc>?</target>
  </segment>
</unit>

```

Processing Requirements

- Modifiers MUST NOT clone a code that has its [canCopy](#) attribute is set to no.
- The [copyOf](#) attribute MUST be used when, and only when, the base code has no associated original data.

3.7.2.4.2 Creating a brand-new code

Another way to add a code is to create it from scratch. For example, this can happen when the translated text requires additional formatting.

For example, in the following unit, the UI text needs to stay in English, and is also translated into French as a hint for the French user. The French translation for the UI text is formatted in italics:

```

<unit id="1">
  <originalData>
    <data id="d1">&lt;b></data>
    <data id="d2">&lt;/b></data>
    <data id="n1">&lt;i></data>
    <data id="n2">&lt;/i></data>
  </originalData>
  <segment>
    <source>Press the <pc id="1" dataRefStart="d1" dataRefEnd="d2">
      Emergency Stop</pc> button to interrupt the count-down
      sequence. </source>
    <target>Appuyez sur le bouton <pc id="1" dataRefStart="d1"
      dataRefEnd="d2">Emergency Stop</pc> (<pc id="2"
      dataRefStart="n1" dataRefEnd="n2">Arrêt d'urgence
      </pc>) pour interrompre le compte à rebours. </target>
  </segment>
</unit>

```

3.7.2.4.3 Converting text into a code

Another way to add a code is to convert part of the extracted text into code. In some cases the inline code can be created after extraction, using part of the text content. This can be done, for instance, to get better matches from an existing Translation Memory, or better candidates from a Machine Translation system.

For example, it can happen that a tool extracting a Java properties file to XLIFF is not sophisticated enough to treat HTML or XML snippets inside the extracted text as inline code:

```
# text property for the widget 'next'
nextText: Click <ui>Next</ui>
```

Resulting XLIFF content:

```
<unit id="1">
  <segment>
    <source>Click &lt;ui>Next&lt;/ui></source>
  </segment>
</unit>
```

But another tool, later in the process, can be used to process the initial XLIFF document and detect additional inline codes. For instance here the XML elements such as `<ui>`.

The original data of the new code is the part of the text content that is converted as inline code.

```
<unit id="1">
  <originalData>
    <data id="d1">&lt;ui></data>
    <data id="d2">&lt;/ui></data>
  </originalData>
  <segment>
    <source>Click <pc id="1" dataRefStart="d1" dataRefEnd="d2">
      Next</pc></source>
    </segment>
  </unit>
```



Warning

Converting XLIFF text content into original data for inline code might need a tool-specific process as the tool which did the initial extraction could have applied some conversion to the original content to create the XLIFF content (e.g. un-escape special characters).

3.7.2.5 Removing Codes

When processing content, there are some possible cases when existing inline codes need to be removed.

For an example the translation of a sentence can result in grouping of several formatted parts into a single one. For instance, the following sentence in English uses bold on the names of two animals:

```
Do <b>cats</b> eat <b>mice</b>?
```

But the Swedish translation group the two names and therefore needs only a single bolded part.

```
Äter <b>katter möss</b>?
```

Processing Requirements

- User agents MAY remove a given inline code only if its `canDelete` attribute is set to yes.
- When removing a given inline code, the user agents MUST remove its associated original data, except if the original data is shared with another inline code that remains in the unit.

Note that having to delete the original data is unlikely because such original data is likely to be associated to an inline code in the source content.

There are several ways to remove codes:

3.7.2.5.1 Deleting a code

One way to remove a code is to delete it from the extracted content. For example, in the following unit, the translated text does not use the italics formatting. It is removed from the target content, but the original data are preserved because they are still used in the source content.

```
<unit id="1">
  <originalData>
    <data id="d1">&lt;i></i></data>
    <data id="d2">&lt;/i></data>
  </originalData>
  <segment>
    <source>I read <pc id="1" dataRefStart="d1" dataRefEnd="d2">Little
      House on the Prairie</pc> to my children.</source>
    <target>
      </target>
    </segment>
  </unit>
```

3.7.2.5.2 Converting a code into text

Another way to remove an inline code is to convert it into text content. This is likely to be a rare use case. It is equivalent to deleting the code, with the addition to place the original data for the given code into the content, as text. This can be done, for example, to get better matches from an existing Translation Memory, or better candidates from a Machine Translation system.

For instance, the following unit has an inline code corresponding to a variable place-holder. A tool can temporarily treat this variable as text to get better matches from an existing Translation Memory.

```
<unit id="1">
  <originalData>
    <data id="d1">%s</data>
  </originalData>
  <segment>
    <source>Cannot find '<ph id="1" dataRef="d1"/>'.</source>
    </segment>
  </unit>
```

The modified unit would end up like as shown below. Note that because the original data was not associated with other inline code it has been removed from the unit:

```
<unit id="1">
  <segment>
    <source>Cannot find '%s'.</source>
  </segment>
</unit>
```



Warning

Converting the original data of an inline code into text content might need a tool-specific process as the tool which did the initial extraction could have applied some conversion to the original content.

3.7.2.6 Editing Hints

XLIFF provides some information about what editing operations are applicable to inline codes:

- A code can be deleted: That is, the code element as well as its original data (if any are attached) are removed from the document. This hint is represented with the `canDelete` attribute. The default value is yes: deletion is allowed.

For example, the following extracted C string has the code `<ph id='1' />` set to be not deletable because removing the original data (the variable placeholder %s) from the string would result in an error when running the application:

- A code can be copied: That is, the code is used as a base code for adding another inline code. See [Section 3.7.2.4.1, “Duplicating an existing code”](#) for more details. This hint is represented with the `canCopy` attribute. The default value is yes: copy is allowed.
- A code can be re-ordered: That is, a given code can be moved before or after another inline code. This hint is represented with the `canReorder` attribute. The default value is yes: re-ordering is allowed.



Note

Note that often those properties are related and appear together. For example, the code in the first unit shown below is a variable placeholder that has to be preserved and cannot be duplicated, and when several of such variables are present, as in the second unit, they cannot be re-ordered:

```
<unit id="1">
  <originalData>
    <data id="d1">%s</data>
  </originalData>
  <segment>
    <source>Can't open '<ph id="1" dataRef="d1" canCopy="no"
      canDelete="no"/>'.</source>
  </segment>
</unit>
<unit id="2">
  <originalData>
    <data id="d1">%s</data>
    <data id="d2">%d</data>
  </originalData>
  <segment>
    <source>Number of <ph id="1" dataRef="d1" canCopy="no"
      canDelete="no" canReorder="firstNo"/>: <ph id="2" dataRef="d2"
      canCopy="no" canDelete="no" canReorder="no"/>. </source>
  </segment>
</unit>
```

See the [Target Content Modification section](#) for additional details on editing.

Constraints

- When the attribute `canReorder` is set to no or firstNo, the attributes `canCopy` and `canDelete` MUST also be set to no.
- Inline codes re-ordering within a source or target content MAY be limited by defining non-reorderable sequences. Such sequence is made of a first inline code with the attribute `canReorder` set to firstNo and zero or more following codes with `canReorder` set to no.

- A non-reorderable sequence of codes MUST NOT start with a code with the attribute `canReorder` set to No and zero or more following codes with `canReorder` set to no



Note

A non-reorderable sequence made of a single code with `canReorder` set to firstNo are allowed just for extraction convenience and are equivalent to a code with the attribute `canReorder` set to yes.

Processing Requirements

- Extractors SHOULD set the `canDelete`, `canCopy` and `canReorder` attributes for the codes that need to be treated differently than with the default settings.
- Modifiers MUST NOT change the number and order of the inline codes making up a non-reorderable sequence.
- Modifiers MAY move a whole non-reorderable sequence before or after another non-reorderable sequence.
- When a non-reorderable sequence is made of a single non-reorderable code, Modifiers MAY remove the `canReorder` attribute of that code or change its value to yes.
- Modifiers MUST NOT delete inline codes that have their attribute `canDelete` set to no.
- Modifiers MUST NOT replicate inline codes that have their attribute `canCopy` set to no.



Note

Conformance of codes to [Editing Hints](#) Processing Requirements within translations can only be checked on existing `<target>` elements, i.e. non-conformance is not reported on `<segment>` or `<ignorable>` elements without `<target>` children.

3.7.3 Annotations

An annotation is an element that associates a section of the content with some metadata information.

Annotations MAY be created by an Extractor that generated the initial XLIFF document, or by any other Modifier or Enricher later in the process. For example, after an Extractor creates the document, an Enricher can annotate the source content with terminological information.

Annotations are represented using either the `<mrk>` element, or the pair of `<sm>` and `<em>` elements.

3.7.3.1 Type of Annotations

There are several pre-defined types of annotation and definition of `custom types` is also allowed.

3.7.3.1.1 Translate Annotation

This annotation is used to indicate whether a span of content is translatable or not.

Usage:

- The `id` attribute is REQUIRED
- The `translate` attribute is REQUIRED and set to yes or no
- The `type` attribute is OPTIONAL and set to generic (this is the default value)

For example:

He saw his <mrk id="m1" translate="no">doppelgänger</mrk>.



Note

This annotation overrides the `translate` attribute set or inherited at the `<unit>` level.



Note

The `translate` attribute can also be used at the same time as another type of annotation. For example:

```
He saw his <mrk id="m1" translate="no" type="term">doppelgänger</mrk>.
```

3.7.3.1.2 Term Annotation

This annotation is used to mark up a term in the content, and possibly associate information to it.

Usage:

- The `id` attribute is REQUIRED
- The `type` attribute is REQUIRED and set to `term`
- The `value` attribute is OPTIONAL and contains a short definition of the term
- The `ref` attribute is OPTIONAL and contains a URI pointing to information on the term
- The `translate` attribute is OPTIONAL and set to yes or no

For example:

```
<file id="f-t_a">
  <unit id="1">
    <segment>
      <source>He is my <mrk id="m1" type="term"
        ref="http://dbpedia.org/page/Doppelgänger">
        doppelgänger</mrk>. </source>
    </segment>
  </unit>
</file>
```

3.7.3.1.3 Comment Annotation

This annotation is used to associate a span of content with a comment.

Usage:

- The `id` attribute is REQUIRED
- The `type` attribute is REQUIRED and set to `comment`
- If the `value` attribute is present it contains the text of the comment. If and only if the `value` attribute is not present, the `ref` attribute MUST be present and contain the URI of a `<note>` element within the same enclosing `<unit>` element that holds the comment.
- The `translate` attribute is OPTIONAL and set to yes or no

For example, here with the `value` attribute:

```
The <mrk id="m1" type="comment"
  value="Possible values: Printer or Stacker"><ph id="1" dataRef="d1"/>
</mrk>
has been enabled.
```

And here using the [ref](#) attribute:

```
<unit id="1">
  <notes>
    <note id="n1" appliesTo="target">Please check the translation for
      'namespace'. One also can use 'espace de nom', but I think most
      technical manuals use the English term.</note>
  </notes>
  <segment>
    <source>You use your own namespace.</source>
    <target>Vous pouvez utiliser votre propre <mrk id="m1"
      type="comment" ref="#n=n1">namespace</mrk>.</target>
  </segment>
</unit>
```

3.7.3.1.4 Custom Annotation

The [<mrk>](#) element can be used to implement custom annotations.

A custom annotation MUST NOT provide the same functionality as a pre-defined annotation.

Usage:

- The [id](#) attribute is REQUIRED
- The [type](#) attribute is REQUIRED and set to a unique user-defined value.
- The [translate](#) attribute is OPTIONAL and set to yes or no
- The use and semantics of the [value](#) and [ref](#) attributes are user-defined.

For example:

```
One of the earliest surviving works of literature is
<mrk id="m1" type="myCorp:isbn" value="978-0-14-44919-8">The
Epic of Gilgamesh</mrk>.
```

3.7.3.2 Splitting Annotations

Annotations can overlap spanning inline codes or other annotations. They also can be split by segmentation. Because of this, a single annotation span can be represented using a pair of [<sm>](#) and [](#) elements instead of a single [<mrk>](#) element.

For example, one can have the following content:

```
<unit id="1">
  <segment>
    <source>Sentence A. <mrk id="m1" type="comment" value="Comment for B
      and C">Sentence B. Sentence C.</mrk></source>
  </segment>
</unit>
```

After a user agent performs segmentation, the annotation element [<mrk>](#) is changed to a pair of [<sm>](#) and [](#) elements:

```

<unit id="1">
  <segment>
    <source>Sentence A. </source>
  </segment>
  <segment>
    <source><sm id="m1" type="comment" value="Comment for B and C"/>
      Sentence B. </source>
  </segment>
  <segment>
    <source>Sentence C.<em startRef="m1"/></source>
  </segment>
</unit>

```

3.7.4 Sub-Flows

A sub-flow is a section of text embedded inside an inline code, or inside another section of text.

For example, the following HTML content includes two sub-flows: The first one is the value of the `title` attribute ("Start button"), and the second one is the value of the `alt` attribute ("Click here to start!"):

```

Click to start: 

```

Another example is the following DITA content where the footnote "A Palouse horse is the same as an Appaloosa." is defined at the middle of a sentence:

```

Palouse horses<fn>A Palouse horse is the same as
an Appaloosa.</fn> have spotted coats.

```

In XLIFF, each sub-flow is stored in its own `<unit>` element, and the `subFlows` attribute is used to indicate the location of the embedded content.

Therefore the HTML content of the example above can be represented like below:

```

<unit id="1">
  <segment>
    <source>Start button</source>
  </segment>
</unit>
<unit id="2">
  <segment>
    <source>Click here to start!</source>
  </segment>
</unit>
<unit id="3">
  <segment>
    <source>Click to start: <ph id="1" subFlows="1 2"/></source>
  </segment>
</unit>

```

Constraints

- An inline code containing or delimiting one or more sub-flows MUST have an attribute `subFlows` that holds a list of the identifiers of the `<unit>` elements where the sub-flows are stored.
- Sub-flows MUST be in the same `<file>` element as the `<unit>` element from which they are referenced.

Processing Requirements

- Extractors SHOULD store each sub-flow in its own `<unit>` element.
- Extractors MAY order the `<unit>` elements of the sub-flows and the `<unit>` element, from where the sub-flows are referenced, as they see fit.



Note

Note that the static structure encoded by `<file>`, `<group>`, and `<unit>` elements is principally immutable in XLIFF Documents and hence the unit order initially set by the Extractor will be preserved throughout the roundtrip even in the special case of sub-flows.

3.7.5 White Spaces

While white spaces can be significant or insignificant in the original format, they are always treated as significant when stored as original data in XLIFF. See the definition of the `<data>` element.

Processing Requirements

- For the inline content and all non empty inline elements: The white spaces MUST be preserved if the value for `xml:space` set or inherited at the enclosing `<unit>` level is `preserve`, and they MAY be preserved if the value is `default`.

3.7.6 Bidirectional Text

Text directionality in XLIFF content is defined by inheritance. Source and target content can have different directionality.

The initial directionality for both the source and the target content is defined in the `<file>` element, using the OPTIONAL attributes `srcDir` for the source and `trgDir` for the target. The default value for both attributes is `auto`.

The `<group>` and `<unit>` elements also have the two OPTIONAL attributes `srcDir` and `trgDir`. The default value of the `srcDir` is inherited from the value of the `srcDir` attribute of the respective parent element. The default value of the `trgDir` attribute is inherited from the value of the `trgDir` attribute of the respective parent element.

The `<pc>`, `<sc>`, and isolated `<ec>` elements have an OPTIONAL attribute `dir` with a value `ltr`, `rtl`, or `auto`. The default value is inherited from the parent `<pc>` element. In case the inline element is a child of a `<source>` element, the default value is inherited from the `srcDir` value of the enclosing `<unit>` element. In case the inline element is a child of a `<target>` element, the default value is inherited from the `trgDir` value of the enclosing `<unit>` element.



Warning

While processing isolated `<ec>` elements with explicitly set directionality, please beware that unlike directionality set on the `<pc>` and `<sc>`, this method decreases the stack level as per [UAX #9].

In addition, the `<data>` element has an OPTIONAL attribute `dir` with a value `ltr`, `rtl`, or `auto` that is not inherited. The default value is `auto`.

Directionality of source and target text contained in the `<source>` and `<target>` elements is fully governed by [UAX #9], whereas explicit XLIFF-defined structural and directionality markup is a higher-level protocol in the sense of [UAX #9]. The XLIFF-defined value `auto` determines the directionality based on the first strong directional character in its scope and XLIFF-defined inline directionality markup behaves exactly as Explicit Directional Isolate Characters, see [UAX #9], http://www.unicode.org/reports/tr9/#Directional_Formatting_Characters.



Note

Note that this specification does not define explicit markup for inline directional Overrides or Embeddings; in case those are needed. Extractors and Modifiers will need to use [UAX #9] defined Directional Formatting Characters.

For instance, HTML elements `<bdi>` and `<bdo>` need both extracted as a `<pc>` or `<sc>` / `<ec/>` pair with the `dir` attribute set respectively.

All XLIFF defined inline directionality markup isolates and `<sc>` / `<ec/>` isolated spans can reach over segment (but not unit) boundaries. This needs to be taken into account when splitting or joining segments (see [Segmentation Modification](#)) that contain inline directionality markup. Albeit It is not advisable to split segments, so that corresponding inline directionality markup start and end would fall into different segments, such a situation is not too confusing. If this happens, the "watertight" BiDi box will simply span two or more segments. This is not too confusing because no XLIFF defined directionality markup is allowed on `<source>`, `<target>`, or `<segment>`, so all higher level protocol inheritance of directionality in such cases is from `<unit>` or higher.

3.7.7 Target Content Modification

This section defines the rules Writers need to follow when working with the target content of a given segment in order to provide interoperability throughout the whole process.

The Extractor MAY create the initial target content as it sees fit.

The Merger is assumed to have the same level of processing and native format knowledge as the Extractor. Providing an interoperable way to convert native documents into XLIFF with one tool and back to the native format with another tool without the same level of knowledge is outside the scope of this specification.

The Writers modifying the target content of an XLIFF Document between the Extractor and the Merger ensure interoperability by applying specific rules. These rules are separated into two cases: When there is an existing target and when there is no existing target.

3.7.7.1 Without an Existing Target

When there is no existing target, the processing requirements for a given segment are the following:

Processing Requirements

- Writers MAY leave the segment without a target.
- Modifiers MAY create a new target as follows:
 - Modifiers MAY add translation of the source text.
 - Modifiers MUST put all [non-removable](#) inline codes in the target.

- Modifiers MUST preserve the order of all the [non-reorderable](#) inline codes.
- Modifiers MAY put any [removable](#) inline code in the target.
- Modifiers MAY add inline codes.
- Modifiers MAY add or remove annotations.
- Modifiers MAY convert any [<pc>](#) element into a pair of [<sc>](#) and [<ec>](#) elements.
- Modifiers MAY convert, if it is possible, any pair of [<sc>](#) and [<ec>](#) elements into a [<pc>](#) element.

3.7.7.2 With an Existing Target

When working with a segment with content already in the target, Writers MUST choose one of the three behaviors described below:

Processing Requirements

- Writers MAY leave the existing target unchanged.
- Modifiers MAY modify the existing target as follow:
 - Modifiers MAY add or modify translatable text.
 - Writers MUST preserve all [non-removable](#) inline codes, regardless whether or not they exist in the source.
 - Writers MUST preserve any [non-reorderable](#) inline codes in the existing target.
 - Writers MUST NOT add any [non-reorderable](#) inline codes to the target.
 - Modifiers MAY remove any [removable](#) inline codes in the target.
 - Modifiers MAY add inline codes (including copying any [cloneable](#) inline codes of the existing target).
 - Modifiers MAY add or remove annotations.
 - Modifiers MAY convert any [<pc>](#) element into a pair of [<sc>](#) and [<ec>](#) elements.
 - Modifiers MAY convert, if it is possible, any pair of [<sc>](#) and [<ec>](#) elements into a [<pc>](#) element.
- Modifiers MAY delete the existing target and start over as if working without an existing target.

3.7.8 Content Comparison

This specification defines two types of content equality:

- Equality type A: Two contents are equal if their normalized forms are equal.
- Equality type B: Two contents are equal if, in their normalized forms and with all inline code markers replaced by the value of their [equiv](#) attributes, the resulting strings are equal.

A content is normalized when:

- The text nodes are in Unicode Normalized Form C defined in the Unicode Annex #15: Unicode Normalization Forms [\[UAX #15\]](#).
- All annotation markers are removed.
- All pairs of [<sc>](#) and [<ec>](#) elements that can be converted into a [<pc>](#) element, are converted.

- All adjacent text nodes are merged into a single text node.
- For all the text nodes with the white space property set to default, all adjacent white spaces are collapsed into a single space.

3.8 Segmentation

In the context of XLIFF, a segment is content which is either a unit of extracted text, or has been created from a unit of extracted text by means of a segmentation mechanism such as sentence boundary detection. For example, a segment can be a title, the text of a menu item, a paragraph or a sentence in a paragraph.

In the context of XLIFF, other types representations sometimes called "segmentation" can be represented using annotations. For example: the terms in a segment can be identified and marked up using the [term annotation](#).

XLIFF does not specify how segmentation is carried out, only how to represent its result. Material provisions regarding segmentation can be found for instance in the Segmentation Rules eXchange standard [SRX] or [UAX #29].

3.8.1 Segments Representation

In XLIFF each segment of processed content is represented by a `<segment>` element.

A `<unit>` can comprise a single `<segment>`.

Each `<segment>` element has one `<source>` element that contains the source content and one OPTIONAL `<target>` element that can be empty or contain the translation of the source content at a given state.

Content parts between segments are represented with the `<ignorable>` element, which has the same content model as `<segment>`.

For example:

```
<unit id="1">
  <segment>
    <source>First sentence.</source>
    <target>Première phrase.</target>
  </segment>
  <ignorable>
    <source> </source>
  </ignorable>
  <segment>
    <source>Second sentence.</source>
  </segment>
</unit>
```

3.8.2 Segments Order

Some Agents (e.g. aligner tools) can segment content, so that the target segments are not in the same order as the source segments.

To be able to map order differences, the `<target>` element has an OPTIONAL `order` attribute that indicates its position in the sequence of segments (and inter-segments). Its value is an integer from 1 to N, where N is the sum of the numbers of the `<segment>` and `<ignorable>` elements within the given enclosing `<unit>` element.



Warning

When Writers set explicit [order](#) on `<target>` elements, they have to check for conflicts with implicit [order](#), as `<target>` elements without explicit [order](#) correspond to their sibling `<source>` elements. Beware that moving one `<target>` element is likely to cause a re-numbering domino effect throughout the enclosing `<unit>` element.

For example, the following HTML documents have the same paragraph with three sentences in different order:

```
<p lang='en'>Sentence A. Sentence B. Sentence C.</p>
```

```
<p lang='fr'>Phrase B. Phrase C. Phrase A.</p>
```

The XLIFF representation of the content, after segmentation and alignment, would be:

```
<unit id="1">
  <segment id="1">
    <source>Sentence A.</source>
    <target order="5">Phrase A.</target>
  </segment>
  <ignorable>
    <source> </source>
  </ignorable>
  <segment id="2">
    <source>Sentence B.</source>
    <target order="1">Phrase B.</target>
  </segment>
  <ignorable>
    <source> </source>
  </ignorable>
  <segment id="3">
    <source>Sentence C.</source>
    <target order="3">Phrase C.</target>
  </segment>
</unit>
```

3.8.3 Segmentation Modification

When modifying segmentation of a `<unit>`, Modifiers MUST meet the Constraints and follow the Processing Requirements defined below:

Constraints

- Integrity of the inline codes MUST be preserved. See the section on [Inline Codes](#) and on [Annotations](#) for details.
- The entire source content of any one `<unit>` element MUST remain logically unchanged: `<segment>` elements or their data MUST NOT be moved or joined across units.



Warning

Note that when splitting or joining segments that have both source and target content it is advisable to keep the resulting segments linguistically aligned, which is likely to require human linguistic expertise and hence manual re-segmentation. If the linguistically correct

alignment cannot be guaranteed, discarding the target content and retranslating the resulting source segments is worth considering.

Processing Requirements

- When the Modifiers perform a split operation:
 - Only `<segment>` or `<ignorable>` elements that have their `canResegment` value resolved to yes MAY be split.
 - All new `<segment>` or `<ignorable>` elements created and their `<source>` and `<target>` children MUST have the same attribute values as the original elements they were created from, as applicable, except for the `id` attributes and, possibly, for the `order`, `state` and `subState` attributes.
 - Any new `id` attributes MUST follow the `<segment>` or `<ignorable>` `id` constraints.
 - If there was a target content in the original segment and if the `state` attribute of the original segment was not `initial`, the `state` attributes of the segments resulting from the split (and possibly their corresponding `subState` attributes) MAY be changed to reflect the fact that the target content MAY need to be verified as the new segmentation MAY have desynchronized the alignment between the source and target contents.
- When the Modifiers perform a join operation:
 - Only `<segment>` or `<ignorable>` elements that have their `canResegment` value resolved to yes MAY be join with other elements.
- When the Modifiers or Mergers perform a join operation:
 - Two elements (`<segment>` or `<ignorable>`) MUST NOT be joined if their `<target>` have resolved `order` values that are not consecutive.
 - The attributes of the elements to be joined (`<segment>` or `<ignorable>`) and the attributes of their `<source>` and `<target>` MUST be carried over in the resulting joined elements.
 - If attributes of elements to be joined (`<segment>` or `<ignorable>`) differ, or if the attributes of their `<source>` or `<target>` differ, the resulting joined elements MUST comply with following rules:
 - If the `state` attributes of the `<segment>` elements differ: the `state` attribute of the joined `<segment>` MUST be set to the "earliest" of the values specified in the original `<segment>` elements. The sequence of `state` values are defined in the following order: 1: `initial`, 2: `translated`, 3: `reviewed`, and 4: `final`.
 - The `subState` attribute MUST be the one associated with the `state` attribute selected to be used in the joined `<segment>`. If no `subState` attribute is associated with that `state`, the joined `<segment>` MUST NOT have a `subState`.
 - If the `xml:space` attributes differ: The `<source>` and `<target>` of the joined element MUST be set to `xml:space="preserve"`.
- When the Modifiers or Mergers perform a join or a split operation:
 - If any `<segment>` or `<ignorable>` element of the `<unit>` had a `<target>` child with an `order` attribute prior to the segmentation modification, the `<target>` child of all `<segment>` and `<ignorable>` elements in the `<unit>` MUST be examined and if necessary their `order` attributes updated to preserve the ordering of the target content prior the segmentation modification.

3.8.4 Best Practice for Mergers (Informative)

Since a typical simple corporate implementation of XLIFF 2 is a localization tool that is at the same time an Extractor and a Merger with the full knowledge of the extraction mechanism, the community requested a non-normative best practice for merging after an XLIFF Round-trip.

First of all, it needs to be noted that Mergers are not advised to rely on their knowledge of the extraction mechanism in terms of [segmentation](#). Modifiers are free to [change segmentation](#) during the roundtrip and even [to change order of target content held in different segments of the same unit](#). Therefore, it can be advised as a best practice before merging to look for all segments within each unit, even and especially when the Extractor had created only one segment per unit.

When joining segments, Mergers need to observe all Processing Requirements for [joining segments](#) and [joining or splitting segments](#)

When joining segments it can happen that not all `<segment>` or `<ignorable>` elements actually have their `<target>` element children. This situation can be legal depending on a specific workflow set up. The `<target>` child within an `<ignorable>` element is always optional, but at the same time can be created any time by simply copying the content of the sibling `<source>`, see [Content Modification Without Target](#). The presence of `<target>` children can be better governed in `<segment>` elements that have the `state` attribute. The `state` attribute is strictly optional with the default `initial`, yet it is advisable for a corporate localization operation to request that their service providers progress that attribute through `translated` and `reviewed` to `final`. This attribute cannot be progressed from the `initial` state without a `<target>` child and all violations of [Editing Hints](#) will become validation errors only in the `final` state. Usage of `state` also allows for fine-tuning of a specific workflow State Machine with the dependent `subState` attribute. With the attribute `subState`, implementers can create an arbitrary number of private state machine under their prefix authorities. It is advisable to register such authority prefixes with the XLIFF TC and publish their documentation.

When Mergers need to perform the merge in a non-final state, when the presence of targets cannot be guaranteed, they are free to create preliminary targets again following the [Processing Requirements for Content Modification Without Target](#)

3.9 Extension Mechanisms

XLIFF 2.3 offers two mechanisms for storing custom data in an XLIFF document:

1. Using the [Metadata module](#) for storing custom data in elements defined by the official XLIFF specification.
2. Using the standard XML namespace mechanism for storing data in elements or attributes defined in a custom XML Schema.

Both mechanisms can be used simultaneously.

3.9.1 Extension Points

The following XLIFF Core elements allow storing custom data in `<mda:metadata>` elements or in elements from a custom XML namespace:

- `<xliff>`
- `<file>`
- `<group>`
- `<unit>`

The following XLIFF Core elements accept custom attributes:

- `<xliff>`

- `<file>`
- `<group>`
- `<unit>`
- `<note>`
- `<mrk>`
- `<sm>`

3.9.1.1 Extensibility of XLIFF Modules

For extensibility of XLIFF Modules please refer to the relevant Module Sections.

3.9.2 Constraints

- When using identifiers, an extension MUST use either an attribute named `id` or the attribute `xml:id` to specify them.
- Extensions identifiers MUST be unique within their immediate `<file>`, `<group>` or `<unit>` enclosing element.
- Identifier values used in extensions MUST be of type `xs:NMTOKEN` or compatible with `xs:NMTOKEN` (e.g. `xs:NAME` and `xs:ID` are compatible).

These constraints are needed for the [fragment identification mechanism](#).

3.9.3 Processing Requirements

- A user extension, whether implemented using `<mda:metadata>` or using a custom namespace, MUST NOT provide the same functionality as an existing XLIFF core or module feature, however it MAY complement an extensible XLIFF core feature or module feature or provide a new functionality at the provided extension points.
- Mergers MUST NOT rely on custom namespace extensions, other than the ones possibly defined in `<skeleton>`, to create the translated version of the original document.
- Writers that do not support a given custom namespace based user extension SHOULD preserve that extension without modification.

4 The Modules Specifications

This section specifies the OPTIONAL Modules that MAY be used along with Core for advanced functionality.

4.1 Translation Candidates Module

4.1.1 Introduction

The source text of a document can be pre-processed against various translation resources (Translation Memory, Machine Translation, etc.) to provide translation candidates. This module provides an XLIFF capability to store lists of possible translations along with information about the similarity of the match, the quality of the translation, its provenance, etc.

4.1.2 Module Namespace and Validation Artifacts

The namespace for the Translation Candidates module is: `urn:oasis:names:tc:xliff:matches:2.0`

XML Schema for this module is available at <https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/matches.xsd>.

4.1.3 Module Fragment Identification Prefix

The fragment identification prefix for the Translation Candidates module is: `mtc`

4.1.4 XLIFF Core Reuse

The [Translation Candidates Module](#) reuses several XLIFF Core elements, most of them have mandatory `xliff:id`. The uniqueness scopes for the reused `xliff:id` attributes are separate from the XLIFF Core. The following states the exact normative Constraints for the validation purposes:

Constraints

- When the `xliff:id` attribute is used on `<xliff:mrk>`, `<xliff:sm>`, `<xliff:pc>`, `<xliff:sc>`, `<xliff:ec>`, or `<xliff:ph>` elements reused within the [Translation Candidates Module](#):
 - The inline elements enclosed by a `<xliff:target>` element MUST use the duplicate `xliff:id` values of their corresponding inline elements enclosed within the sibling `<xliff:source>` element if and only if those corresponding elements exist.
 - Except for the above exception, the value MUST be unique among all of the above within the enclosing `<match>` element.
- When used on `<xliff:data>` elements reused within the [Translation Candidates Module](#):

The value MUST be unique among all `<xliff:data>` `xliff:id` attribute values within the enclosing `<match>` element.
- When the `xliff:dataRef`, `xliff:datarefstart`, and `xliff:dataRefEnd` attributes are used on `<xliff:pc>`, `<xliff:sc>`, `<xliff:ec>`, or `<xliff:ph>` elements reused within the [Translation Candidates Module](#), their NMTOKEN values MUST identify `<data>` elements within the enclosing `<match>` element. Those attributes MUST NOT be used without corresponding `<data>` elements within the enclosing `<match>` element.

4.1.5 Translation Candidate Annotation

This annotation can be used to mark up the scope of a translation candidate within the content of a unit. This module can reference any source or even target spans of content that are referenceable via the XLIFF [Fragment Identification](#) mechanism, however in case the corresponding fragment is not suitably delimited, the best way how to mark the relevant span is to use the following annotation.

Usage:

- The **id** attribute is REQUIRED
- The **type** attribute is REQUIRED and set to `mtc:match`
- The **ref** attribute is not used.
- The **translate** attribute is OPTIONAL

For example:

```
<unit id="1">
  <mtc:matches>
    <mtc:match ref="#m1" similarity="68">
      <source>He is my friend.</source>
      <target>Il est mon ami.</target>
    </mtc:match>
    <mtc:match ref="#m1" similarity="60">
      <source>He is my best friend.</source>
      <target>Il est mon meilleur ami.</target>
    </mtc:match>
  </mtc:matches>
  <segment>
    <source>Paul <mrk id="m1" type="mtc:match">is my friend</mrk>.</source>
  </segment>
  <segment>
    <source>Yet, I barely see him.</source>
  </segment>
</unit>
```

4.1.6 Module Elements

The elements defined in the Translation Candidates module are: `<matches>` and `<match>`.

4.1.6.1 Tree Structure

Legend:

1 = one
+ = one or more
? = zero or one
* = zero or more

```
<matches>
|
+---<match> +
|
|   +---<mda:metadata> ?
|   |
|   +---<xlf:originalData> ?
|   |
```

```
+---<xliff:source> 1
|
+---<xliff:target> 1
|
+---<other> *
```

4.1.6.2 matches

Collection of matches retrieved from any leveraging system (Machine Translation, Translation Memory, etc.)

Contains:

- One or more `<match>` elements

4.1.6.3 match

A potential translation suggested for a part of the source content of the enclosing `<unit>` element.

Contains:

- Zero or one `<mda:metadata>` element followed by
- Zero or one `<originalData>` element followed by
- One `<source>` element followed by
- One `<target>` element followed by
- elements from other namespaces, OPTIONAL

Attributes:

- `id`, OPTIONAL
- `matchQuality`, OPTIONAL
- `matchSuitability`, OPTIONAL
- `origin`, OPTIONAL
- `ref`, REQUIRED
- `reference`, OPTIONAL
- `similarity`, OPTIONAL
- `subType`, OPTIONAL
- `type`, OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- When a `<target>` element is a child of `<match>` and the `reference` attribute is set to yes, the OPTIONAL `xml:lang` attribute's value is not REQUIRED to be equal to the value of the `trgLang` attribute of the enclosing `<xliff>` element.
- The following XLIFF Module attributes are explicitly allowed by the wildcard other:
 - attributes from the namespace `http://www.w3.org/2005/11/its`, OPTIONAL, provided that the Constraints specified in the *ITS Module* are met.
 - attributes from the namespace `urn:oasis:names:tc:xliff:itsm:2.1`, OPTIONAL, provided that the Constraints specified in the *ITS Module* are met.

4.1.7 Module Attributes

The attributes defined in the Translation Candidates module are: [id](#), [matchQuality](#), [matchSuitability](#), [origin](#), [ref](#), [reference](#), [similarity](#), [subType](#), and [type](#).

4.1.7.1 id

Identifier - a character string used to identify a [<match>](#) element.

Value description: NMTOKEN.

Default value: undefined

Used in: [<match>](#).

Constraints

- The [id](#) value MUST be unique within the enclosing [<matches>](#) element.

4.1.7.2 matchQuality

Match quality - indicates the quality of the [<target>](#) child of a [<match>](#) element based on an external benchmark or metric.

Value description: A decimal number between 0.0 and 100.0.

Default value: undefined

Used in: [<match>](#).



Note

This attribute can carry a human review based metrics score, a Machine Translation self-reported confidence score etc.

4.1.7.3 matchSuitability

Match suitability - indicates the general suitability and relevance of its [<match>](#) element based on various external benchmarks or metrics pertaining to both the [<source>](#) and the [<target>](#) children of the [<match>](#).

This attribute is intended to carry a value that can be combined from values provided in [similarity](#) and [matchQuality](#) attributes based on an externally provided algorithm.

Value description: A decimal number between 0.0 and 100.0.

Default value: undefined

Used in: [<match>](#).



Note

This attribute is also useful for mapping match-quality as specified in XLIFF 1.2 because 1.2 is not capable of discerning between the source similarity and the target quality.

Processing Requirements

- Agents processing this module MUST make use of [matchSuitability](#) for match ordering purposes if the attribute is specified.

4.1.7.4 origin

Match origin - indicates the tool, system or repository that generated a `<match>` element. This is a free text short informative description. For example, 'Microsoft Translator Hub' or 'tm-client123-v456', or 'MSTH (52217d25-d9e7-54a2-af44-3d4e4341d112_healthc).'

Value description: Text.

Default value: undefined

Used in: `<match>`.

4.1.7.5 ref

Reference - points to a span of text within the same unit, to which the translation candidate is relevant.

Value description: IRI

Default value: undefined

Used in: `<match>`.

Constraints

- The value of the `ref` attribute MUST point to a span of text within the same `<unit>` element where the `<match>` is located.



Note

When the `ref` attribute points to a `<segment>` element, by default the `<match>` content applies to its `<source>` child.

4.1.7.6 reference

Reference - indicates that the `<target>` child of the `<match>` element contains a translation into a reference language rather than into the target language. For example, a German translation can be used as reference by a Luxembourgish translator.

Value description: yes or no.

Default value: no.

Used in: `<match>`

4.1.7.7 similarity

Similarity - indicates the similarity level between the content of the `<source>` child of a `<match>` element and the translatable text being matched.

Value description: A decimal number between 0.0 and 100.0.

Default value: undefined

Used in: `<match>`.

4.1.7.8 subType

Sub-type - indicates the sub-type, i.e. a secondary level type, of a `<match>` element.

Value description:

The value is composed of a prefix and a sub-value separated by a character : (U+003A). The prefix is a string uniquely identifying a collection of values for a specific authority. The sub-value is any string value defined by an authority.

The prefix xlf is reserved for this specification, but no sub-values are defined for it at this time. Other prefixes and sub-values MAY be defined by the users.

- Default value: undefined

Used in: `<match>`

Constraints

- If the attribute `subType` is used, the attribute `type` MUST be explicitly set.

Processing Requirements

- Writers updating the attribute `type` MUST also update or delete `subType`.

4.1.7.9 type

Type - indicates the type of a `<match>` element, it gives the value providing additional information on how the match was generated or qualifying further the relevance of the match. The list of pre-defined values is general and user-specific information can be added using the `subType` attribute.

Value description:

Table 3. Values

Value	Description
am	Assembled Match: candidate generated by assembling parts of different translations. For example: constructing a candidate by using the known translations of various spans of content of the source.
mt	Machine Translation: candidate generated by a machine translation system.
icm	In Context Match: candidate for which the content context of the translation was the same as the one of the current source. For example: the source text for both contents is also preceded and/or followed by an identical source segment, or both appear as e.g. level 2 headings.
idm	Identifier-based Match: candidate that has an identifier identical to the one of the source content. For example: the previous translation of a given UI component with the same ID. match that has an identifier identical to the source content.
tb	Term Base: candidate obtained from a terminological database, i.e. the whole source segment matches with a source term base entry.
tm	Translation Memory: candidate based on a simple match of the source content.
other	Candidate of a top level type not covered by any of the above definitions.

- Default value: tm

Used in: `<match>`

Processing Requirements

- Writers updating the attribute `type` MUST also update or delete `subType`.

4.1.8 Example

```
<unit id="2">
```

```

<mtc:matches>
  <mtc:match origin="Self" type="tm" ref="#2-0" matchQuality="74">
    <source xml:lang="es">Esta es una línea de
      <ph id="ph1"/>texto<ph id="ph2"/> que necesita traducción.</source>
    <target xml:lang="en">This is a line of
      <ph id="ph1"/>text<ph id="ph2"/> that needs translation.</target>
  </mtc:match>
  <mtc:match origin="Self" type="tm" ref="#2-0" matchQuality="66">
    <source xml:lang="es">Esta es la primera línea de texto que
      necesita traducción.</source>
    <target xml:lang="en">This is the first line of text that needs
      translation.</target>
  </mtc:match>
</mtc:matches>
<originalData>
  <data id="ph1">&lt;start/></data>
  <data id="ph2">&lt;end/></data>
</originalData>
<segment state="final" id="2-0">
  <source>Otra línea de <ph id="ph1"/>texto<ph id="ph2"/> que
    necesita traducción.</source>
  <target>Another line of <ph id="ph1"/>text<ph id="ph2"/> that
    needs translation.</target>
</segment>
</unit>

```

4.2 Glossary Module

4.2.1 Introduction

Simple glossaries, consisting of a list of terms with a definition or translation, can be optionally embedded in an XLIFF document using the namespace mechanism to include elements from the Glossary module.

4.2.2 Module Namespace and Validation Artifacts

The namespace for the Glossary module is: `urn:oasis:names:tc:xliff:glossary:2.0`

XML Schema for this module is available at <https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/glossary.xsd>.

4.2.3 Module Fragment Identification Prefix

The fragment identification prefix for the Glossary module is: `gls`

4.2.4 Module Elements

The elements defined in the Glossary module are: `<glossary>`, `<glossEntry>`, `<term>`, `<translation>` and `<definition>`.

4.2.4.1 Tree Structure

Legend:

1 = one
 + = one or more
 ? = zero or one

* = zero, one or more

```
<glossary>
|
+---<glossEntry> +
|
|   +---<term> 1
|   |
|   +---<translation> *
|   |
|   +---<definition> ?
|   |
|   +---<other> *
```

4.2.4.2 glossary

Container for a list of glossary terms.

Contains:

- One or more `<glossEntry>` elements.

4.2.4.3 glossEntry

Glossary entry.

Contains:

- One `<term>` element followed by
- Zero, one or more `<translation>` elements followed by
- Zero or one `<definition>` element followed by
- elements from other namespaces, OPTIONAL

Attributes:

- `id`, OPTIONAL
- `ref`, OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- A `<glossEntry>` element MUST contain a `<translation>` or a `<definition>` element to be valid.
- The following XLIFF Module elements are explicitly allowed by the wildcard `other`:
 - Zero or one `<mda:metadata>` elements

4.2.4.4 term

A term in the glossary, expressed in the source language of the enclosing `<xliff>` element.

Contains:

- Text

Attributes:

- [source](#), OPTIONAL
- attributes from other namespaces, OPTIONAL

4.2.4.5 translation

A translation of the sibling [term](#) element expressed in the target language of the enclosing [xliff](#) element. Multiple translations can be specified as synonyms.

Contains:

- Text

Attributes:

- [id](#), OPTIONAL
- [ref](#), OPTIONAL
- [source](#), OPTIONAL
- attributes from other namespaces, OPTIONAL

4.2.4.6 definition

Optional definition in plain text for the term stored in the sibling [term](#) element.

Contains:

- Text

Attributes:

- [source](#), OPTIONAL
- attributes from other namespaces, OPTIONAL

4.2.5 Module Attributes

The attributes defined in the Glossary module are: [id](#), [ref](#), and [source](#)

4.2.5.1 id

Identifier - a character string used to identify a [glossEntry](#) or [translation](#) element.

Value description: NMTOKEN

Default value: undefined

Used in: [glossEntry](#) and [translation](#)

Constraints

- The values of [id](#) attributes MUST be unique among all [glossEntry](#) and [translation](#) elements within the given enclosing [glossary](#) element.

4.2.5.2 ref

Reference - points to a span of source or target text within the same unit, to which the glossary entry is relevant.

Value description: IRI

Default value: undefined

Used in: [<glossEntry>](#) and [<translation>](#).

Constraints

- The value of the [ref](#) attribute MUST point to a span of text within the same [<unit>](#) element, where the enclosing [<glossary>](#) element is located.

4.2.5.3 source

Source - indicates the origin of the content of the element where the attribute is defined.

Value description: Text.

Default value: undefined

Used in: [<term>](#), [<translation>](#), and [<definition>](#).

4.2.6 Example

```
<unit id="1">
  <gls:glossary>
    <gls:glossEntry ref="#m1">
      <gls:term source="publicTermbase">TAB key</gls:term>
      <gls:translation id="1" source="myTermbase">Tabstopptaste
        </gls:translation>
      <gls:translation ref="#t=m1" source="myTermbase">TAB-TASTE
        </gls:translation>
      <gls:definition source="publicTermbase">A keyboard key that is
        traditionally used to insert tab characters into a document.
        </gls:definition>
    </gls:glossEntry>
  </gls:glossary>
  <segment>
    <source>Press the <mrk id="m1" type="term">TAB key</mrk>.</source>
    <target>Drücken Sie die <mrk id="m1" type="term">TAB-TASTE</mrk>.</target>
  </segment>
</unit>
```

4.3 Format Style Module

4.3.1 Introduction

This is intended as a namespace mechanism to carry inside an XLIFF document information needed for generating a quick at a glance HTML preview of XLIFF content using a predefined set of simple HTML formatting elements.

4.3.2 Module Namespace and Validation Artifacts

The namespace for the Format style module is: `urn:oasis:names:tc:xliff:fs:2.0`

XML Schema is available at <https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/fs.xsd> .

4.3.3 Module Fragment Identification Prefix

Format Style module does not have a fragment identification prefix. Prefix `fs` is reserved in case it became needed in the future developments of this module.

4.3.4 Module Specification

Format Style module consists of just two attributes: `fs` and `subFs`. It does not specify any elements.

Format Style allows most structural and inline XLIFF core elements to convey basic formatting information using a predefined subset of HTML formatting elements. It primarily enables the generation of HTML pages or snippets for preview and review purposes. It MUST NOT be used to prescribe a roundtrip to a source document format.

The `fs` attribute holds the name of an HTML formatting element. If additional style information is needed, the OPTIONAL `subFs` attribute is provided.

Constraints

- The Format Style attributes MUST be configured in such a way that the HTML [HTML5] snippet resulting at the `<file>` level is valid.

Processing Requirements

- Extractors and Enrichers SHOULD use the following method to validate their HTML snippets:
 1. Parse the snippet with the [HTML5] fragment parsing algorithm, see <http://www.w3.org/TR/html5/syntax.html#parsing-html-fragments>.
 2. the result MUST be a valid DOM tree as per [HTML5], see <http://www.w3.org/TR/html5/infrastructure.html#tree-order>.



Note

The above constraint and validation method will make sure that the snippets are renderable by standard HTML browsers.

4.3.5 Module Attributes

The attributes defined in the `Format Style` module are: `fs`, `subFs`.

4.3.5.1 `fs`

Format style attribute, `fs` - allows most structural and inline XLIFF core elements to convey basic formatting information using a predefined subset of HTML formatting elements (for example, HTML elements names like `<script>` are not included). It enables the generation of HTML pages or snippets for preview and review purposes. If additional style information is needed, the OPTIONAL `subFs` attribute is provided.

Value description:

Table 4. Values

a	anchor
b	bold text style
bdo	l18N BiDi over-ride
big	large text style
blockquote	long quotation
body	document body
br	forced line break
button	push button
caption	table caption
center	shorthand for DIV align=center
cite	citation
code	computer code fragment
col	table column
colgroup	table column group
dd	definition description
del	deleted text
div	generic language/style container
dl	definition list
dt	definition term
em	emphasis
h1	heading
h2	heading
h3	heading
h4	heading
h5	heading
h6	heading
head	document head
hr	horizontal rule
html	document root element
i	italic text style
img	image
label	form field label text
legend	fieldset legend
li	list item
ol	ordered list
p	paragraph
pre	preformatted text
q	short inline quotation
s	strike-through text style
samp	sample program output, scripts, etc.
select	option selector

small	small text style
span	generic language/style container
strike	strike-through text
strong	strong emphasis
sub	subscript
sup	superscript
table	
tbody	table body
td	table data cell
tfoot	table footer
th	table header cell
thead	table header
title	document title
tr	table row
tt	teletype or monospaced text style
u	underlined text style
ul	unordered list

Default value: undefined.

Used in: [<file>](#), [<unit>](#), [<note>](#), [<sc>](#), [<ec>](#), [<ph>](#), [<pc>](#), [<mrk>](#), and [<sm>](#).



Warning

The [fs](#) attribute is not intended to facilitate merging back into the original format.

Constraints

- The [fs](#) MUST only be used with [<ec>](#) in cases where the `isolated` attribute is set to 'yes'.

Processing Requirements

- Writers updating the attribute [fs](#) MUST also update or delete [subFs](#).

Example: To facilitate HTML preview, [fs](#) can be applied to XLIFF like this like:

```
<xliff xmlns:fs="urn:oasis:names:tc:xliff:fs:2.0">
  <file fs:fs="html">
    <unit id="1" fs:fs="p">
      <segment>
        <source>Mick Jones renewed his interest in the Vintage <pc id="1"
          fs:fs="strong">'72 Telecaster Thinline </pc> guitar.
          <ph id="ph2" fs:fs="br" />He says <pc fs:fs="q">I love 'em
          </pc><ph id="ph1" fs:fs="img"
            fs:subFs="src,smileface.png" /></source>
        </segment>
      </unit>
    </file>
  </xliff>
```

With an XSL stylesheet like this:

```

<xsl:template match="*" priority="2"
  xmlns:fs="urn:oasis:names:tc:xliff:fs:2.0">
  <xsl:choose>
    <xsl:when test="@fs:fs">
      <xsl:element name="{@fs:fs}">
        <xsl:if test="@fs:subFs">
          <xsl:variable name="att_name"
            select="substring-before(@fs:subFs, ', ')" />
          <xsl:variable name="att_val"
            select="substring-after(@fs:subFs, ', ')" />
          <xsl:attribute name="{ $att_name }">
            <xsl:value-of select="$att_val" />
          </xsl:attribute>
        </xsl:if>
        <xsl:apply-templates />
      </xsl:element>
    </xsl:when>
    <xsl:otherwise>
      <xsl:apply-templates />
    </xsl:otherwise>
  </xsl:choose>
</xsl:template>

```

You can generate a an HTML page like this:

```

<html>
  <p>Mick Jones renewed his interest in the Vintage <strong>'72
    Telecaster Thinline </strong> guitar. <br/>He says <q>I love 'em
    </q></p>
</html>

```

4.3.5.2 subFs

Sub-format style, subFs - allows extra metadata, like URL for example, to be added in concert with the [fs](#) attribute.

Value description: The subFs attribute is used to specify the HTML attributes to use along with the HTML element declared in the [fs](#) attribute. It is a list of name/value pairs. Each pair is separated from the next with a backslash (\). The name and the value of a pair are separated with a comma (.). Both literal backslash and comma characters are escaped with a backslash prefix.

Default value: undefined.

Used in: [<file>](#), [<unit>](#), [<note>](#), [<source>](#), [<target>](#), [<sc>](#), [<ec>](#), [<ph>](#), [<pc>](#), [<mrk>](#), and [<sm>](#).



Warning

The [subFs](#) attribute is not intended to facilitate merging back into the original format.

Constraints

- Commas (,) and backslashes (\) in the value parts of the [subFs](#) MUST be escaped with a backslash (\).
- If the attribute [subFs](#) is used, the attribute [fs](#) MUST be specified as well.

- The `subFs` MUST only be used with `<ec>` in cases where the isolated attribute is set to 'yes'.

Processing Requirements

- Writers updating the attribute `fs` MUST also update or delete `subFs`.

Example: For complex HTML previews that require more than one attribute on an HTML preview element, attribute pairs are separated by backslashes (\). Any literal comma or backslash in an attribute value MUST be escaped with a backslash.

For example, we would use this convention:

```
<ph id="p1" fs="img" subFs="src,c:\\docs\\images\\smile.png\\alt,
    My Happy Smile\\title,Smiling faces\\, are nice" />
```

To produce this HTML preview:

```

```

4.4 Metadata Module

4.4.1 Introduction

The Metadata module provides a mechanism for storing custom metadata using elements that are part of the official XLIFF specification.

4.4.2 Module Namespace and Validation Artifacts

The namespace for the Metadata module is: `urn:oasis:names:tc:xliff:metadata:2.0`

XML Schema for this module is available at <https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/metadata.xsd>.

4.4.3 Module Fragment Identification Prefix

The fragment identification prefix for the Metadata module is: `mda`

4.4.4 Module Elements

The elements defined in the Metadata module are: `<metadata>`, `<metaGroup>`, and `<meta>`.

4.4.4.1 Tree Structure

Legend:

+ = one or more

```
<metadata>
|
+---<metaGroup> +
|
+---At least one of (<metaGroup> OR <meta>)
|
+---<meta>
```

4.4.4.2 metadata

Container for metadata associated with the enclosing element.

Contains:

- One or more `<metaGroup>` elements

Attributes:

- `id`, OPTIONAL

Example: Metadata can be used to store XML attribute names and values for XLIFF documents that do not use a skeleton. The following XML sample contains attributes on the `<document>` and `<row>` elements.

```
<document version="3" phase="draft">
  <table>
    <row style="head">
      <cell>Name</cell>
      <cell>Position</cell>
    </row>
    <row>
      <cell>Patrick K.</cell>
      <cell>Right Wing</cell>
    </row>
    <row>
      <cell>Bryan B.</cell>
      <cell>Left Wing</cell>
    </row>
  </table>
</document>
```

The Metadata module can be used to preserve these attributes for a round trip without using a skeleton:

```
<?xml version="1.0" encoding="utf-8"?>
<xliiff xmlns="urn:oasis:names:tc:xliiff:document:2.3"
  xmlns:fs="urn:oasis:names:tc:xliiff:fs:2.0"
  xmlns:mda="urn:oasis:names:tc:xliiff:metadata:2.0" version="2.3"
  srcLang="en">
  <file id="f1">
    <group id="g1" name="document">
      <mda:metadata>
        <mda:metaGroup category="document_xml_attribute">
          <mda:meta type="version">3</mda:meta>
          <mda:meta type="phase">draft</mda:meta>
        </mda:metaGroup>
      </mda:metadata>
    </group id="g1" name="document">
    <group id="g2" name="table">
      <group id="g3" name="row">
        <mda:metadata>
          <mda:metaGroup category="row_xml_attribute">
            <mda:meta type="style">head</mda:meta>
          </mda:metaGroup>
        </mda:metadata>
      </group id="g3" name="row">
    </group id="g2" name="table">
  </group id="f1">
</xliiff>
```

```

        </mda:metaGroup>
    </mda:metadata>
    <unit id="u1" name="cell">
        <segment>
            <source>Name</source>
        </segment>
    </unit>
    <unit id="u2" name="cell">
        <segment>
            <source>Position</source>
        </segment>
    </unit>
</group>
<group id="g4" name="row">
    <unit id="u3" name="cell">
        <segment>
            <source>Patrick K.</source>
        </segment>
    </unit>
    <unit id="u4" name="cell">
        <segment>
            <source>Right Wing</source>
        </segment>
    </unit>
</group>
<group id="g5" name="row">
    <unit id="u5" name="cell">
        <segment>
            <source>Bryan B.</source>
        </segment>
    </unit>
    <unit id="u6" name="cell">
        <segment>
            <source>Left Wing</source>
        </segment>
    </unit>
</group>
</group>
</group>
</file>
</xliff>

```

4.4.4.3 metaGroup

Provides a way to organize metadata into a structured hierarchy.

Contains:

- One or more [<metaGroup>](#) or [<meta>](#) elements in any order.

Attributes:

- [id](#), OPTIONAL
- [category](#), OPTIONAL
- [appliesTo](#), OPTIONAL

4.4.4.4 meta

Container for a single metadata component.

Contains:

- Non-translatable text

Attributes:

- [type](#), REQUIRED

4.4.5 Module Attributes

The attributes defined in the Metadata module are: [appliesTo](#), [category](#), [id](#), and [type](#).

4.4.5.1 appliesTo

Indicates the element to which the content of the metagroup applies.

Value description: source, target, or ignorable.

Default value: undefined.

Used in: [<metaGroup>](#).

4.4.5.2 category

[category](#) - indicates a category for metadata contained in the enclosing [<metaGroup>](#) element.

Value description: Text.

Default value: undefined.

Used in: [<metaGroup>](#).

4.4.5.3 id

Identifier - a character string used to identify a [<metadata>](#) or [<metaGroup>](#) element.

Value description: NMTOKEN

Default value: undefined

Used in: [<metadata>](#) and [<metaGroup>](#)

Constraints

- The values of [id](#) attributes MUST be unique among all [<metaGroup>](#) and [<metadata>](#) elements within the given enclosing [<metadata>](#) element.

4.4.5.4 type

[type](#) - indicates the type of metadata contained by the enclosing element.

Value description: Text.

Default value: undefined.

Used in: [<meta>](#).

4.5 Resource Data Module

4.5.1 Introduction

The Resource Data module provides a mechanism for referencing external resource data that MAY need to be modified or used as contextual reference during translation.

4.5.2 Module Namespace and Validation Artifacts

The namespace for the Resource Data module is: `urn:oasis:names:tc:xliff:resourcedata:2.0`

XML Schema for this module is available at https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/resource_data.xsd.

4.5.3 Module Fragment Identification Prefix

The fragment identification prefix for the Resource Data module is: `res`

4.5.4 Module Elements

The elements defined in the Resource Data module are: `<resourceData>`, `<resourceItemRef>`, `<resourceItem>`, `<source>`, `<target>`, and `<reference>`.

4.5.4.1 Tree Structure

Legend:

? = zero or one

* = zero, one or more

```
<resourceData>
|
+---<resourceItemRef> *
|
+---<resourceItem> *
|
|   +---<notes> *
|   |
|   |   +===<note> ?
|   |
|   +---<source> ?
|   |
|   |   +---<other> *
|   |
|   +---<target> ?
|   |
|   |   +---<other> *
|   |
|   +---<reference> *
```

4.5.4.2 resourceData

Parent container for resource data associated with the enclosing element.

Contains:

At least one of the following

- Zero, one or more `<resourceItemRef>` elements.
- Zero, one or more `<resourceItem>` elements.

4.5.4.3 resourceItemRef

Specifies a reference to an associated `<resourceItem>` element located at the `<file>` level.

Contains:

This element is always empty.

Attributes:

- `id`, OPTIONAL
- `ref`, REQUIRED
- attributes from other namespaces, OPTIONAL

Constraints

- The value of the OPTIONAL `id` attribute MUST be unique among all `<resourceItem>` and `<resourceItemRef>` elements of the enclosing `<resourceData>` element.

Processing Requirements

- Modifiers MUST remove `<resourceItemRef>` when removing the referenced `<resourceItem>`.

4.5.4.4 resourceItem

Container for specific resource data that is either intended for modification, or to be used as contextual reference during translation.

Contains:

At least one of the following

- Zero or one `<notes>` element followed by
- Zero or one `<source>` element followed by
- Zero or one `<target>` element followed by
- Zero, one or more `<reference>` elements

Attributes:

- `contentType`, OPTIONAL
- `id`, OPTIONAL
- `context`, OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- The `contentType` attribute is REQUIRED if `<target>` and `<source>` child elements are empty, otherwise it is OPTIONAL.
- The value of the OPTIONAL `id` attribute MUST be unique among all `<resourceItem>` and `<resourceItemRef>` elements of the enclosing `<resourceData>` element.

Processing Requirements

- If a Modifier does not understand how to process the `mimeType` attribute, or the file it references, the `<resourceItem>` element MAY be ignored, but still MUST be preserved.
- The `mimeType` attribute SHOULD only be modified or removed if the referenced files are modified or removed.
- For each instance of `<resourceItem>` containing only `<source>`:
 - Modifiers MAY leave `<resourceItem>` unchanged, i.e. they are not REQUIRED to create `<target>` or `<reference>`.
 - Modifiers MAY create `<target>` or `<reference>` as a siblings of `<source>`.

4.5.4.5 source

References the actual resource data that is either intended for modification, or to be used as contextual reference during translation.

Contains:

Either

- elements from other namespaces

or

- is empty.

Attributes:

- `href` , OPTIONAL
- `xml:lang` , OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- The attribute `href` is REQUIRED if and only if `<source>` is empty.
- When the OPTIONAL `xml:lang` attribute is present, its value MUST be equal to the value of the `srcLang` attribute of the enclosing `<xliff>` element.

Processing Requirements

- When the `context` attribute of `<resourceItem>` is set to yes:
 - Modifiers MAY create `<source>` if not already present.
 - Modifiers SHOULD NOT change `<source>`.
 - Modifiers MAY remove `<source>`.
- When the `context` attribute of `<resourceItem>` is set to no:
 - `<source>` MUST be present.
 - Modifiers MUST NOT change `<source>`.
 - Modifiers MUST NOT remove `<source>`.

4.5.4.6 target

References the localized counterpart of the sibling `<source>` element.

Contains:

Either

- elements from other namespaces

or

- is empty.

Attributes:

- `href`, OPTIONAL
- `xml:lang`, OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- The attribute `href` is REQUIRED if and only if `<target>` is empty.
- When the OPTIONAL `xml:lang` attribute is present, its value MUST be equal to the value of the `trgLang` attribute of the enclosing `<xliff>` element.

Processing Requirements

- When the `context` attribute of `<resourceItem>` is set to yes:
 - Modifiers MAY create `<target>` if not already present.
 - Modifiers SHOULD NOT change `<target>`.
 - Modifiers MAY remove `<target>`.
- When the `context` attribute of `<resourceItem>` is set to no:
 - Modifiers MAY create `<target>` if not already present.
 - Modifiers MAY leave `<target>` unchanged.
 - Modifiers MAY change `<target>`.
 - Modifiers MAY replace an existing `<target>`, i.e. the previously populated `<target>` MUST NOT be left blank.

4.5.4.7 reference

References contextual data relating to the sibling `<source>` and `<target>` elements, such as a German screenshot for a Luxembourgish translator.

Contains:

- This element is always empty.

Attributes:

- `href`, REQUIRED
- `xml:lang`, OPTIONAL

- attributes from other namespaces, OPTIONAL

Constraints

- When the OPTIONAL `xml:lang` attribute is present, its value does not need to be equal to the value of the `srcLang` or `trgLang` attribute of the enclosing `<xliff>` element.

Processing Requirements

- Writers MAY create `<reference>` if not already present.
- Modifiers SHOULD NOT change `<reference>`.
- Modifiers MAY remove `<reference>`.

4.5.5 Module Attributes

The attributes defined in the Resource Data module are: `id`, `xml:lang`, `mimeType`, `context`, `href`, and `ref`.

4.5.5.1 id

Identifier - A character string used to identify a `<resourceData>` element.

Value description: NMTOKEN

Default value: undefined

Used in: `<resourceItem>` and `<resourceItemRef>`

4.5.5.2 xml:lang

Language - The `xml:lang` attribute specifies the language variant of the text of a given element. For example: `xml:lang="fr-FR"` indicates the French language as spoken in France.

Value description: A language code as described in [BCP 47].

Default value: undefined

Used in: `<source>`, `<target>`, and `<reference>`.

4.5.5.3 mimeType

MIME type, `mimeType` - indicates the type of a resource object. This generally corresponds to the content type of [RFC 2045], the MIME specification; e.g. `mimeType="text/xml"` indicates the resource data is a text file of XML format.

Value description: A MIME type. An existing MIME type MUST be used from a [list of standard values](#).

Default value: undefined

Used in: `<resourceItem>`



Note

If you cannot use any of the standard MIME type values as specified above, a new MIME type can be registered according to [RFC 2048].

4.5.5.4 context

Contextual Information - Indicates whether an external resource is to be used for context only and not modified.

Value description: yes or no

Default value: yes

Used in: [<resourceItem>](#)

4.5.5.5 href

Hypertext Reference, href - IRI referencing an external resource.

Value description: IRI.

Default value: undefined

Used in: [<source>](#), [<target>](#), and [<reference>](#)

4.5.5.6 ref

Resource Item Reference - holds a reference to an associated [<resourceItem>](#) element located at the [<file>](#) level.

Value description: An [\[XML Schema Datatypes\]](#) NMTOKEN

Default value: undefined

Used in: [<resourceItemRef>](#)

Constraints

- The [ref](#) attribute value MUST be the value of the [id](#) attribute of the [<resourceItem>](#) element being referenced.

4.5.6 Examples

In this example, the [<resourceData>](#) module at [<file>](#) level references external XML that contains resource data for a user interface control. The control is the container for the text “Load Registry Config” and needs to be resized to accommodate the increased length of the string due to translation. The [<resourceItemRef>](#) element contained in the [<resourceData>](#) module at [<unit>](#) level provides the reference between them. The name attribute of the [<unit>](#) element could serve as the key for an editor to associate [<source>](#) and [<target>](#) text with the resource data contained in the referenced XML and display it for modification.

```
<file id="f1">
  <res:resourceData>
    <res:resourceItem id="r1" mimeType="text/xml" context="no">
      <res:source href="resources\en\registryconfig.resources.xml" />
      <res:target href="resources\de\registryconfig.resources.xml" />
    </res:resourceItem>
  </res:resourceData>
  <unit id="1" name="130;WIN_DLG_CTRL_">
    <res:resourceData>
      <res:resourceItemRef ref="r1" />
    </res:resourceData>
  </unit>
</file>
```

```

    <segment id="1" state="translated">
      <source>Load Registry Config</source>
      <target>Registrierungskonfiguration laden</target>
    </segment>
  </unit>
</file>

```

In this example, the `<resourceData>` module at the `<unit>` level contains elements from another namespace (abc), which could be displayed for modification in an editor that understands how to process the namespace.

```

<file id="f2" xmlns:abc="urn:abc">
  <unit id="1">
    <res:resourceData>
      <res:resourceItem id="r1" context="no">
        <res:source>
          <abc:resourceType>button</abc:resourceType>
          <abc:resourceHeight>40</abc:resourceHeight>
          <abc:resourceWidth>75</abc:resourceWidth>
        </res:source>
        <res:target>
          <abc:resourceType>button</abc:resourceType>
          <abc:resourceHeight>40</abc:resourceHeight>
          <abc:resourceWidth>150</abc:resourceWidth>
        </res:target>
      </res:resourceItem>
    </res:resourceData>
    <segment id="1" state="translated">
      <source>Load Registry Config</source>
      <target>Registrierungskonfiguration laden</target>
    </segment>
  </unit>
</file>

```

In this example, the `<resourceData>` module references multiple static images that an editor can make use of as context while translating or reviewing.

```

<file id="f3">
  <res:resourceData>
    <res:resourceItem id="r1" mimeType="image/jpeg" context="yes">
      <notes>
        <note>Registry configuration UI screen shot</note>
      </notes>
      <res:source xml:lang="en-us"
        href="resources\en\registryconfig1.resources.jpg" />
      <res:target xml:lang="lb-lu"
        href="resources\lb\registryconfig1.resources.jpg" />
      <res:reference xml:lang="de-de"
        href="resources\de\registryconfig1.resources.jpg" />
    </res:resourceItem>
    <res:resourceItem id="r2" mimeType="image/jpeg" context="yes">
      <res:source xml:lang="en-us"

```

```

        href="resources\en\registryconfig2.resources.jpg" />
        <res:target xml:lang="lb-lu"
        href="resources\lb\registryconfig2.resources.jpg" />
    </res:resourceItem>
</res:resourceData>
<unit id="1">
    <res:resourceData>
        <res:resourceItemRef ref="r1" />
        <res:resourceItemRef ref="r2" />
    </res:resourceData>
    <segment id="1" state="translated">
        <source>Remove Registry Config</source>
        <target>Registrierungskonfiguration entfernen</target>
    </segment>
</unit>
</file>

```

4.6 Size and Length Restriction Module

4.6.1 Introduction

The Size and Length Restriction module provides a mechanism to annotate the XLIFF content with information on storage and general size restrictions.

The restriction framework has support for two distinct types of restrictions; storage size restrictions and general size restriction. The reason for this is that it is often common to have separate restrictions between storage and display / physical representation of data. Since it would be impossible to define all restrictions here a concept of restriction profile is introduced. The profiles for storage size and general size are independent. The information related to restriction profiles are stored in the processing invariant part of the XLIFF file like the `<xlf:file>`, `<xlf:group>` and `<xlf:unit>` elements and contained within elements defined in this module. The information regarding the specific restrictions are stored on the processing invariant parts and on the inline elements as attributes or attributes referencing data in the elements defined in this module. To avoid issues with segmentation no information regarding size restrictions is present on `<xlf:segment>`, `<xlf:source>` and `<xlf:target>` elements. The module defines a namespace for all the elements and attributes it introduces, in the rest of the module specification elements and attributes are in this namespace unless stated otherwise. In other parts of the XLIFF specification the prefix "slr" is used to refer to this module's namespace. For clarity the prefix "xlf" will be used for XLIFF Core elements and attributes. Profile names use the same namespace-like naming convention as user defined values in the XLIFF Core specification. The names SHOULD be composed of two components separated by a colon. `<authority>:<name>`. The authority "xliff" is reserved for profiles defined by the OASIS XLIFF Technical Committee.

4.6.2 Module Namespace and Validation Artifacts

The namespace for the Size and Length restriction module is: `urn:oasis:names:tc:xliff:sizerestriction:2.0`

XML Schema for this module is available at https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/size_restriction.xsd.

4.6.3 Module Fragment Identification Prefix

The fragment identification prefix for the Size and Length restriction module is: `slr`

4.6.4 Module Elements

The elements defined in the Size and Length restriction module are: `<profiles>`, `<normalization>` and `<data>`.

4.6.4.1 Tree Structure

Legend:

? = zero or one

```
<profiles>
|
+---<normalization> ?
|
+---<other> *
```

4.6.4.2 profiles

This element selects the restriction profiles to use in the document. If no storage or general profile is specified the default values (empty) of those elements will disable restriction checking in the file.

Contains:

- Zero or one `<normalization>` element followed by
- elements from other namespaces, OPTIONAL

Attributes:

- `generalProfile`, OPTIONAL
- `storageProfile`, OPTIONAL

Processing Requirements

- Any overall configuration or settings related to the selected profile MUST be placed in child elements of this element.
- Data not related to the configuration of the selected profiles MUST NOT be placed in this element.

4.6.4.3 normalization

This element is used to hold the attributes specifying the normalization form to apply to storage and size restrictions defined in the standard profiles.

Contains:

This element is always empty.

Attributes:

- `general`, OPTIONAL
- `storage`, OPTIONAL

Processing Requirements

- If this element is not present no normalization SHOULD be performed for the standard profiles.

- Other profiles MAY use this element in its specified form but MUST NOT add new extensions to it.

4.6.4.4 data

This elements act as a container for data needed by the specified profile to check the part of the XLIFF document that is a sibling or descendant of a sibling of this element. It is not used by the default profiles.

Contains:

- elements from other namespaces, OPTIONAL

Attributes:

- [profile](#), REQUIRED
- attributes from other namespaces, OPTIONAL

Processing Requirements

- Third party profiles MUST place all data in this element instead of using other extension points if the data serves no other purpose in the processing of the document.
- Data not used by the specified profile MUST NOT be placed in this element.

4.6.5 Module Attributes

The attributes defined in the Size and Length restriction module are: [storageProfile](#), [generalProfile](#), [storage](#), [general](#), [profile](#), [storageRestriction](#), [sizeRestriction](#), [equivStorage](#), [sizeInfo](#) and [sizeInfoRef](#).

4.6.5.1 storageProfile

This attribute specifies, which profile to use while checking storage size restrictions. Empty string means that no restrictions are applied.

Value description: Name of restriction profile to use for storage size restrictions.

Default value: empty string

Used in: [<profiles>](#).

4.6.5.2 generalProfile

This attribute specifies, which profile to use while checking the general size restrictions. Empty string means that no restrictions apply.

Value description: Name of restriction profile to use for general size restrictions.

Default value: empty string

Used in: [<profiles>](#).

4.6.5.3 storage

This attribute specifies the normalization form to apply for storage size restrictions. Only the normalization forms C and D as specified by the Unicode Consortium are supported, see [Unicode Standard Annex #15](#).

Value description: Normalization to apply.

Table 5. Values

Value	Description
none	No additional normalization SHOULD be done, content SHOULD be used as represented in the document. It is possible that other Agents have already done some type of normalization when modifying content. This means that this setting could give different results depending on what Agents are used to perform a specific action on the XLIFF document.
nfc	Normalization Form C MUST be used
nfd	Normalization Form D MUST be used

Default value: none

Used in: [<normalization>](#).

4.6.5.4 general

This attribute specifies the normalization to apply for general size restrictions. Only the normalization forms C and D as specified by the Unicode Consortium are supported, see [Unicode Standard Annex #15](#).

Value description: Normalization to apply.

Table 6. Values

Value	Description
none	No additional normalization SHOULD be done, content SHOULD be used as represented in the document. It is possible that other Agents have already done some type of normalization when modifying content. This means that this setting could give different results depending on what Agents are used to perform a specific action on the XLIFF document.
nfc	Normalization Form C MUST be used
nfd	Normalization Form D MUST be used

Default value: none

Used in: [<normalization>](#).

4.6.5.5 profile

This attribute is used on the `<data>` element to indicate what profile the contents of that element apply to.

Value description: Name of a restriction profile

Default value: undefined

Used in: [<data>](#).

4.6.5.6 storageRestriction

This attribute specifies the storage restriction to apply to the collection descendants of the element it is defined on.

Value description: Interpretation of the value is dependent on selected [storageProfile](#). It MUST represent the restriction to apply to the indicated sub part of the document.

Default value: undefined

Used in: [<file>](#), [<group>](#), [<unit>](#), [<mrk>](#), [<sm>](#), [<pc>](#) and [<sc>](#).

4.6.5.7 sizeRestriction

This attribute specifies the size restriction to apply to the collection descendants of the element it is defined on.

Value description: Interpretation of the value is dependent on selected [generalProfile](#). It MUST represent the restriction to apply to the indicated sub part of the document.

Default value: undefined

Used in: [<file>](#), [<group>](#), [<unit>](#), [<mrk>](#), [<sm>](#), [<pc>](#) and [<sc>](#).

4.6.5.8 equivStorage

This attribute provides a means to specify how much storage space an inline element will use in the native format. This size contribution is then added to the size contributed by the textual parts. This attribute is only allowed on the [<ec>](#) element if that element has the [isolated](#) attribute set to yes. Otherwise the attribute on the paired [<sc>](#) element also cover its partner [<ec>](#) element.

Value description: Interpretation of the value is dependent on selected [storageProfile](#). It MUST represent the equivalent storage size represented by the inline element.

Default value: undefined

Used in: [<pc>](#), [<sc>](#), [<ec>](#), [<ph>](#) and

4.6.5.9 sizeInfo

This attribute is used to associate profile specific information to inline elements so that size information can be decoupled from the native format or represented when the native data is not available in the XLIFF document. It can be used on both inline elements and structural elements to provide information on things like GUI dialog or control sizes, expected padding or margins to consider for size, what font is used for contained text and so on. This attribute is only allowed on the [<ec>](#) element if that element has the [isolated](#) attribute set to yes. Otherwise the attribute on the paired [<sc>](#) element also cover its partner [<ec>](#) element.

Value description: Interpretation of the value is dependent on selected [generalProfile](#). It MUST represent information related to how the element it is attached to contributes to the size of the text or entity in which it occurs or represents.

Default value: undefined

Used in: [<file>](#), [<group>](#), [<unit>](#), [<pc>](#), [<sc>](#), [<ec>](#), and [<ph>](#).

Constraints

- This attribute MUST NOT be specified if and only if [sizeInfoRef](#) is used. They MUST NOT be specified at the same time.

4.6.5.10 sizeInfoRef

This attribute is used to point to data that provide the same function as the [sizeInfo](#) attribute does, but with the data stored outside the inline content of the XLIFF segment. This attribute is only allowed on the [<ec>](#) element if that element has the [isolated](#) attribute set to yes. Otherwise the attribute on the paired [<sc>](#) element also cover its partner [<ec>](#) element.

Value description: A reference to data that provide the same information that could be otherwise put in a [sizeInfo](#) attribute. The reference MUST point to an element in a [<data>](#) element that is a sibling to the element this attribute is attached to or a sibling to one of its ancestors.

Default value: undefined

Used in: [<file>](#), [<group>](#), [<unit>](#), [<pc>](#), [<sc>](#), [<ec>](#), and [<ph>](#),

Constraints

- This attribute MUST NOT be specified if and only if [sizeInfo](#) is used. They MUST NOT be specified at the same time.

4.6.6 Standard profiles

4.6.6.1 General restriction profile "xliff:codepoints"

This profile implements a simple string length restriction based on the number of Unicode code points. It is OPTIONAL to specify if normalization is to be applied using the [<normalization>](#) element and the [general](#) attribute. This profile makes use of the following attributes from this module:

4.6.6.1.1 sizeRestriction

The value of this attribute holds the "maximum" or "minimum and maximum" size of the string. Either size MUST be an integer. The maximum size MAY also be '*' to denote that there is no maximum restriction. If only a maximum is specified it is implied that the minimum is 0 (empty string). The format of the value is the OPTIONAL minimum size and a coma followed by a maximum size ("[minsize,]maxsize"). The default value is '*' which evaluates to a string with unbounded size.

4.6.6.1.2 sizeInfo

The value of this attribute is an integer representing how many code points the element it is set on is considered to contribute to the total size. If empty, the default for all elements is 0.

4.6.6.2 Storage restriction profiles "xliff:utf8", "xliff:utf16" and "xliff:utf32"

These three profiles define the standard size restriction profiles for the common Unicode character encoding schemes. It is OPTIONAL to specify if normalization is to be applied using the [<normalization>](#) element and the [storage](#). All sizes are represented in 8bit bytes. The size of text for these profiles is the size of the text converted to the selected encoding without any byte order marks attached. The encodings are specified by the Unicode Consortium in [chapter 2.5 of the Unicode Standard \[Unicode\]](#).

Table 7. Profiles

Name	Description
xliff:utf8	The number of 8bit bytes needed to represent the string encoded as UTF-8 as specified by the Unicode consortium.
xliff:utf16	The number of 8bit bytes needed to represent the string encoded as UTF-16 as specified by the Unicode consortium.
xliff:utf32	The number of 8bit bytes needed to represent the string encoded as UTF-32 as specified by the Unicode consortium.

These profiles make use of the following attributes from this module:

4.6.6.2.1 storageRestriction

The value of this attribute holds the "maximum" or "minimum and maximum" size of the string. Either size MUST be an integer. The maximum size MAY also be '*' to denote that there is no maximum restriction. If only a maximum is specified it is implied that the minimum is 0 (empty string). The format of the value is the OPTIONAL minimum size and a coma followed by a maximum size ("[minsize,]maxsize"). The default value is '*' which evaluates to a string with unbounded size.

4.6.6.2 equivStorage

The value of this attribute is an integer representing how many bytes the element it is set on is considered to contribute to the total size. If empty the default is 0. The `<cp>` is always converted to its representation in the profiles encoding and the size of that representation is used as the size contributed by the `<cp>`.

4.6.7 Third party profiles

The general structure of this module together with the extensibility mechanisms provided has been designed with the goal to cater for all practically thinkable size restriction schemes. For example, to represent two dimensional data, a profile can adopt a coordinate style for the values of the general restriction attributes. For instance `{x, y}` to represent width and height, or `{{x1, y1}, {x2, y2}}` to represent a bounding box. It is also possible to embed information necessary to drive for instance a display simulator and attach that data to text in order to be able to perform device specific checking. Providing font information and checking glyph based general size are other feasible options.

4.6.8 Conformance

To claim conformance to the XLIFF size and length restriction module an Agent MUST meet the following criteria:

- MUST be compliant with the schema of the XLIFF Core specification and its extensions provided in this module.
- MUST follow all processing requirements set forth in this module specification regarding the general use of elements and attributes.
- MUST support all standard profiles with normalization set to none.
- SHOULD support all standard profiles with all modes of normalization.
- MAY support additional third party profiles for storage or general restrictions.
- MUST provide at least one of the following:
 - add size and length restriction information to an XLIFF Document
 - if it supports the profile(s) specified in the XLIFF Document it MUST provide a way to check if the size and length restrictions in the document are met according to the profile(s) requirements.

4.6.9 Example

A short example on how this module can be used is provided here with inline XML comments explaining the usage of the module features.

```
<xliff version="2.3" srcLang="en-us"
  xmlns="urn:oasis:names:tc:xliff:document:2.3"
  xmlns:slr="urn:oasis:names:tc:xliff:sizerestriction:2.0">
  <file id="f1">
    <slr:profiles generalProfile="xliff:codepoints"
      storageProfile="xliff:utf8">
      <!-- Select standard UTF-8 storage encoding and standard codepoint
        size restriction both with NFC normalization-->
      <slr:normalization general="nfc" storage="nfc" />
    </slr:profiles>
    <!-- The group should not require more than 255 bytes of storage And
      have at most 90 codepoints. Note that the sum of the unit sizes
      are larger than this the total content of the group must still
      be at most 90 codepoints. -->
    <group id="g1" slr:storageRestriction="255" slr:sizeRestriction="90">
      <!-- This unit must not contain more than 60 code points -->
      <unit id="u1" slr:sizeRestriction="60">
        <segment>
          <!-- The spanning <pc> element require 7 bytes of storage in the
```

```

        native format. Its content must not have more than 25
        codepoints -->
        <source>This is a small <pc equivStorage="7" id="pc1"
        slr:sizeRestriction="25">size restriction</pc>
        example. </source>
    </segment>
</unit>
<!-- This unit must not have more than 35 codepoints -->
<unit id="u2" slr:sizeRestriction="35">
    <segment>
        <source>With a group structure.</source>
    </segment>
</unit>
</group>
</file>
</xliff>

```

4.7 Validation Module

4.7.1 Introduction

This module defines a specific set of validation rules that can be applied to target text both globally and locally. Further constraints can be defined that allow rules to be applied to target text based on conditions in the source text or disabled to override a global scope.

4.7.2 Module Namespace and Validation Artifacts

The namespace for the Validation module is: `urn:oasis:names:tc:xliff:validation:2.0`

XML Schema for this module is available at <https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/validation.xsd>.

4.7.3 Module Fragment Identification Prefix

The fragment identification prefix for the Validation module is: `val`

4.7.4 Module Elements

The elements defined in the Validation module are: `<validation>` and `<rule>`.

4.7.4.1 Tree Structure

Legend:

+ = one or more

```

<validation>
|
+---<rule> +

```

4.7.4.2 validation

Parent container for a list of rules and constraints to apply to the target text of the enclosing element.

Contains:

- One or more `<rule>` elements.

Attributes:

- attributes from other namespaces, OPTIONAL

Processing Requirements

- When the `<validation>` element occurs at the `<file>` level, rules MUST be applied to all `<target>` elements within the scope of that `<file>` element, except where overrides are specified at the `<group>` or `<unit>` level.
- When `<validation>` occurs at the `<group>` level, rules MUST be applied to all `<target>` elements within the scope of that `<group>`, except where overrides are specified in a nested `<group>` element, or at the `<unit>` level.
- When `<validation>` occurs at the `<unit>` level, rules MUST be applied to all `<target>` elements within the scope of that `<unit>`.

4.7.4.3 rule

A specific rule and constraint to apply to the target text of the enclosing element.

Contains:

- This element is always empty.

Attributes:

- `isPresent`, OPTIONAL
- `occurs`, OPTIONAL
- `isNotPresent`, OPTIONAL
- `startsWith`, OPTIONAL
- `endsWith`, OPTIONAL
- `existsInSource`, OPTIONAL
- `caseSensitive`, OPTIONAL
- `normalization`, OPTIONAL
- `disabled`, OPTIONAL
- attributes from other namespaces, OPTIONAL

Constraints

- Exactly one of the following attributes:
 - `isPresent`
 - `isNotPresent`
 - `startsWith`
 - `endsWith`
- a custom rule defined by attributes from any namespace is REQUIRED in any one `<rule>` element.

Processing Requirements

- Writers MAY create and add new `<rule>` elements, provided that the new rules do not contradict rules already present.
- Modifiers MUST NOT change attributes defined in this module that are already present in any `<rule>` element.
- Modifiers MUST NOT remove either `<rule>` elements or their attributes defined in this module.

4.7.5 Module Attributes

The attributes defined in the Validation module are: `isPresent`, `occurs`, `isNotPresent`, `startsWith`, `endsWith`, `existsInSource`, `caseSensitive`, `normalization`, and `disabled`.

4.7.5.1 isPresent

This rule attribute specifies that a string MUST be present in the target text at least once.

For example, the following is valid:

```
<unit id="1">
  <val:validation>
    <val:rule isPresent="online" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store.</source>
    <target>Escolha uma opção na loja online.</target>
  </segment>
</unit>
```

Whereas the following is invalid:

```
<unit id="1">
  <val:validation>
    <val:rule isPresent="loja" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store.</source>
    <target>Escolha uma opção na online store.</target>
  </segment>
</unit>
```

Other rule attributes can be combined with `isPresent` to produce the following results:

`isPresent="loja"` - loja is found in the target text at least once.

`isPresent="loja" occurs="1"` - loja is found in the target text exactly once.

`isPresent="loja" existsInSource="yes"` - loja is found in both source and target text the same number of times.

`isPresent="loja" existsInSource="yes" occurs="1"` - loja is found in both source and target text and occurs in target text exactly once.

Value description: Text.

Default value: none

Used in: [<rule>](#)

4.7.5.2 occurs

This rule attribute is used with the [isPresent](#) rule attribute to specify the exact number of times a string MUST be present in the target text. When this rule attribute is not used, then the string MUST be present in the target text at least once.

For example, the following is valid:

```
<unit id="1">
  <val:validation>
    <val:rule isPresent="loja" occurs="2" />
  </val:validation>
  <segment id="1">
    <source>Choose a store option in the online store.</source>
    <target>Escolha uma opção de loja na loja online.</target>
  </segment>
</unit>
```

Whereas the following is invalid:

```
<unit id="1">
  <val:validation>
    <val:rule isPresent="loja" occurs="2" />
  </val:validation>
  <segment id="1">
    <source>Choose a store option in the online store.</source>
    <target>Escolha uma opção de loja na online store.</target>
  </segment>
</unit>
```

Value description: A number of 1 or greater.

Default value: none

Used in: [<rule>](#)

4.7.5.3 isNotPresent

This rule attribute specifies that a string MUST NOT be present in the target text.

For example, the following is valid:

```
<unit id="1">
  <val:validation>
    <val:rule isNotPresent="store" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store.</source>
    <target>Escolha uma opção na loja online.</target>
  </segment>
</unit>
```


Whereas the following is invalid:

```
<unit id="1">
  <val:validation>
    <val:rule isNotPresent="store" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store.</source>
    <target>Escolha uma opção na online store.</target>
  </segment>
</unit>
```

Value description: Text.

Default value: none

Used in: [<rule>](#)

4.7.5.4 startsWith

This rule attribute specifies that a string **MUST** start with a specific value.

For example, the following is valid:

```
<unit id="1">
  <val:validation>
    <val:rule startsWith="*" />
  </val:validation>
  <segment id="1">
    <source>*Choose an option in the online store.</source>
    <target>*Escolha uma opção na loja online.</target>
  </segment>
</unit>
```

Whereas the following is invalid:

```
<unit id="1">
  <val:validation>
    <val:rule startsWith="*" />
  </val:validation>
  <segment id="1">
    <source>*Choose an option in the online store.</source>
    <target>Escolha uma opção na loja online.</target>
  </segment>
</unit>
```

Value description: Text.

Default value: none

Used in: [<rule>](#)

4.7.5.5 endsWith

This rule attribute specifies that a string **MUST** end with a specific value.

For example, the following is valid:

```
<unit id="1">
  <val:validation>
    <val:rule endsWith=":" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store:</source>
    <target>Escolha uma opção na loja online:</target>
  </segment>
</unit>
```

Whereas the following is invalid:

```
<unit id="1">
  <val:validation>
    <val:rule endsWith=":" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store:</source>
    <target>Escolha uma opção na online store.</target>
  </segment>
</unit>
```

Value description: Text.

Default value: none

Used in: [<rule>](#)

4.7.5.6 existsInSource

When this rule attribute is used with another rule attribute and is set to yes, it specifies that for the rule to succeed, the condition **MUST** be satisfied in both source and target text. This rule attribute is valid only when used with one of the following rule attributes: [isPresent](#), [startsWith](#), or [endsWith](#).

When [existsInSource](#) is set to no, it will have no impact on execution of rules, except for overriding rules where [existsInSource](#) is set to yes on a higher level.

For example, the following are valid:

```
<unit id="1">
  <val:validation>
    <val:rule endsWith=":" existsInSource="yes" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store:</source>
    <target>Escolha uma opção na loja online:</target>
  </segment>
</unit>
```

```

...
<unit id="2">
  <val:validation>
    <val:rule endsWith=":" existsInSource="no" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store.</source>
    <target>Escolha uma opção na loja online:</target>
  </segment>
</unit>

```

Whereas the following is invalid:

```

<unit id="1">
  <val:validation>
    <val:rule endsWith=":" existsInSource="yes" />
  </val:validation>
  <segment id="1">
    <source>Choose an option in the online store.</source>
    <target>Escolha uma opção na loja online:</target>
  </segment>
</unit>

```

Value description: yes or no

Default value: no

Used in: [<rule>](#)

Constraints

- When [existsInSource](#) is specified, exactly one of
 - [isPresent](#)
 - [startsWith](#)
 - [endsWith](#)
 is REQUIRED in the same [<val:rule>](#) element.

4.7.5.7 caseSensitive

This rule attribute specifies whether the test defined within that rule is case sensitive or not.

Value description: yes if the test is case sensitive, no if the test is case insensitive.

Default value: yes.

Used in: [<rule>](#)

4.7.5.8 normalization

This rule attribute specifies the normalization type to apply when validating a rule. Only the normalization forms C and D as specified in [\[UAX #15\]](#).

Value description: The allowed values are listed in the table below along with their corresponding types of normalization to be applied.

Table 8. Values

Value	Description
none	No normalization SHOULD be done.
nfc	Normalization Form C MUST be used.
nfd	Normalization Form D MUST be used.

Default value: nfc

Used in: [<rule>](#)

4.7.5.9 disabled

This rule attribute determines whether a rule MUST or MUST NOT be applied within the scope of its enclosing element. For example, a rule defined at the [<file>](#) level can be disabled at the [<unit>](#) level.

This attribute is provided to allow for overriding execution of rules set at higher levels, see [<val:validation>](#).

In the following example, the isNotPresent rule is applied in its entirety to the first unit, but not to the second.

```
<file id="f1">
  <val:validation>
    <val:rule isPresent="store" />
  </val:validation>
  <unit id="1">
    <segment id="1">
      <source>Choose an option in the online store:</source>
      <target>Escolha uma opção na loja online:</target>
    </segment>
  </unit>
  <unit id="2">
    <val:validation>
      <val:rule isPresent="store" disabled="yes" />
    </val:validation>
    <segment id="1">
      <source>Choose an option in the application store:</source>
      <target>Escolha uma opção na application store:</target>
    </segment>
  </unit>
</file>
```

Value description: yes or no

Default value: no

Used in: [<rule>](#)

4.8 Plural, Gender, and Select Module

4.8.1 Introduction

This module provides an XLIFF capability to store information needed to represent and process messages with variants. This includes plural & gender variants, and a generic select.

We have all seen messages like this: “You have 12 day(s).”

This was a common way to deal with plural in English, but didn’t sound natural in English, and didn’t translate well in other languages.

Sometimes developers tried to make things more natural with some help from code:

```
if (count == 1)
    message = "You have {count} day"
else
    message = "You have {count} days"
```

However, this does not work for all languages. The plural rules are different between locales. French uses singular for zero and one (“0 jour”, “1 jour”, “2 jours”), most Slavic languages have four plural forms (Czech: “1 den”, “2 dny”, “1,5 dne”, “5 dní”), Chinese has no plurals at all, etc.

This sounds very unusual to many English speakers, but it is in fact very similar to the way English handles ordinals: “21st place”, “32nd place”, “23rd place”, “25th place”.

The [\[CLDR Plural Spec\]](#) project collected the plural rules (both cardinal and ordinal) for many languages.

Many technologies and internationalization libraries already support such advanced plural rules, including: [gettext](#), [macOS and iOS](#), [Android](#), [ICU \(International Components for Unicode\)](#) and [ECMAScript Intl](#).

The main challenge with these kinds of messages for localization (and XLIFF) is how to map between the plural variants, when the source and target language don’t have the same number of variants.

When the two English forms (“{count} day” / “{count} days”) to four Czech forms (“{count} den”, “{count} dny”, “{count} dne”, “{count} dní”? Or Arabic (5 forms), or Chinese one form?

This kind of n-to-m mapping is difficult, because many localization systems are designed with a one-to-one mapping in mind (one string in the source language results in one string in the target language).

Gender represents a similar challenge (when translating “red {item}” the form of “red” should change depending on the gender (and number, and case) of the item.

Combinations of these two concepts are even more challenging:

- “{host_name} invited {count} guest to **her** party”
- “{host_name} invited {count} guest to **his** party”
- “{host_name} invited {count} guest to **their** party”
- “{host_name} invited {count} guests to **her** party”
- “{host_name} invited {count} guests to **his** party”
- “{host_name} invited {count} guests to **their** party”

This XLIFF extension is designed to represent such concepts, already supported by several technologies.

4.8.2 Module Namespace and Validation Artifacts

The namespace for the Plural, Gender, and Select module is: `urn:oasis:names:tc:xliff:pgs:1.0`.

XML Schema for this module is available at https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/plural_gender_select.xsd.

4.8.3 Module Fragment Identification Prefix

The fragment identification prefix for the Plural, Gender, and Select module is: `pgs`.

4.8.4 Module Elements

None.

4.8.5 Module Attributes

The attributes defined in the Plural, Gender, and Select module are: `switch`, and `case`.

4.8.5.1 switch

Indicates the variable(s) used to select the message variant, and the kind of “selector” that will be used.

Value description: Text.

The text contains a space-separated list of items, each item containing a selector keyword, followed by colon (:), and followed by the variable name used for selection.

List of allowed selector keywords: `plural`, `ordinal`, `gender`, `select`. No other selector keywords are allowed.

Used in: `<unit>`.

Example:

```
<unit id="tu1" pgs:switch="plural:count gender:host_gender">
...
</unit>
```

4.8.5.2 case

Indicates the value(s) that the `switch` variable(s) should have in order to select the message variant “annotated” with this element.

Value description: Text.

The text contains a space-separated list of values forming a “tuple” used for selection.

Used in: `<segment>`.

Example:

```
<unit id="tu1" pgs:switch="plural:count gender:host_gender">
  <segment id="seg1" pgs:case="1 feminine">...</segment>
</unit>
```

Constraints: the number of space-separated items (variables) in the `switch` attribute of the unit MUST match the number of space-separated values in the `case` attribute.

Also, each value in case should also match the type of selector in the switch attribute.

Valid matching:

- plural and ordinal: a numeric_value or a plural_keyword
 - numeric_value: integer (0, 7, 365) or decimal (3.14) value
 - plural_keyword: zero, one, two, few, many, other
- gender: feminine, masculine, neuter, other, anything else (see "More than three grammatical genders" section at [\[Grammatical Genders\]](#))
- select: the values can be anything, or the other keyword

4.8.6 Examples

4.8.6.1 Plural

[\[ICU MessageFormat\]](#) example:

```
{file_count, plural,
    =0 {You deleted no file.}
    =1 {You deleted one file.}
    other {You deleted # files.}
}
```

XLIFF:

```
<unit id="tu1" pgs:switch="plural:file_count">
  <segment id="seg1" pgs:case="0">
    <source>You deleted no file.</source>
  </segment>
  <segment id="seg2" pgs:case="1">
    <source>You deleted one file.</source>
  </segment>
  <segment id="seg3" pgs:case="other">
    <source>You deleted <ph id="1" disp="file_count"/> files.</source>
  </segment>
</unit>
```

4.8.6.2 Ordinal

```
<unit id="tu1" pgs:switch="ordinal:place">
  <!-- For English ordinals "one" is NOT the same as "1" -->
  <segment id="seg1" pgs:case="one">
    <source>You won <ph id="ph1" disp="place"/>st place</source>
  </segment>
  <segment id="seg2" pgs:case="two">
    <source>You won <ph id="ph2" disp="place"/>nd place</source>
  </segment>
  <segment id="seg3" pgs:case="few">
    <source>You won <ph id="ph3" disp="place"/>rd place</source>
  </segment>
  <segment id="seg4" pgs:case="other">
    <source>You won <ph id="ph4" disp="place"/>th place</source>
  </segment>
</unit>
```

4.8.6.3 Gender

```
<unit id="tu1" pgs:switch="gender:host_gender">
  <segment id="seg1" pgs:case="feminine">
    <source>You are invited to her party</source>
  </segment>
  <segment id="seg2" pgs:case="masculine">
    <source>You are invited to his party</source>
  </segment>
  <segment id="seg3" pgs:case="other">
    <source>You are invited to their party</source>
  </segment>
</unit>
```

Gender also allows for cases other than feminine / masculine / neuter / other in order to support languages with more than 3 genders (see "More than three grammatical genders" section at [\[Grammatical Genders\]](#)).

4.8.6.4 Combinations

```
<unit id="tu1" pgs:switch="gender:host_gender plural:guest_count">
  <segment id="seg1" pgs:case="feminine 0">
    <source><ph id="ph1" disp="host_name"/> did not invite
      anyone to her party.</source>
  </segment>
  <segment id="seg2" pgs:case="feminine 1">
    <source><ph id="ph2" disp="host_name"/> invited
      one guest to her party.</source>
  </segment>
  <segment id="seg3" pgs:case="feminine other">
    <source><ph id="ph3" disp="host_name"/> invited
      <ph id="ph4" disp="guest_count"/> guests to her party.</source>
  </segment>
  <segment id="seg4" pgs:case="masculine 0">
    <source><ph id="ph5" disp="host_name"/> did not invite
      anyone to his party.</source>
  </segment>
  <segment id="seg5" pgs:case="masculine 1">
    <source><ph id="ph6" disp="host_name"/> invited
      one guest to his party.</source>
  </segment>
  <segment id="seg6" pgs:case="masculine other">
    <source><ph id="ph7" disp="host_name"/> invited
      <ph id="ph8" disp="guest_count"/> guests to his party.</source>
  </segment>
  <segment id="seg7" pgs:case="other 0">
    <source><ph id="ph9" disp="host_name"/> did not invite
      anyone to their party.</source>
  </segment>
  <segment id="seg8" pgs:case="other 1">
    <source><ph id="ph10" disp="host_name"/> invited
      one guest to their party.</source>
  </segment>
  <segment id="seg9" pgs:case="other other">
    <source><ph id="ph11" disp="host_name"/> invited
      <ph id="ph12" disp="guest_count"/> guests to their party.</source>
  </segment>
</unit>
```


4.8.7 Recommended practices

There are a few things that an implementer can do to help applications that are not aware of this module.

4.8.7.1 Generate “translator-friendly” identifiers

This can be done by combining a “normal” message ID with information from the case.

Using the short representation (and # for identifiers):

```
segment "$host_name invited @subFlow(tu2 tu3)
        to @subFlow(tu4 tu5 tu6) party." #msgid
unit #g_msgid_plural_gender @switch(plural:guest_count gender:host_gender)
  unit #g_msgid_plural_guest_count @switch(plural:guest_count)
    segment "one guest" #msgid_guest_count_1 @case(1)
    segment "$guest_count guests" #msgid_guest_count_other @case(other)
  unit #g_msgid_gender_host_gender @switch(gender:host_gender)
    segment: "her" #msgid_host_gender_feminine @case(feminine)
    segment: "his" #msgid_host_gender_masculine @case(masculine)
    segment: "their" #msgid_host_gender_other @case(other)
```

So if the older system shows the identifiers to the translators then they can get an idea of what is going on.

Also, using a consistent way of generating identifiers one can improve the leverage.

Translation Memories that give priority to a text-id match vs text-only match will be able to properly leverage even if the English string is the same.

For example:

```
<unit id="tu1" pgs:switch="plural:file_count">
  <segment id="tu1_file_count_1" pgs:case="1">
    <source>You deleted one file.</source>
    <target xml:lang="ro">A i ters un fi ier.</target>
  </segment>
  <segment id="tu1_file_count_few" pgs:case="few">
    <source>You deleted <ph id="1" disp="file_count"/> files.</source>
    <target xml:lang="ro">A i ters
      <ph id="1" disp="file_count"/> fi iere.</target>
  </segment>
  <segment id="tu1_file_count_other" pgs:case="other">
    <source>You deleted <ph id="1" disp="file_count"/> files.</source>
    <target xml:lang="ro">A i ters
      <ph id="1" disp="file_count"/> de fi iere.</target>
  </segment>
</unit>
```

4.8.7.2 Keep the same order of the selectors

By keeping the generated text segments in the same order we can improve Translation Memory leveraging that relies on context (the text before and after the current segment).

Proposed order:

- Plural, ordinal
 - exact selectors firsts, sorted by numerical value (=0, =1, =2, ...)

- the predefined keywords in this order: zero, one, two, few, many, other
- Gender
 - the “free form” selectors in alpha order
 - followed by the predefined keywords in this order: feminine, masculine, neuter, other
- Selection
 - the selectors in alpha order

Keeping a consistent order (the one suggested above or a different one) will improve the leveraging of Translation Memory tools that rely on context (what is before and after the segment) to improve the result.

4.8.7.3 Generate “translator-friendly” notes for plurals

Native speakers intuitively know what the correct plural form is.

But will have a difficult time explaining what the rules are. And even fewer will be able to map those rules to the predefined plural keywords (zero, one, two, few, many).

So generating a note can help a lot.

The notes refer to the segments, so they need a ref attribute pointing to the segment. And since they contain examples for the target language, not the source, they also need an appliesTo="target" attribute.

Romanian example:

```
<unit id="tu1" pgs:switch="plural:file_count">
  <notes>
    <note appliesTo="target" ref="tu1_file_count_1"
      category="plural_examples">1</note>
    <note appliesTo="target" ref="tu1_file_count_few"
      category="plural_examples">0, 2~16, 102, 1002</note>
    <note appliesTo="target" ref="tu1_file_count_other"
      category="plural_examples">20~35, 100, 1000, 10000, 100000</note>
  </notes>
  <segment id="tu1_file_count_1" pgs:case="1">
    <source>You deleted one file.</source>
    <target xml:lang="ro">A i ters un fi ier.</target>
  </segment>
  <segment id="tu1_file_count_few" pgs:case="few">
    <source>You deleted <ph id="1" disp="file_count"/> files.</source>
    <target xml:lang="ro">A i ters
      <ph id="1" disp="file_count"/> fi iere.</target>
  </segment>
  <segment id="tu1_file_count_other" pgs:case="other">
    <source>You deleted <ph id="1" disp="file_count"/> files.</source>
    <target xml:lang="ro">A i ters
      <ph id="1" disp="file_count"/> de fi iere.</target>
  </segment>
</unit>
```

The examples are available from CLDR, in XML formal, or using the ICU library.

A browser friendly table with all the supported rules for all languages, including examples, is available at [\[CLDR Plural Lang\]](#).

Warning: these examples are language dependent. This means one cannot just send a single XLIFF to be translated into several languages (see next item).

4.8.7.4 Generate separate localization packages for each language

Depending on the degree of automation this might mean just a small change in a configuration, or a lot of manual work.

Some companies already use workflows doing this because they pre-populate the <target>, pre-leverage, include translation candidates or glossary info, partial Translation Memories, etc.

Having such a workflow means that the tools can add the missing plural cases depending on the target language.

So even if the source language only has =1 / other cases, the tooling can add the missing ones (few for Romanian, one / few / many for Russian, etc.)

The list of selectors needed for each locale are available from CLDR or ICU APIs.

4.9 Provenance Module

4.9.1 Introduction

This module describes how to include provenance information related to the nature and origin of the localization data or process.

Provenance information is important for the localization process itself, as it helps to document the different steps involved. It can also facilitate the potential reuse of localized material. Such reuse may take the form of a translation memory, where key provenance details, such as the translation date or the agents involved (people and/or tools) can be retrieved. This information may also be crucial for training other products (such as LLMs) that rely on XLIFF files as part of their training data.

4.9.2 Module Namespace and Validation Artifacts

The namespace for the Provenance module is: urn:oasis:names:tc:xliff:pvn:2.3

XML Schema is available at ...

4.9.3 Module Fragment Identification Prefix

The fragment identification prefix for the Provenance module is: pvn

4.9.4 Module Specification

The Provenance module defines X elements and X attributes that enable the embedding of provenance metadata directly within XLIFF CORE elements. In the XLIFF context, provenance data is usually associated with two key aspects: the origin of the source content and the information about the localisation process applied to it. For example, it will allow you to include information about when the original content was extracted, when it was translated or who undertook that task.

4.9.5 Module Elements

The elements defined in the provenance module are <change>, <provenance>, <usage> and <value>

4.9.5.1 Tree Structure

Legend:

1 = one
+ = one or more
? = zero or one
* = zero, one or more

```
<provenance>
  |
  +---<change>
  |
  ?---<usage>
  |
  ?---<value>
```

4.9.6 provenance

Generic container for storing provenance data.

Contains:

-One or more <change> elements.

Attributes:

ref (see <https://docs.oasis-open.org/xliff/xliff-core/v2.2/xliff-extended-v2.2-part2.html#ref>)

4.9.7 change

Information associated with a specific modification or update

Contains:

-Zero or one <usage> element.

-Zero or one <value> element.

Attributes:

-agentName, OPTIONAL

-appliesTo, REQUIRED

-date, OPTIONAL

-intent, OPTIONAL

-orgName, OPTIONAL

-personName, OPTIONAL

-toolName, OPTIONAL

4.9.8 usage

Data generated or used by the processor.

Contains:

-TEXT

Attributes:

-reference, OPTIONAL

-unit, REQUIRED

4.9.9 value

Concrete data reflecting the actual modification .E.g. if this was a quality score it's the new score. If this was the entire target it's the entire target

Contains:

-TEXT?

Attributes:

....

Full example:

```
<xliff srcLang="en" trgLang="es" version="2.2"
  xmlns:pvn="urn:oasis:names:tc:xliff:pvn:2.3"
  xmlns:its="http://www.w3.org/2005/11/its"
  xmlns="urn:oasis:names:tc:xliff:document:2.2">
  <file id="f1">
    <unit id="1" its:locQualityRatingScore="0.968">
      <pvn:provenance>
        <pvn:change agentName="Google Translate" appliesTo="target"
          date="2026-05-05T21:06:00Z">
          <pvn:usage unit="characters">66</pvn:usage>
          <pvn:new>0 seu certificado anual de impuestos está disponible para
            descarga.</pvn:new>
        </pvn:change>
        <pvn:change agentName="TAUS MTQE" appliesTo="its:locQualityRatingScore"
          date="2026-05-05T21:07:00Z">
          <pvn:usage unit="characters">66</pvn:usage>
          <pvn:new>0.968</pvn:new>
        </pvn:change>
        <pvn:change agentName="gpt-5.5" appliesTo="target"
          date="2026-05-05T21:08:00Z"
          intent="This version sounds slightly smoother and more idiomatic: 'para descarga'
            often preferred over 'para descarga' in everyday usage.">
          <pvn:usage unit="tokens">45</pvn:usage>
          <pvn:previous>0 seu certificado anual de impuestos está disponible para
            descarga.</pvn:previous>
          <pvn:new>0 seu certificado anual de impuestos xa está disponible para
            descargar.</pvn:new>
        </pvn:change>
      </pvn:provenance>
      <segment>
        <source>Your annual tax certificate is now available for
          download.</source>
        <target>0 seu certificado anual de impuestos xa está disponible para
          descargar.</target>
      </segment>
    </unit>
  </file>
```

</xliff>

4.9.10 Module Attributes

The attributes defined in the Provenance module are: agentName, appliesTo, date, intent, orgName, personName, toolName, reference, ref and unit.

4.9.11 agentName

This attribute identifies the agent that has performed a specific localization task. The agent in this context is the system that was used to perform a specific task in the localization process, e.g. an MT engine that has automatically translated the content of a given translation unit. If the task was carried out by a human, please use the attribute personName instead.

// the first part was adapted from the definition of "contact-name" in the 1.2 spec. <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#contact-name>. In ITS, there is the org attribute: "The text string is supposed to identify an organizational translation agent as per [Organizational provenance information](https://docs.oasis-open.org/xliff/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org)." https://docs.oasis-open.org/xliff/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org

There is also the "person" attribute defined as: "The text string is supposed to identify a human translation agent as per [Human provenance information](https://docs.oasis-open.org/xliff/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_person)." https://docs.oasis-open.org/xliff/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_person

//we could also just use "agent" and "person" instead of agentName and personName.

//please note that we have a definition for agent in XLIFF 2.X: Agent

any application or tool that generates (creates), reads, edits, writes, processes, stores, renders or otherwise handles XLIFF documents.

Agent is the most general application conformance target that subsumes all other specialized user agents disregarding whether they are defined in this specification or not.

//Another idea would be to use the generic "tool/toolName" instead of "agentName" for this case as well.

Value description:

Text.

Default value:

Undefined.

Used in:

<change>

//In 1.2 "contact-name" was allowed in <phase>.

Example:

```
<unit id="1">
  <pvn:provenance>
    <pvn:change agentName="Google Translate">
      <pvn:change>
        </pvn:provenance>
      <segment>
        <source>Your annual tax certificate is now available for
```

```
download.</source>
<target>0 seu certificado anual de impostos está disponível para
  descarga.</target>
</segment>
</unit>
```

4.9.12 appliesTo

Indicates the element or attribute to which the content of the change applies.

Value description:

Text.

Default value:

Undefined.

Used in:

<change>

Example:

```
<unit id="1">
  <pvn:provenance>
    <pvn:change agentName="Google Translate" appliesTo="target">
    </pvn:change>
  </pvn:provenance>
  <segment>
    <source>Your annual tax certificate is now available for
      download.</source>
    <target>0 seu certificado anual de impostos está disponível para
      descarga.</target>
  </segment>
</unit>
```

4.9.13 date

This attribute indicates when a given element was created or modified.

//original definition in 1.2 of "date": "The date attribute indicates when a given element was created or modified." <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#date>

Value description:

Pattern described in [ISO 8601]: CCYY-MM-DDThh:mm:ssZ. Where: CCYY is the year (4 digits), MM is the month (2 digits), DD is the day (2 digits), hh is the hours (2 digits), mm is the minutes (2 digits), ss is the second (2 digits), and Z indicates the time is UTC time. For example: date="2026-01-25T21:06:00Z"

//taken from xliff 1.2

Default value:

Undefined.

Used in:

<change>

Example:

```
<unit id="1">
  <pvn:provenance>
    <pvn:change agentName="Google Translate" appliesTo="target" date="2026-01-01T21:00:00Z">
    </pvn:change>
  </pvn:provenance>
  <segment>
    <source>Your annual tax certificate is now available for
      download.</source>
    <target>O seu certificado anual de impostos está disponível para
      descarga.</target>
  </segment>
</unit>
```

4.9.14 intent

Rationale for the change.

Value description:

Text.

Default value:

Undefined.

Used in:

<change>

Example:

4.9.15 orgName

This attribute identifies the organization that has mandated or commissioned a specific localization task.

Value description:

Text.

Default value:

Undefined.

Used in:

<change>

Example:

```
<unit id="1">
  <pvn:provenance>
```



```

<pvn:change agentName="Google Translate" orgName="OASIS">
</pvn:change>
</pvn:provenance>
<segment>
<source>Your annual tax certificate is now available for
download.</source>
<target>O seu certificado anual de impostos está disponível para
descarga.</target>
</segment>
</unit>

```

4.9.16 personName

This attribute identifies the human agent that has performed a specific localization task. The agent in this context is the person that was used to perform a specific task in the localization. If the task was carried out using an automated system (e.g. an MT engine), please use the attribute agentName instead.

// the first part was adapted from the definition of “contact-name” in the 1.2 spec. <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#contact-name>. In ITS, there is the org attribute: “The text string is supposed to identify an organizational translation agent as per [Organizational provenance information](https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org).” https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org

there is also the “person” attribute defined as: “The text string is supposed to identify a human translation agent as per [Human provenance information](https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_person).” https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_person

//we could also just use “agent” and “person” instead of agentName and personName.

Value description:

Text.

Default value:

Undefined.

Used in:

<change>

//In 1.2 “contact-name” was allowed in <phase>.

Example:

```

<unit id="1">
<pvn:provenance>
<pvn:change personName="Anxela Loureiro">
<pvn:change>
</pvn:provenance>
<segment>
<source>Your annual tax certificate is now available for
download.</source>
<target>O seu certificado anual de impostos está disponível para
descarga.</target>
</segment>
</unit>

```

4.9.17 toolName

This attribute identifies the name of the localization tool used to create or modify a specific element.

//see tool, tool-id, tool-name and tool-version in 1.2 <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#tool>

In the ITS module, there is also the “tool” attribute defined as: “The text string is supposed to identify a software tool translation agent as per [Tool-related provenance information.](https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#ism_tool)” https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#ism_tool

//we could also just use “tool” instead of toolName.

Value description:

Text.

Default value:

Undefined.

Used in:

<change>

Example:

```
<unit id="1">
  <pvn:provenance>
    <pvn:change personName="Anxela Loureiro" toolName="Swordfish" ref="#s=s1">
      <pvn:change>
        </pvn:provenance>
      <segment id="s1">
        <source>Your annual tax certificate is now available for
          download.</source>
        <target>O seu certificado anual de impostos está disponível para
          descarga.</target>
      </segment>
    </unit>
```

4.9.18 reference

This attribute is used to identify the agent or model referred in the <usage> element.

Value description:

Text.

Default value:

Undefined.

Used in:

<usage>

Example:

4.9.19 ref

Reference. holds a reference for the associated element. (taken from: <https://docs.oasis-open.org/xliff/xliff-core/v2.2/xliff-extended-v2.2-part2.html#ref>)

Value description:

When used in a <https://docs.oasis-open.org/xliff/xliff-core/v2.2/xliff-extended-v2.2-part2.html#note> <provenance> element, the URI value is referring to a specific [<unit>](#).

Default value:

Undefined.

Used in:

<provenance>

Example:

4.9.20 unit

This attribute describes the unit used by the agent or model referred in the <usage> element.

Value description:

One of the following values: **word**, **character**, **token**, or a user-defined value that is composed of a prefix and a sub-value separated by a character : (U+003A).

The prefix is a string uniquely identifying a collection of sub-values for a specific authority. The sub-value is any string value defined by the authority. The prefix xlf is reserved.

Default value:

Undefined.

Used in:

<usage>

Example:

//Proposal by Mathijs:

<provenance> // container element to store provenance data in. Contains one or more change elements

<change> // element that holds all data pertaining to a particular change. It can contain the elements <usage> and <value>. Contains attributes appliesTo, date, agentName, personName, orgName, toolName, intent

<usage> // element that holds processor data. Is a child of change. Contains reference, unit. The text property contains a number.

<value> // element that holds the actual change. Is a child of change. E.g. if this was a quality score it's the new score. If this was the entire target it's the entire target. Can thus have any child elements that are of a valid XLIFF node subtype depending on the appliesTo attribute.

Who	agentName, personName, orgName, toolName
What (part)	appliesTo

What (into what)	value
When	date
Why	intent
For how much?	usage + reference

4.9.21 Module Attributes

The attributes defined in the Provenance module are: agentName, date, personName, orgName, toolName, usageUnit, appliesTo, intent, reference

Used in: <change>, <file>, <unit>, <group> and <note>.

Table x below provides a summary of the provenance information supported by XLIFF 2.x Core and the Provenance module. For completeness, XLIFF 1.2 is also provided as a legacy reference.

		XLIFF 1.2	XLIFF 2.x Core	Provenance module
What (content)	is it?	original	original	
	is being translated/localised?	source-language	srcLang	
		target-language	trgLang	
	does it do?	category	(please note that the "category" attribute in 2.2 is used with a different purpose and in a different element (note instead of file))	domain
		version	version	
		datatype	type (indicates the type of the element, e.g. image, user interface element, etc.) name (resource name, for example, the ID of a string in a Window string table)	datatype
Who (people/system)	is localising/ translating?	contact-name		agentName,
		contact-email		personName,
Where (network)	is it localised [company/country]?	company-name	trgLang	orgID, orgName
		target-language		
		contact-phone		
When (date)	was the XLIFF document created?	date		date
	was the content extracted?			
	was the translation made?			
Why (motivation)?	Is it being localised?	job-id		
How (function)	Is it localised [tool, process, auxiliary material]?	tool-id		toolName

		tool-name		
	Is it being localised?	state	state, subSate	
		state-qualifier		
	much has it cost?			usage usageUnit

4.9.22 agentName

This attribute identifies the agent that has performed a specific localization task. The agent in this context is the system that was used to perform a specific task in the localization process, e.g. an MT engine that has automatically translated the content of a given translation unit. If the task was carried out by a human, please use the attribute `personName` instead.

// the first part was adapted from the definition of “contact-name” in the 1.2 spec. <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#contact-name>. In ITS, there is the `org` attribute: “The text string is supposed to identify an organizational translation agent as per [Organizational provenance information](https://docs.oasis-open.org/xliff/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org).” https://docs.oasis-open.org/xliff/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org

There is also the “person” attribute defined as: “The text string is supposed to identify a human translation agent as per [Human provenance information](https://docs.oasis-open.org/xliff/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_person).” https://docs.oasis-open.org/xliff/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_person

//we could also just use “agent” and “person” instead of `agentName` and `personName`.

//please note that we have a definition for agent in XLIFF 2.X: Agent

any application or tool that generates (creates), reads, edits, writes, processes, stores, renders or otherwise handles XLIFF documents.

Agent is the most general application conformance target that subsumes all other specialized user agents disregarding whether they are defined in this specification or not.

//Another idea would be to use the generic “tool/toolName” instead of “agentName” for this case as well.

Value description:

Text.

Default value:

Undefined.

Used in:

<target>

//In 1.2 “contact-name” was allowed in <phase>.

Example:

```
<unit id="1">
  <segment>
    <source>Your annual tax certificate is now available for
      download.</source>
    <target pvn:agentName="Google Translate">0 seu
      certificado anual de impuestos está disponible para descarga.</target>
```

```
</segment>
</unit>
```

4.9.23 datatype

The datatype attribute specifies the kind of text contained in the element. Depending on that type, you may apply different processes to the data. For example: datatype="winres" specifies that the content is Windows resources which would allow using the Win32 API in rendering the content.

//original definition taken from 1.2, should we adapt it?, they had a list of predefined values. <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#datatype>

Value description:

Text. //should we have a predefined list of values as in 1.2?

Default value:

Undefined.

Used in:

<file>, <unit>.

//In 1.2 it was allowed in <file>.

Example:

//add example

4.9.24 date

This attribute indicates when a given element was created or modified.

//original definition in 1.2 of "date": "The date attribute indicates when a given element was created or modified." <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#date>

Value description:

Date in [ISO 8601] Format. The recommended pattern to use is: CCYY-MM-DDThh:mm:ssZ

Where: CCYY is the year (4 digits), MM is the month (2 digits), DD is the day (2 digits), hh is the hours (2 digits), mm is the minutes (2 digits), ss is the second (2 digits), and Z indicates the time is UTC time. For example: date="2002-01-25T21:06:00Z"

//taken from xliff 1.2

Default value:

Undefined.

Used in:

<file>, <note>, <unit>, <source>, <target>

//In 1.2 it was allowed in <file> and <phase>.

Example:

<unit id="1">

<segment>

<source pvn:date="2026-01-01T21:06:00Z">Your annual tax certificate is now available for download.</source>

<target pvn:date="2026-01-20T21:06:00Z">O seu certificado anual de impostos está disponível para descarga.</target>

</segment>

</unit>

4.9.25 domain

This attribute identifies the domain area of the content. For instance, domain="finance" can be applied to content from a finance-related product.

//original definition in 1.2 of "category": This provides information on the subject of what is being translated. For example: category="medical" for files from a medical related product. <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#category>

Value description:

Text.

Default value:

Undefined.

Used in:

<file>, <unit>.

//In 1.2 it was allowed in <file>.

Example:

```
<unit id="1" pvn:domain="finance">
  <segment>
    <source>Your annual tax certificate is now available for
      download.</source>
    <target>O seu certificado anual de impostos está disponível para
      descarga.</target>
  </segment>
</unit>
```

4.9.26 orgName

//to be defined. Necessary? Would be enough to just have personName and agentName? See https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org

See also company-name in xliff 1.2 <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#company-name>

4.9.27 personName

This attribute identifies the human agent that has performed a specific localization task. The agent in this context is the person that was used to perform a specific task in the localization. If the task was carried using an automated system (e.g. an MT engine), please use the attribute agentName instead.

// the first part was adapted from the definition of “contact-name” in the 1.2 spec. <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#contact-name>. In ITS, there is the org attribute: “The text string is supposed to identify an organizational translation agent as per [Organizational provenance information](https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org).” https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_org

there is also the “person” attribute defined as: “The text string is supposed to identify a human translation agent as per [Human provenance information](https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_person).” https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_person

//we could also just use “agent” and “person” instead of agentName and personName.

Value description:

Text.

Default value:

Undefined.

Used in:

<target>

//In 1.2 “contact-name” was allowed in <phase>.

Example:

```
<unit id="1">
  <segment>
    <source>Your annual tax certificate is now available for
      download.</source>
    <target pvn:personName="Xurxo Loureiro">O seu
      certificado anual de impostos está dispoñible para descarga.</target>
  </segment>
</unit>
```

4.9.28 toolName

This attribute identifies the name of the localization agent used to create or modify a specific element.

//see tool, tool-id, tool-name and tool-version in 1.2 <https://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html#tool>

In the ITS module, there is also the “tool” attribute defined as: “The text string is supposed to identify a software tool translation agent as per [Tool-related provenance information](https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_tool).” https://docs.oasis-open.org/xliff/xliff-core/v2.2/csd01/xliff-extended-v2.2-csd01-part2.html#itsm_tool

//we could also just use “tool” instead of toolName.

Value description:

Text.

Default value:

Undefined.

Used in:

<file>, <unit>, <source> and <target>

//In 1.2 “tool” was allowed in <file>, <phase>, <alt-trans>. Tool-id was allowed in <file>, <phase>, <alt-trans>, <tool>. Tool-name and tool-version were allowed in <tool>.

Example:

```
<unit id="1">
  <segment>
    <source pvn:toolName="Swordfish">Your annual
      tax certificate is now available for download.</source>
    <target pvn:toolName="OmegaT">0 seu
      certificado anual de impuestos está disponible para descarga.</target>
  </segment>
</unit>
```

4.9.29 usage

This attribute contains information about how the processing cost of the translation or review. This is common practice in (automated) translation or review tools where any processing often has a cost associated with it.

Value description:

A positive integer.

Default value:

Undefined.

Used in:

<target>.

Example:

//add example

4.9.30 usageUnit

This attribute indicates the unit of measurement referenced in the “usage” attribute. For example, characters, words, tokens, seconds.

Value description:

Text. //add a predefined list of values?

Default value:

Undefined.

Used in:

<target>.

Example:

//add example

4.9.31 Definitions [to be included in the section Definition of terms of the main spec]

Provenance information refers to information about the nature and origin of the localisation data or process. For example, the attribute srcLang allows you to define the language of the source content.

Translatable text: Textual content subject to translation during the translation or localisation process. It can derive from multiple sources, for example, user interface strings in software products or the textual content of [websites](#). In the context of XLIFF, textual content is usually extracted from the original products and organised in translation units for its later translation.

5 Conformance

1. Document Conformance

- a. XLIFF is an XML vocabulary, therefore conformant XLIFF Documents MUST be well formed and valid [\[XML\]](#) documents.
- b. Conformant XLIFF documents MUST be valid instances of the official Core XML Schema (https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/xliff_core_2.2.xsd) that is a part of this multipart Work Product.
- c. As not all aspects of the XLIFF specification can be expressed in terms of XML Schemas, conformant XLIFF documents MUST also comply with all relevant elements and attributes definitions, normative usage descriptions, and Constraints specified in this specification document.
- d. XLIFF documents MAY contain custom extensions, as defined in the [Extension Mechanisms](#) section.

2. Application Conformance

- a. XLIFF Writers MUST create conformant XLIFF Documents to be considered XLIFF compliant.
- b. Agents processing conformant XLIFF Documents that contain custom extensions are not REQUIRED to understand and process non-XLIFF elements or attributes. However, conformant applications SHOULD preserve existing custom extensions when processing conformant XLIFF documents, provided that the elements that contain custom extensions are not removed according to XLIFF Processing Requirements or the extension's own processing requirements.
- c. All Agents MUST comply with Processing Requirements for otherwise unspecified Agents or without a specifically set target Agent.
- d. Specialized Agents defined in this specification - this is Extractor, Merger, Writer, Modifier, and Enricher Agents - MUST comply with the Processing Requirements targeting their specifically defined type of Agent on top of Processing Requirements targeting all Agents as per point c. above.
- e. XLIFF is a format explicitly designed for exchanging data among various Agents. Thus, a conformant XLIFF application MUST be able to accept XLIFF Documents it had written after those XLIFF Documents were Modified or Enriched by a different application, provided that:
 - i. The processed files are conformant XLIFF Documents,
 - ii. in a state compliant with all relevant Processing Requirements.

3. Backwards Compatibility

- a. Conformant applications are REQUIRED to support XLIFF 2.0, 2.1 and 2.2.
- b. Conformant applications are NOT REQUIRED to support XLIFF 1.2 or previous versions.



Note

XLIFF documents conformant to this specification are not and cannot be conformant to XLIFF 1.2 or earlier versions. If an application needs to support for whatever business reason both XLIFF 2.x and XLIFF 1.2 or earlier, these will need to be supported as separate functionalities.

Appendix A. References

This appendix contains the normative and informative references that are used in this document. While any hyperlinks included in this appendix were valid at the time of publication, OASIS cannot guarantee their long-term validity.

A.1 Normative References

- [RFC 2119] (BCP 14) S. Bradner, *Key words for use in RFCs to Indicate Requirement Levels*, <https://www.rfc-editor.org/rfc/rfc2119> IETF (Internet Engineering Task Force) RFC 2119, March 1997.
- [RFC 8174] B. Leiba, *Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words*, <https://www.rfc-editor.org/rfc/rfc8174>
- [BCP 47] M. Davis, *Tags for Identifying Languages*, <https://www.rfc-editor.org/info/rfc5646> IETF (Internet Engineering Task Force).
- [HTML5] Ian Hickos, Robin Berjon, Steve Faulkner, Travis Leithead, Erika Doyle Navara, Edward O'Connor, Silvia Pfeiffer *HTML5. A vocabulary and associated APIs for HTML and XHTML*, <http://www.w3.org/TR/html5/> W3C Recommendation 28 October 2014.
- [ITS] David Filip, Shaun McCance, Dave Lewis, Christian Lieske, Arle Lommel, Jirka Kosek, Felix Sasaki, Yves Savourel *Internationalization Tag Set (ITS) Version 2.0*, <http://www.w3.org/TR/its20/> W3C Recommendation 29 October 2013.
- [NOTE-datetime] M. Wolf, C. Wicksteed, *Date and Time Formats*, <http://www.w3.org/TR/NOTE-datetime> W3C Note, 15th September 1997.
- [RFC 3987] M. Duerst and M. Suignard, *Internationalized Resource Identifiers (IRIs)*, <https://www.ietf.org/rfc/rfc3987.txt> IETF (Internet Engineering Task Force) RFC 3987, January 2005.
- [RFC 7303] H. Thompson and C. Lilley, *XML Media Types*, <https://www.tools.ietf.org/html/rfc7303> IETF (Internet Engineering Task Force) RFC 7303, July 2014.
- [UAX #9] M. Davis, A. Lanin, A. Glass, *UNICODE BIDIRECTIONAL ALGORITHM*, <http://www.unicode.org/reports/tr9/tr9-35.html> Unicode Bidirectional Algorithm, May 18, 2016.
- [UAX #15] M. Davis, K. Whistler, *UNICODE NORMALIZATION FORMS*, <http://www.unicode.org/reports/tr15/tr15-44.html> Unicode Normalization Forms, February 24, 2016.
- [Unicode] The Unicode Consortium, *The Unicode Standard*, <http://www.unicode.org/versions/Unicode9.0.0/> Mountain View, CA: The Unicode Consortium, June 21, 2016.
- [XML] W3C, *Extensible Markup Language (XML) 1.0*, <http://www.w3.org/TR/xml/> (Fifth Edition) W3C Recommendation 26 November 2008.
- [XML namespace] W3C, *Schema document for namespace* <http://www.w3.org/XML/1998/namespace> <http://www.w3.org/2001/xml.xsd> [<http://www.w3.org/2009/01/xml.xsd>]. at <https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/informativeCopiesOf3rdPartySchemas/w3c/xml.xsd> in this distribution
- [XML Catalogs] Norman Walsh, *XML Catalogs*, <https://www.oasis-open.org/committees/download.php/14809/xml-catalogs.html> OASIS Standard V1.1, 07 October 2005.

[**XML Schema**] W3C, *XML Schema*, refers to the two part standard comprising [[XML Schema Structures](#)] and [[XML Schema Datatypes](#)] (Second Editions) W3C Recommendations 28 October 2004.

[**XML Schema Datatypes**] W3C, *XML Schema Part 2: Datatypes*, <http://www.w3.org/TR/xmlschema-2/> (Second Edition) W3C Recommendation 28 October 2004.

[**XML Schema Structures**] W3C, *XML Schema Part 1: Structures*, <https://www.w3.org/TR/xmlschema-1/> (Second Edition) W3C Recommendation 28 October 2004.

A.2 Informative References

[**LDML**] *Unicode Locale Data Markup Language* <http://unicode.org/reports/tr35/>

[**SRX**] *Segmentation Rules eXchange* <http://www.unicode.org/uli/pas/srx/>

[**UAX #29**] M. Davis, *UNICODE TEXT SEGMENTATION*, <http://www.unicode.org/reports/tr29/> Unicode text Segmentation.

[**XML I18N BP**] *Best Practices for XML Internationalization*, 13 February 2008, <http://www.w3.org/TR/xml-i18n-bp/> W3C Working Group.

[**ICU MessageFormat**] *ICU MessageFormat Class* https://unicode-org.github.io/icu/user-guide/format_parse/messages/#messageformat

[**Grammatical Genders**] *List of languages by type of grammatical genders* https://en.wikipedia.org/wiki/List_of_languages_by_type_of_grammatical_genders#More_than_three_grammatical_genders

[**CLDR Plural Spec**] *The CLDR spec for Plural Rules*: <https://cldr.unicode.org/index/cldr-spec/plural-rules>

[**CLDR Plural Lang**] *Language Plural Rules (all languages)*: http://www.unicode.org/cldr/charts/latest/supplemental/language_plural_rules.html

[**CLDR Plural**] *CLDR plural files (plurals.xml & ordinals.xml in core.zip)*: <https://www.unicode.org/Public/cldr/44/>

[**ICU4C API**] *ICU4C APIs*: https://unicode-org.github.io/icu-docs/apidoc/released/icu4c/classicu_1_1PluralRules.html

[**ICU4J API**] *ICU4J APIs*: <https://unicode-org.github.io/icu-docs/apidoc/released/icu4j/com/ibm/icu/text/PluralRules.html>

Appendix B. MIME Type for XLIFF Version 2.0 and Later Releases (Normative)

A MIME type (Multipurpose Internet Mail Extensions type) is a two-part identifier for file formats and format content transmitted on the Internet. The MIME type is the mechanism used to tell a client application the type of document being transferred from a server. It is important that servers are set up correctly so that the correct MIME type is transferred with each document.

XLIFF is registered in the [IANA Media Types Registry](#) as `application/xliff+xml`.

Appendix C. XLIFF Grammar Files

The basic grammar and structure of XLIFF 2.3 is defined using several XML Schemas and one XML catalog. The module schemas are specifically referenced from their respective modules.

1. XLIFF Core [\[XML Schema\]](#),
https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/xliff_core_2.2.xsd
2. [\[XML Catalog\]](#) of XLIFF Defined XML Schemas,
<https://docs.oasis-open.org/xliff/xliff-core/v2.3/cs/schemas/catalog.xml>
3. XML Schemas of XLIFF Modules are referenced from those modules.

C.1 XML Schemas Tree

Core XML Schema

- |
- +---Candidates Module XML Schema
- |
- +---Glossary Module XML Schema
- |
- +---Format Style Module XML Schema
- |
- +---Metadata Module XML Schema
- |
- +---Resource Data Module XML Schema
- |
- +---Size and Length Restriction Module XML Schema
- |
- +---Validation Module XML Schema
- |
- +---ITS Module XML Schema (W3C namespace subset)
- |
- +---ITS Module XML Schema (additional attributes)
- |
- +---Plural, Gender, and Select Module

Appendix D. Specification Change Tracking (Informative)

This is to facilitate human tracking of changes between XLIFF Versions 2.3 and 2.2.

1. Produced two versions of the specification:

Part 1: Core - simplified version that does not include optional modules.

Part 2: Extended - complete version, including all modules.

2. Changed namespace for the core module to `urn:oasis:names:tc:xliff:document:2.2`.
3. Added an optional `ref` attribute to `<note>` element.
4. Changed the type of `version` attribute to an enumeration containing 2.0, 2.1 and 2.2 as valid values.
5. Allowed an optional `<notes>` element at the start of `<xliff>`.
6. Allowed an optional `<mda:metadata>` element at the start of `<xliff>`.
7. Removed references to Schematron, NVDL and Test Suite from this specification.
8. Updated import references to XLIFF Core Schema in `matches.xsd`, `resource_data.xsd` and `validation.xsd`.
9. Removed the informative Change Tracking Extension. This module had been already demoted in the previous XLIFF version (2.1) but left in the specification for informative purposes. During the development of XLIFF 2.2, the TC did not work on this item. Therefore a decision was taken to remove it completely from the current specification to reduce its volume.
10. Added new [Plural, Gender, and Select Module](#).
11. Updated [Appendix B](#) with the official MIME type listed in IANA Media Type Registry.
12. Allowed an optional `<notes>` element in `<res:resourceItem>`.
13. Removed references to third-party XML Schemas from [XLIFF Grammar Files](#).

In spite of the above mentioned changes, fixes, clarifications, and additions, the practical workings of the previous versions of the XLIFF Core have not been affected.

All valid XLIFF 2.0 and 2.1 files are valid XLIFF 2.2 files. The changes introduced in version 2.2 are designed to maintain compatibility with versions 2.0 and 2.1.



Note

NVDL and Schematron files used in previous versions of XLIFF are available at <https://github.com/oasis-tcs/xliff-xliff-22/tree/master/xliff-21>.

Appendix E. Acknowledgements

The following individuals have participated in the creation of this specification and are gratefully acknowledged:

- Morado Vázquez, Lucía - University of Geneva
- Nita, Mihai - Google Inc.
- Raya, Rodolfo M. - Individual
- Sonnemans, Mathijs - Individual
- Umaoka, Yoshito - Individual

Appendix F. Revision History

Table F.1. Tracking of changes in response to the 1st Public Review

Revision	Date	Editor	Changes Made
CSD02	12 November, 2024	Rodolfo M. Raya	- Updated Specification Change Tracking appendix with additional information about the removal of the informative Change Tracking Extension. - Removed compact form (non-XLIFF) examples from Plural, Gender, and Select Module module to avoid potential confusions. - Corrected link to grammar file in Plural, Gender, and Select Module.

Table F.2. Tracking of changes in response to the 2nd Public Review

Revision	Date	Editor	Changes Made
CSD03	17 February, 2025	Rodolfo M. Raya	- Updated Specification Change Tracking to add the removal of third-party XML Schemas from this specification. - Corrected links to third-party XML Schemas in the grammar files listed in XLIFF Grammar Files.

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